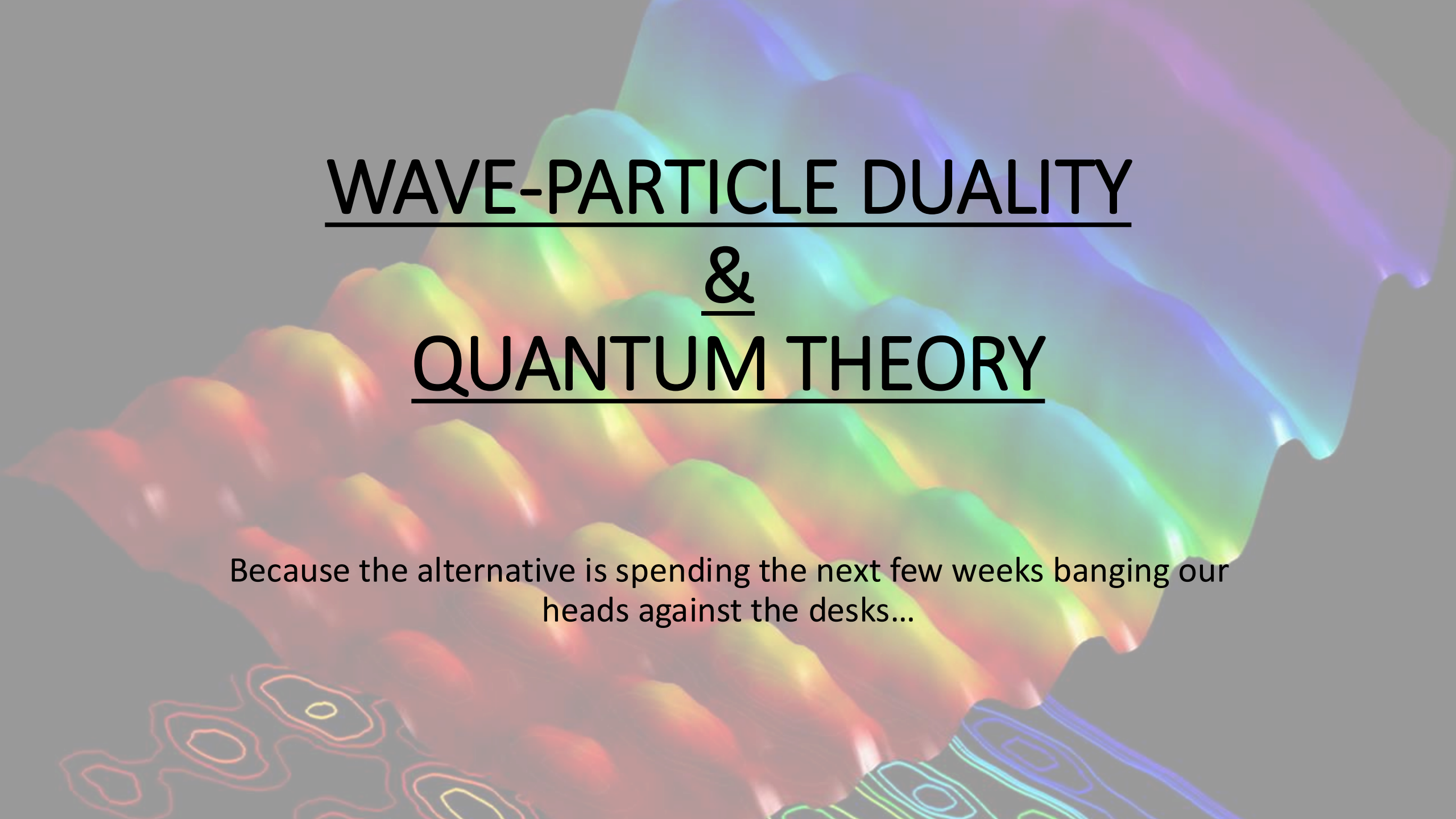




WAVE-PARTICLE DUALITY & QUANTUM THEORY

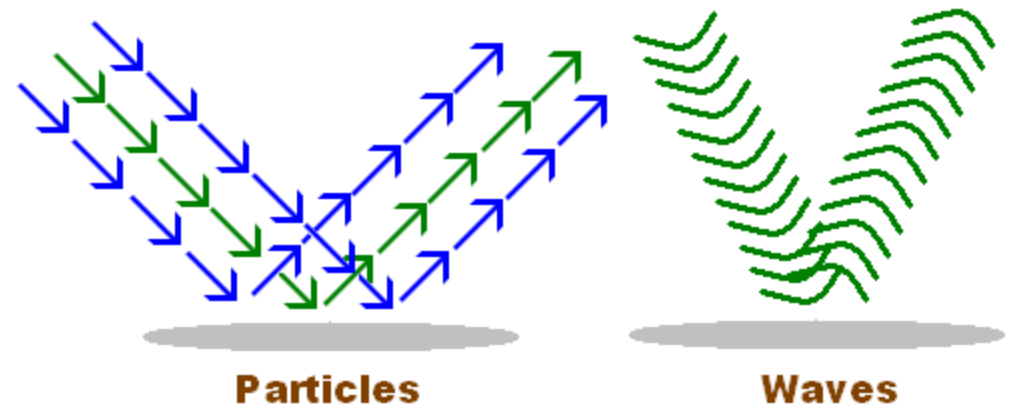
Because the alternative is spending the next few weeks banging our heads against the desks...

LIGHT IS A

WAVE!

WAVE MODEL VS PARTICLE MODEL

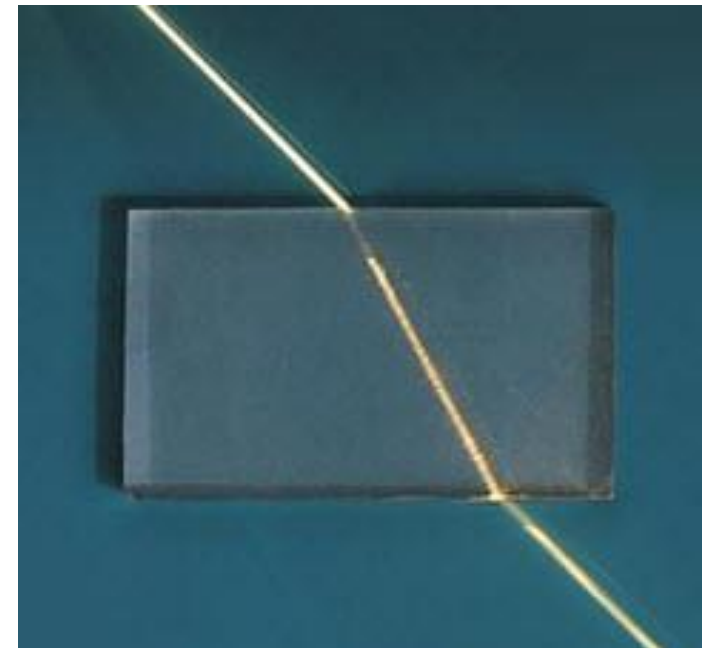
- So, is light a particle or a wave?
- Sir Isaac Newton explained light in terms of particles that he called *corpuscles*, with each different colour of the spectrum representing a different type of particle.
- Christian Huygens proposed a model of light as a wave.
- Both models could adequately describe certain observable phenomena, such as reflection:



Particles and wave Reflected by a Mirror

WAVE MODEL VS PARTICLE MODEL

- However, neither model was quite perfect. For example, Newton's Corpuscular Theory predicted that light would speed up when travelling through a solid material (i.e. glass), whereas the Wave Theory predicted that light would, in fact, slow down.
- Although the technology did not exist at the time for either scientist to test their rival claims, we now know that light does, in fact, slow down as it enters any medium that is more optically dense than a vacuum, resulting in *refraction*.



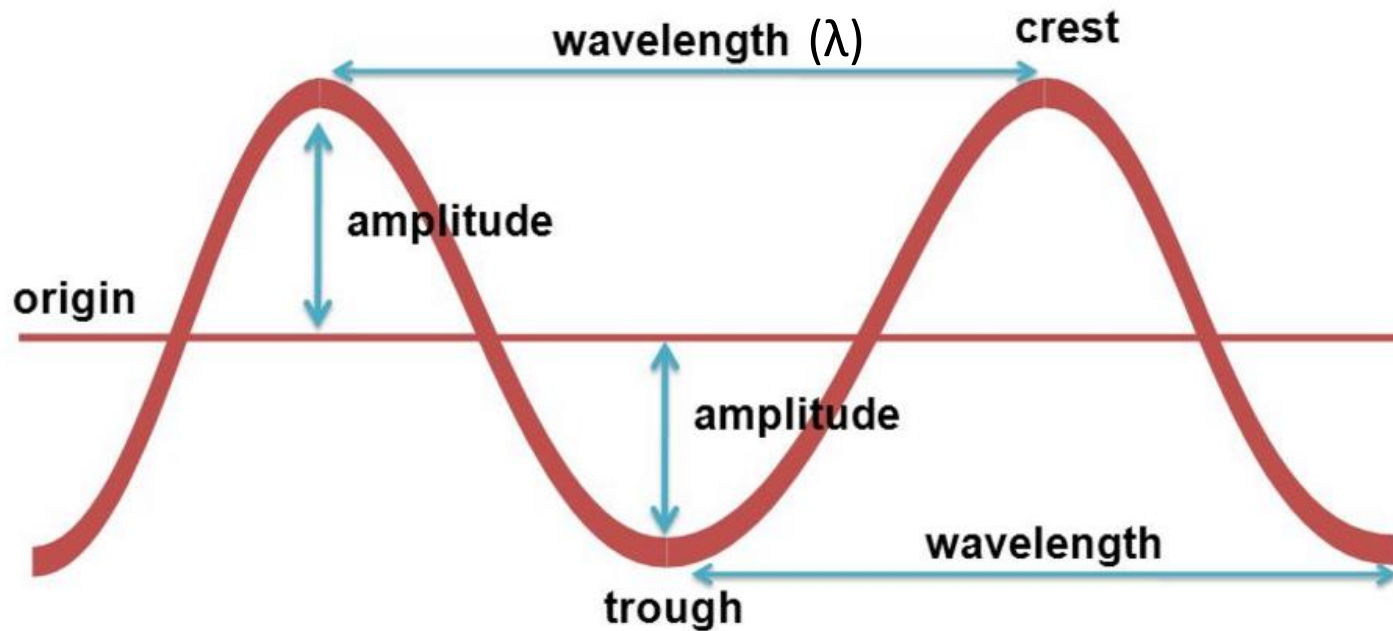
WAVE MODEL VS PARTICLE MODEL

- Nonetheless, each theory has supporting evidence... along with phenomena that they cannot explain (which we shall cover in due course):

Phenomenon	Can be explained in terms of waves.	Can be explained in terms of particles.
<u>Reflection</u>	 ✓	 ✓ ?
<u>Refraction</u>	 ✓	 ✓ ?
<u>Interference</u>	 ✓	 ✗
<u>Diffraction</u>	 ✓	 ✗
<u>Polarization</u>	 ✓	 ✗
<u>Photoelectric effect</u>	 ✗	 ✓

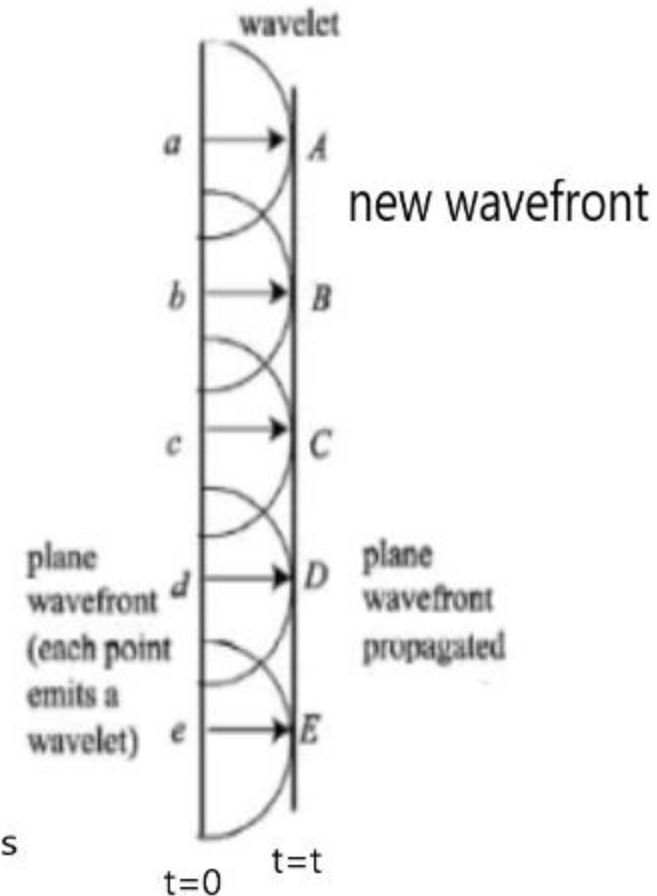
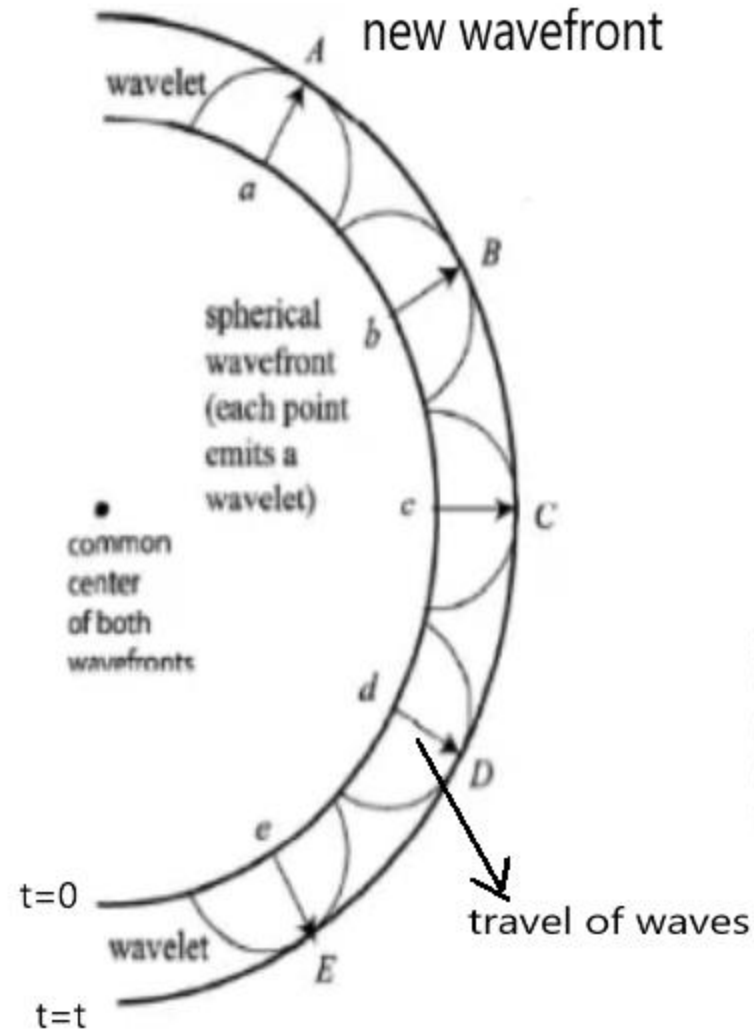
MODEL OF A WAVE

- Before we get too carried away, let's recap some wave-terminology that should hopefully be familiar:



- Period (T) = Time for the wavefront to travel a distance of λ .
- Frequency (f) = $1/T$ = the number of waves to pass a point in 1 second (Hz).
- Amplitude is a measure of the energy carried by the wave.

HUYGENS' PRINCIPLE



DISPERSION

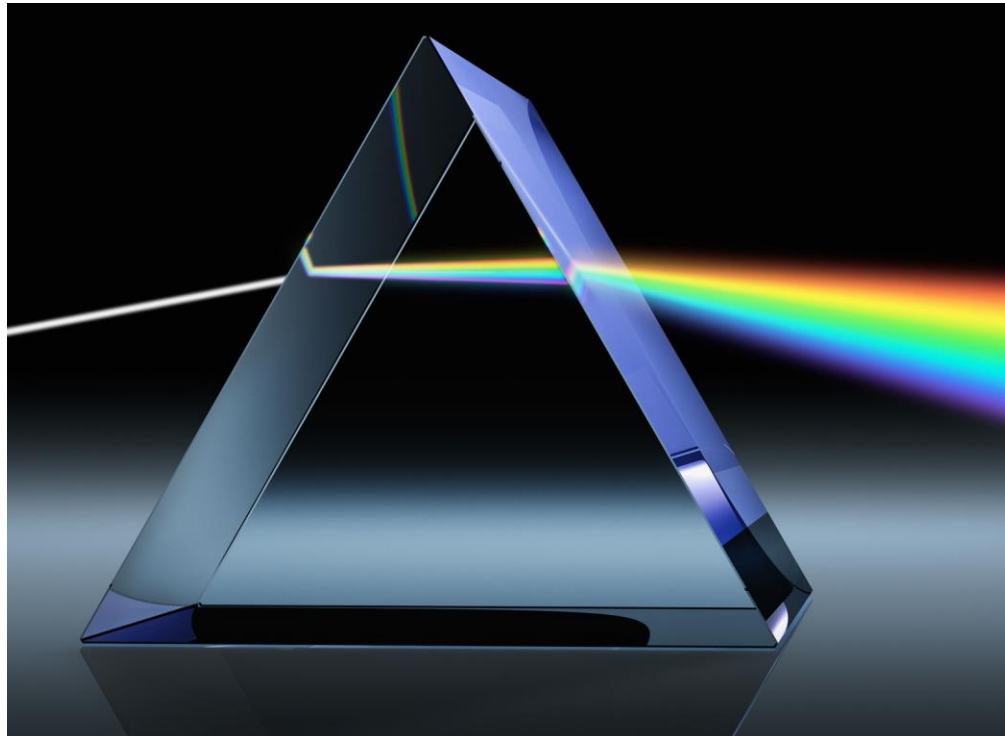
- White light is comprised of the colours of the rainbow. When white light passes from a vacuum to glass, the light waves slow down. Frequency remains unchanged, so according to the Wave Equation, wavelength must decrease.

$$v = f \lambda$$

- The wavelengths of the different colours are compressed by slightly different amounts, so each colour travels at a slightly different speed, and is therefore refracted by a slightly different amount. Consequently, the colours of white light can be separated out. This is *dispersion*.

DISPERSION

- Longer wavelengths (red) travel faster in the new medium and so are refracted least. Shorter wavelengths (violet) travel slower in the new medium and are refracted more.

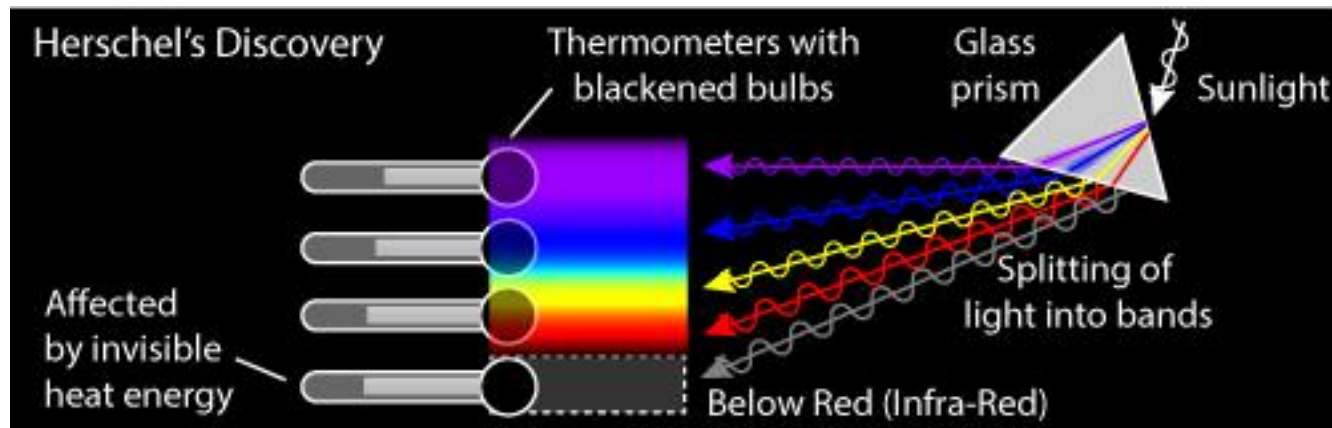


DISPERSION

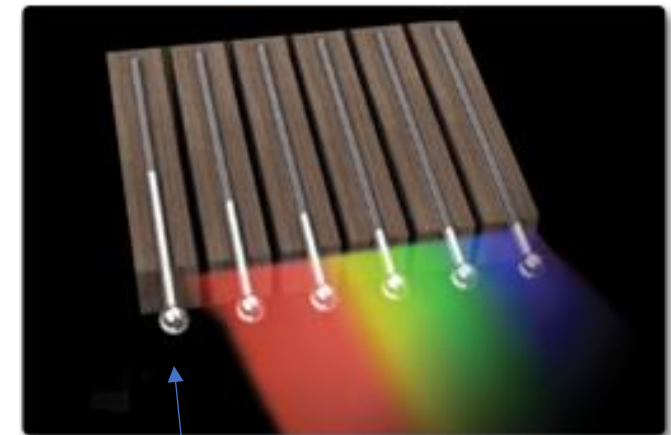
- This was how Sir Isaac Newton was able to demonstrate that white light was comprised of multiple colours. The colours are (as every child knows): Red, Orange, Yellow, Green, Blue, Violet (ROYGBV).
- Newton was a bit of a superstitious man (as were many in his time), and believed that the number 7 was somehow sacred or mystical. For this reason, he slipped another colour into the rainbow that *isn't really there*... Indigo.
- This just goes to show that nobody's perfect. That said, this is the guy who first described gravity and, in so doing, basically invented calculus. Still, the mistake persists in some Physics texts even today.

DISPERSION

- Dispersion actually led to the realisation that the electromagnetic spectrum consisted of more than just the visible spectrum (William Herschel, circa 1800). Herschel was trying to establish which colour of light carried the most energy...



This thermometer, meant as a control, turned out hotter than the thermometers exposed to visible light; it was measuring infrared!



DIFFRACTION

- When a wave passes through a narrow opening, it bends. This bending is significant when the size of the opening/obstacle/aperture is similar in size to the wavelength (or smaller).
*Similar in size = typically within an order of magnitude ($\times 10$).

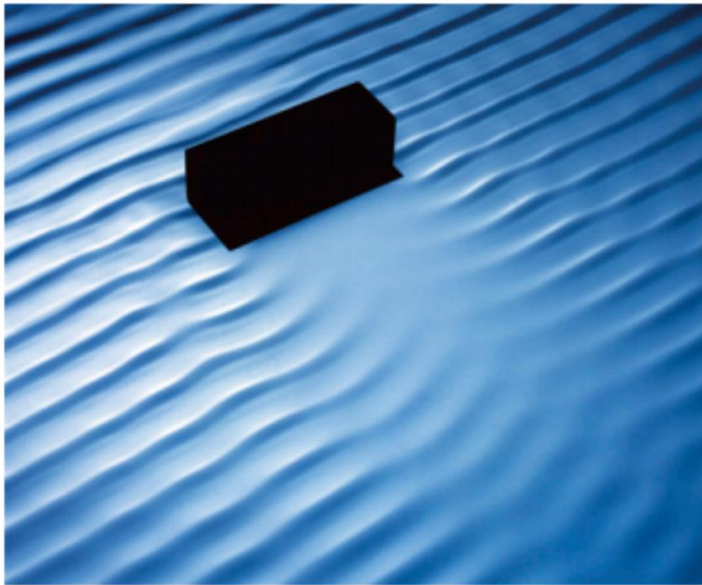


FIGURE 7.1.18 Water waves will bend around an obstacle. Sound waves diffract as well, allowing you to hear around corners.

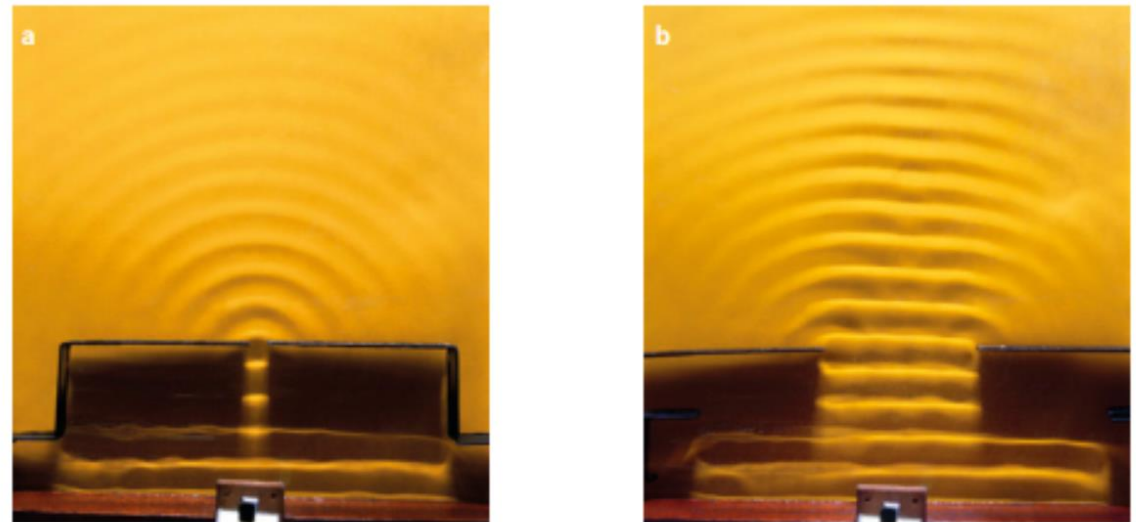
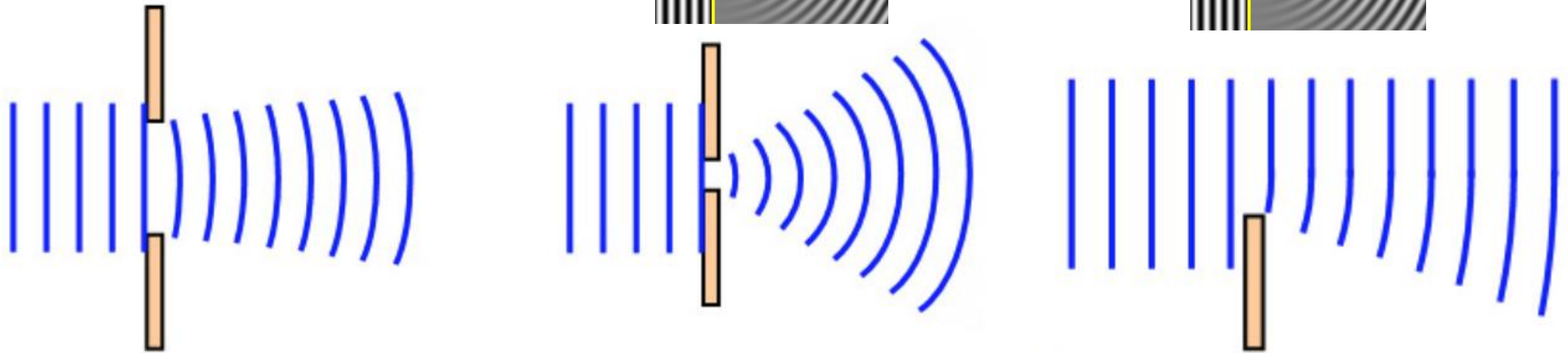


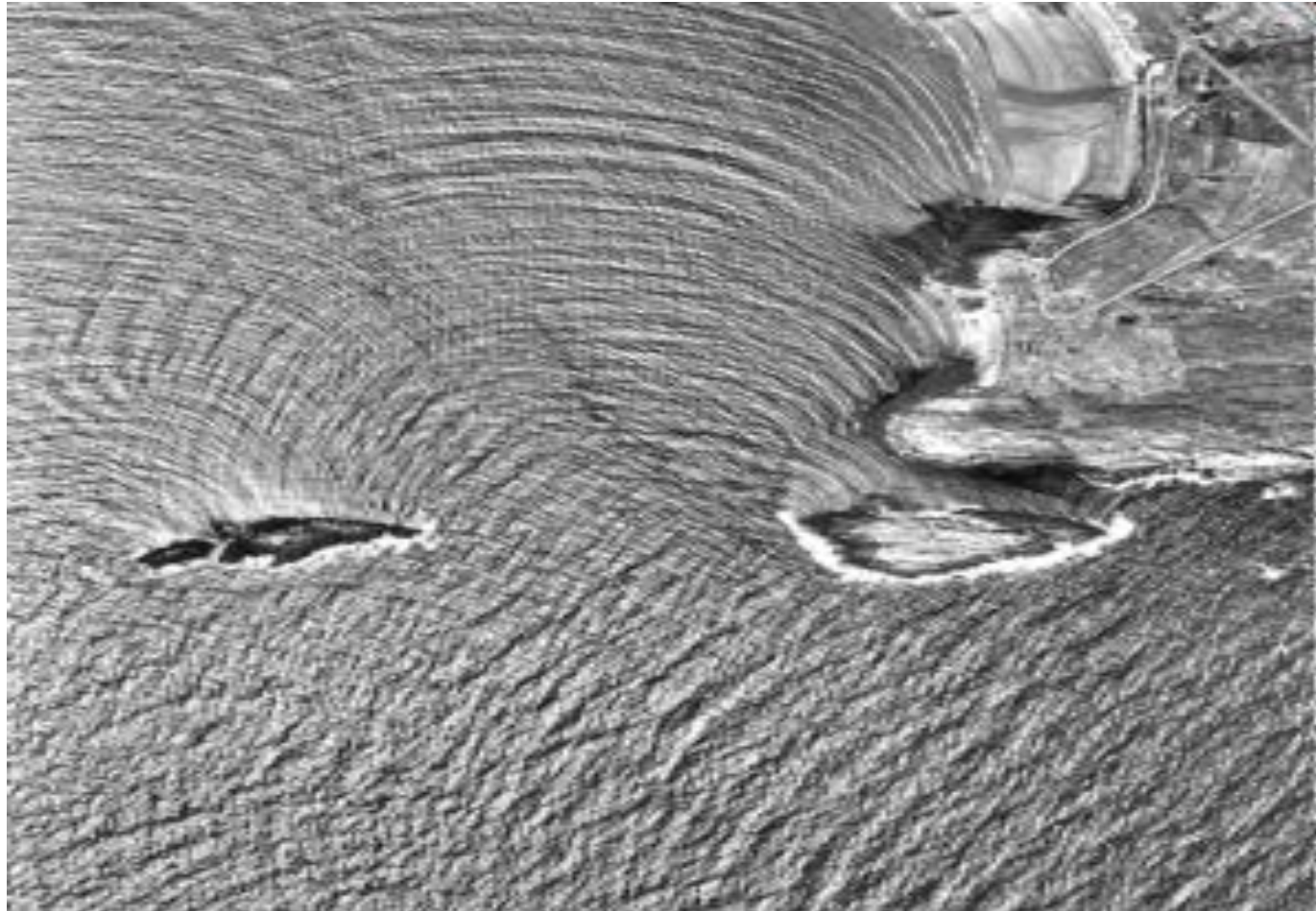
FIGURE 7.1.21 The diffraction of water waves in a ripple tank. (a) Significant diffraction occurs when the wavelength approximates the slit width, i.e. $\lambda = w$. (b) As the gap increases, i.e. $\lambda \ll w$, diffraction becomes less obvious, but is still present.

DIFFRACTION



- Diffraction is useful for explaining why sound can travel around corners (but not light – why not?).

DIFFRACTION (IN THE SEA)



DIFFRACTION

- Diffraction patterns: the wavefronts passing through the gap interfere with themselves, creating patterns of constructive and destructive interference (more on this later).

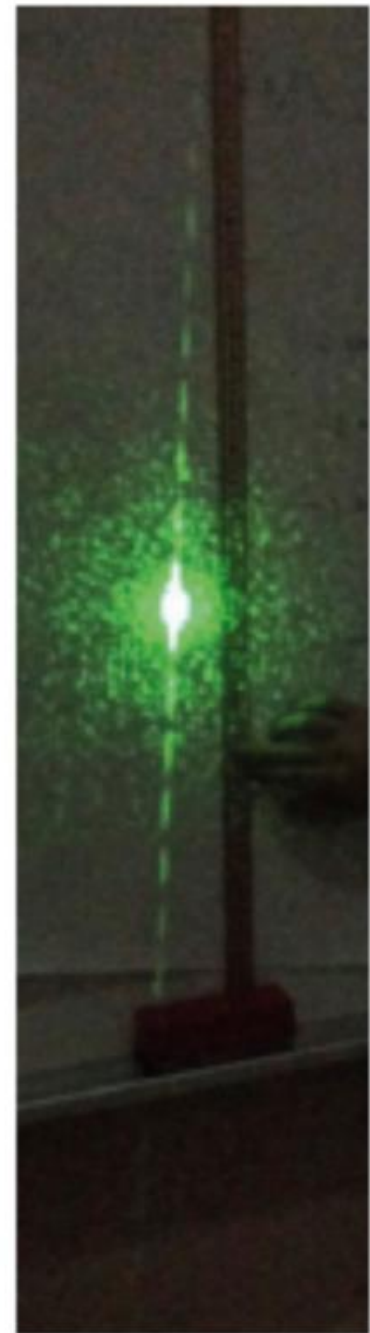
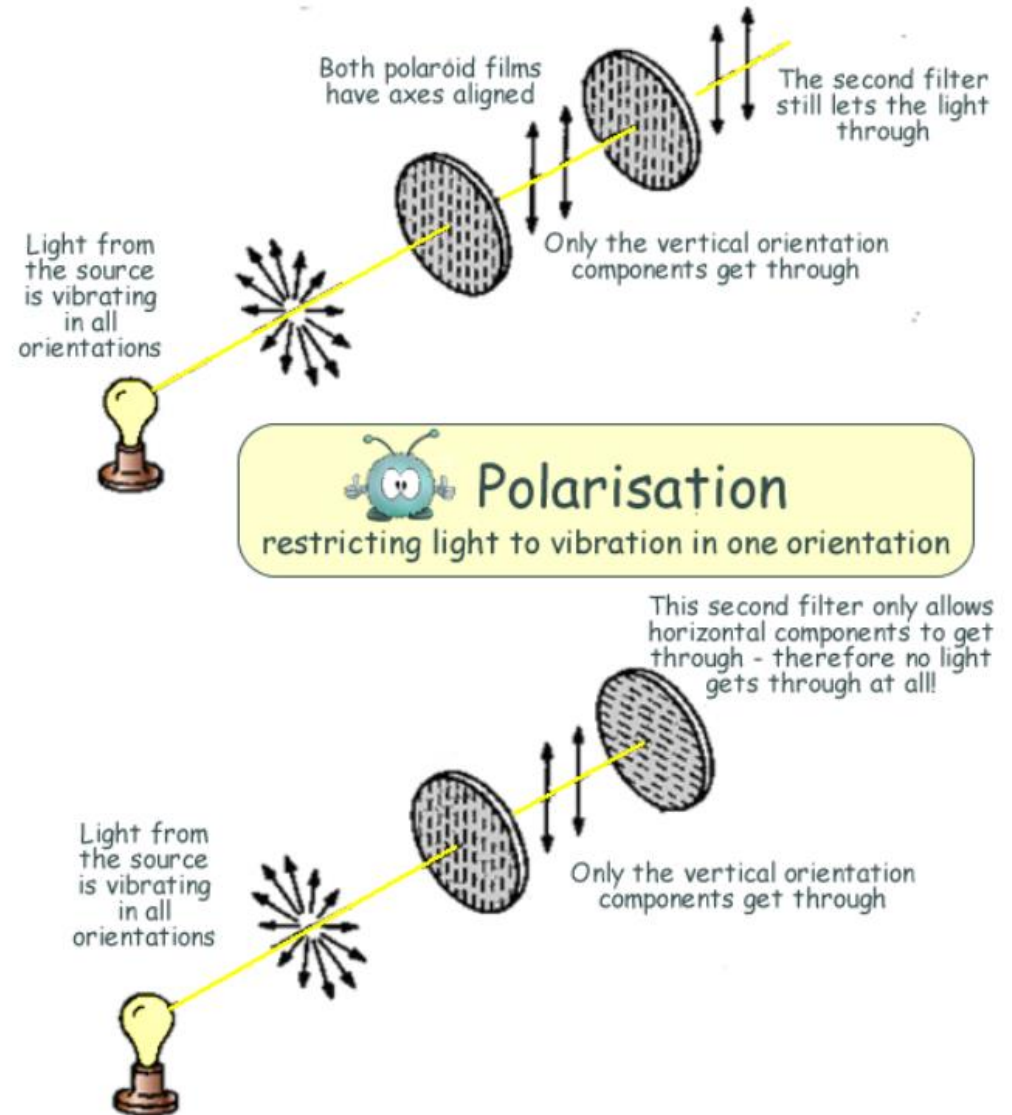


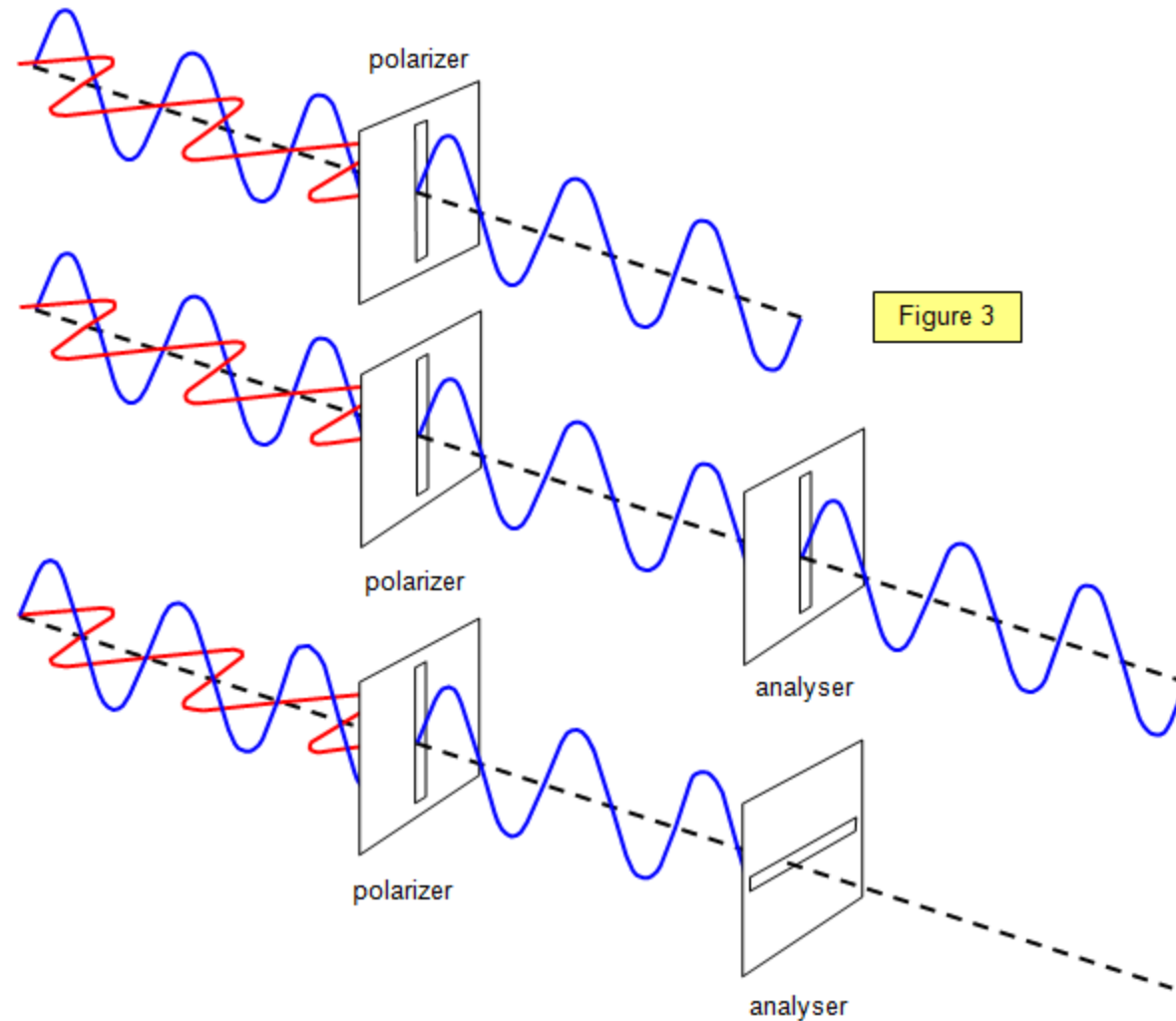
FIGURE 7.1.19 The interference pattern obtained from a cotton thread.

POLARISATION

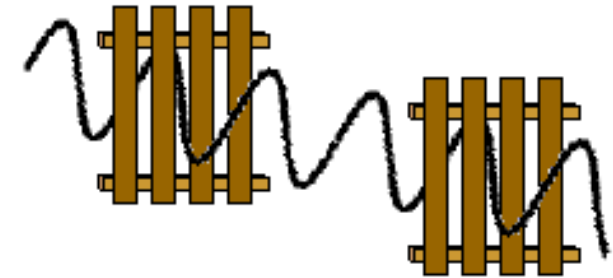
- Polarisation is one of the most convincing pieces of evidence for the wave theory of light.
- Polaroid filters only allow light to pass through in certain orientations. The result is that unpolarised light turns into polarised light.
- A wave can only pass through a filter if it has the correct orientation (or if a vector-component of the wave is in the correct orientation).



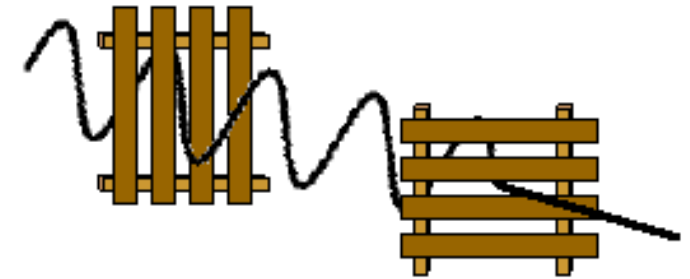
POLARISATION



The Picket Fence Analogy

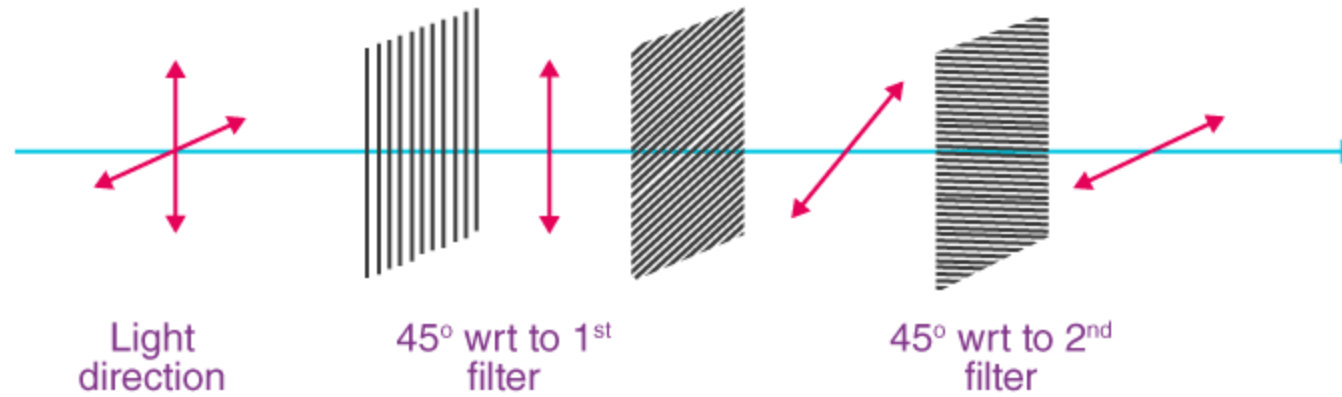


When the pickets of both fences are aligned in the vertical direction, a vertical vibration can make it through both fences.

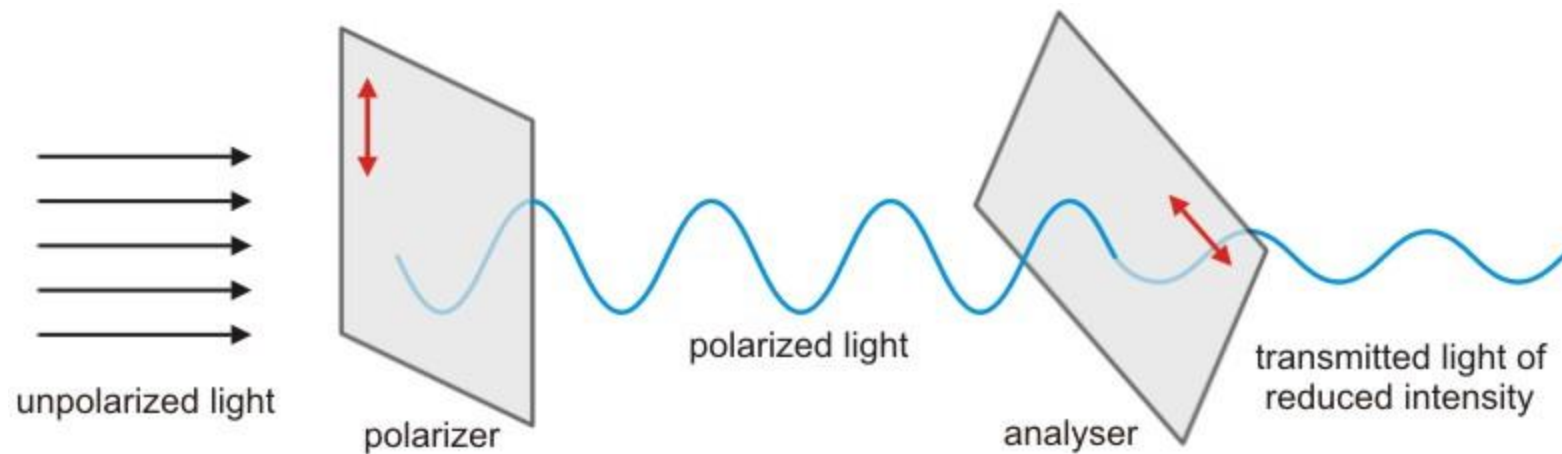


When the pickets of the second fence are horizontal, vertical vibrations which make it through the first fence will be blocked.

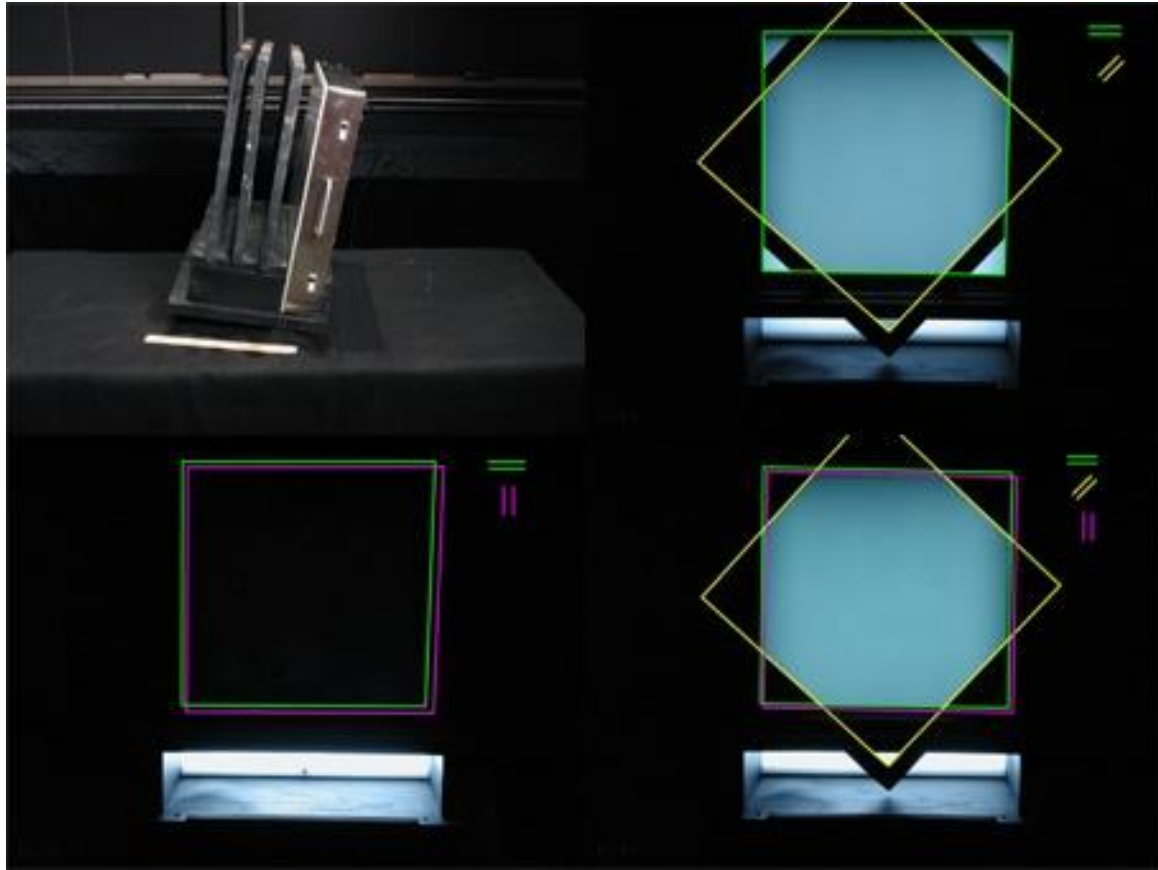
POLARISATION



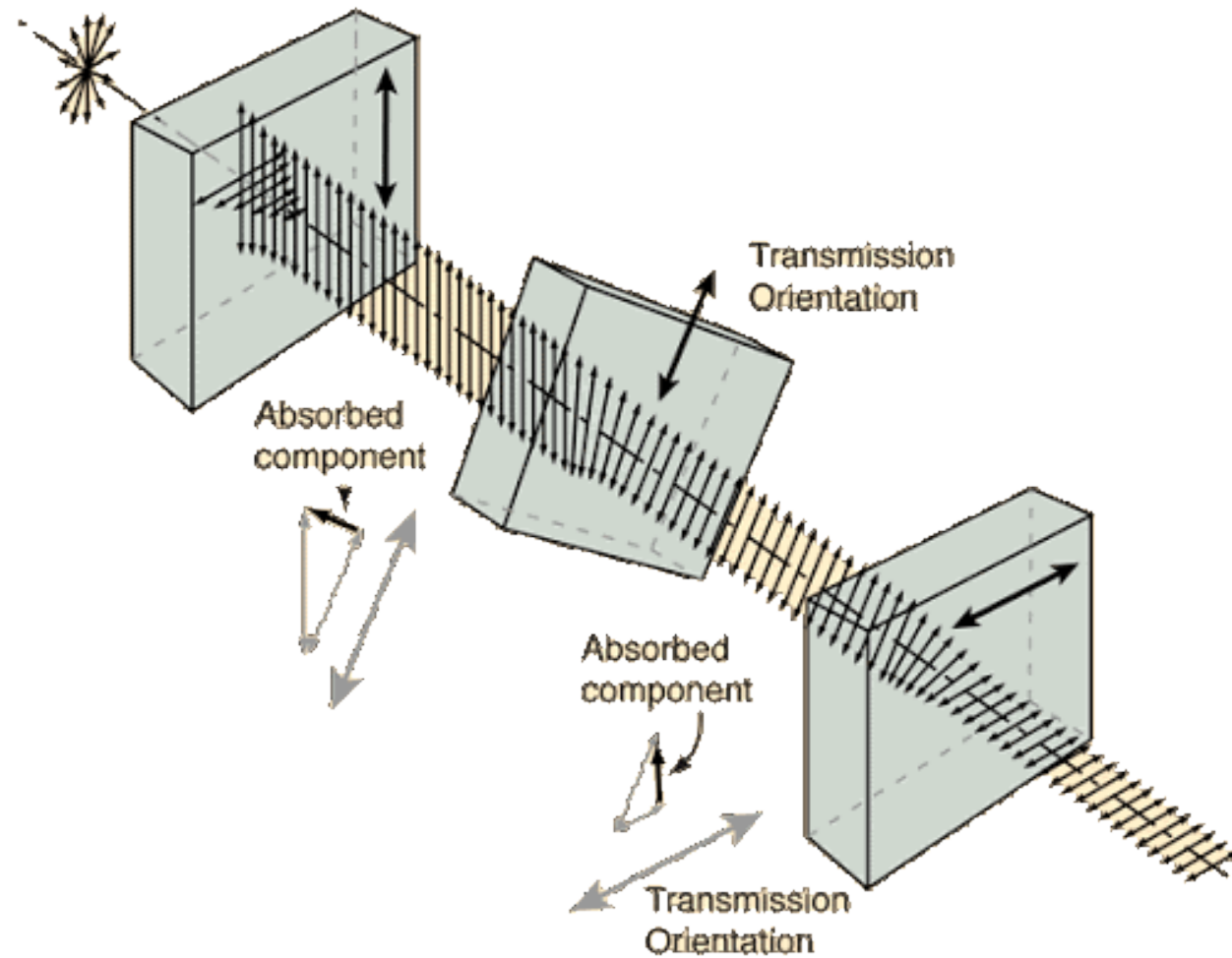
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POLARISATION



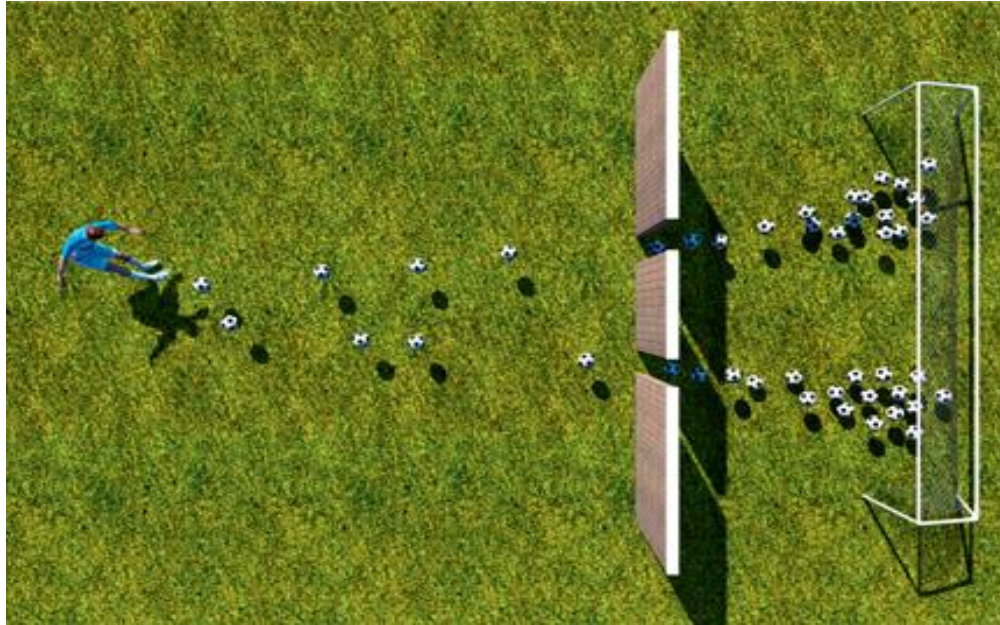
POLARISATION



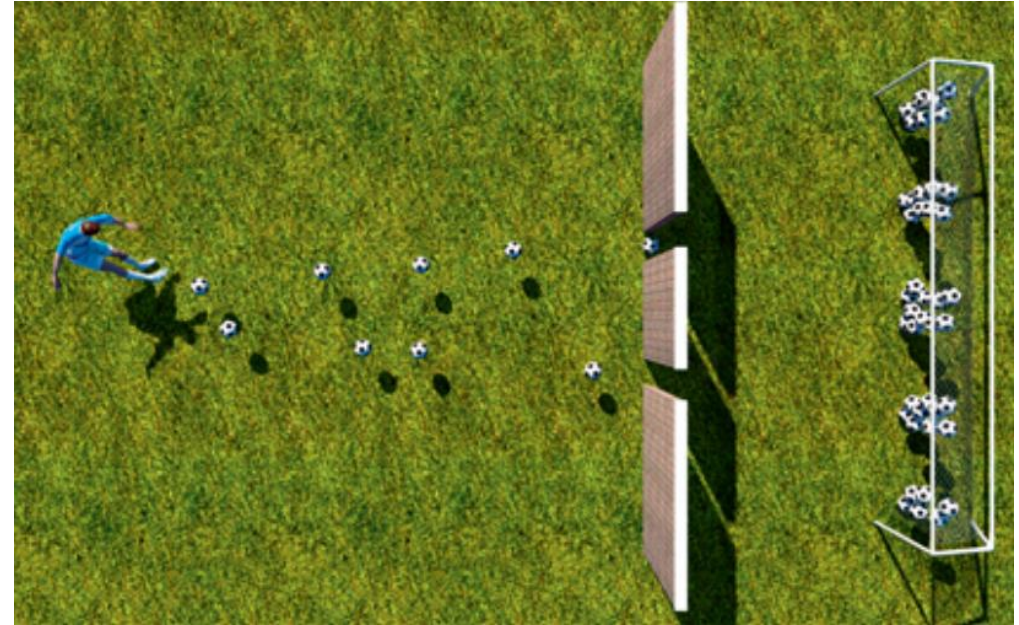
YOUNG'S DOUBLE-SLIT EXPERIMENT

- In 1801, Thomas Young devised a brilliant experiment to test the nature of light; the double-slit experiment. The experiment was predicted to have one of two different outcomes, dependant upon either a particle-model or a wave-model for light.
- A single, monochromatic (single-frequency) light-source is shone through an opaque partition with two uniform slits cut into it, and onto a screen.
- The particle-model predicts two big 'patches' of light on the screen, one in front of each slit.
- The wave-model predicts an alternating pattern of light and dark patches or lines; an interference pattern.

YOUNG'S DOUBLE-SLIT EXPERIMENT

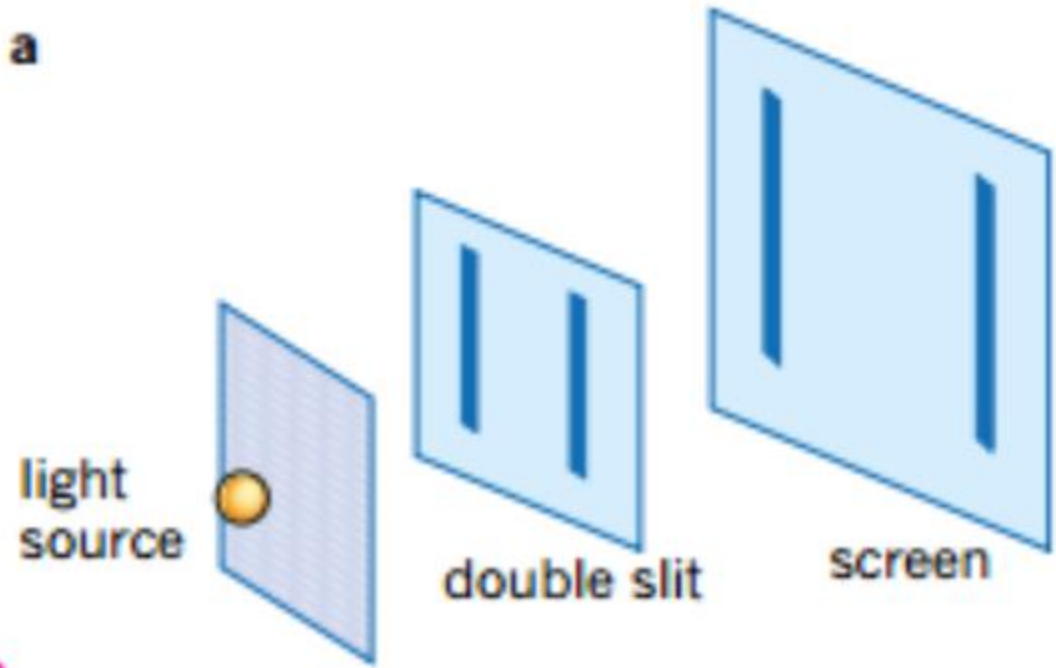


Particle-Model Prediction

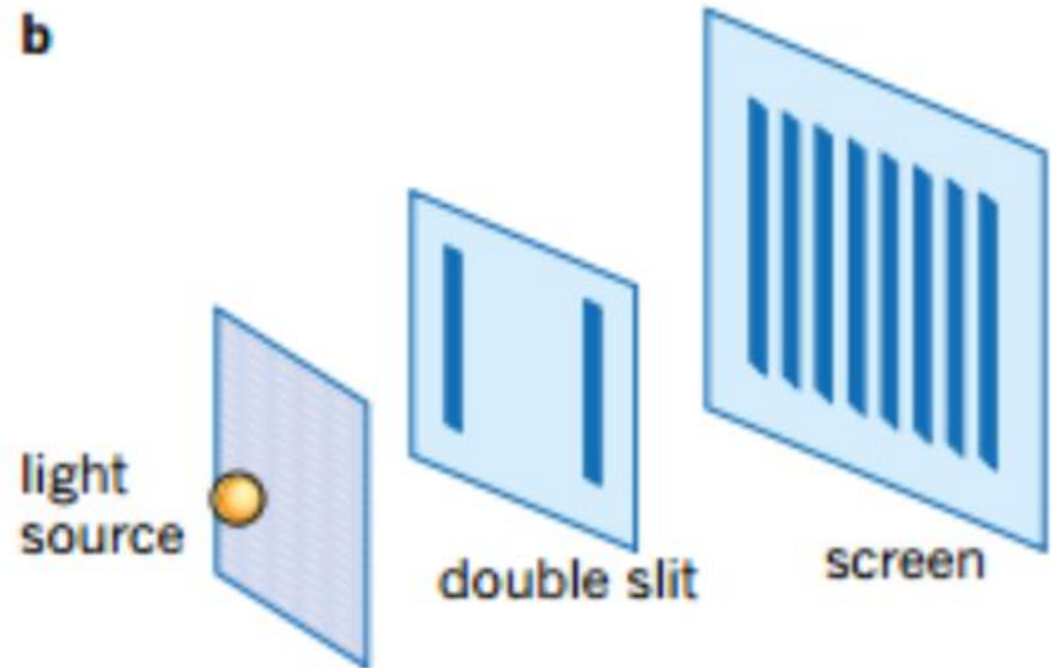


Wave-Model Prediction

YOUNG'S DOUBLE-SLIT EXPERIMENT



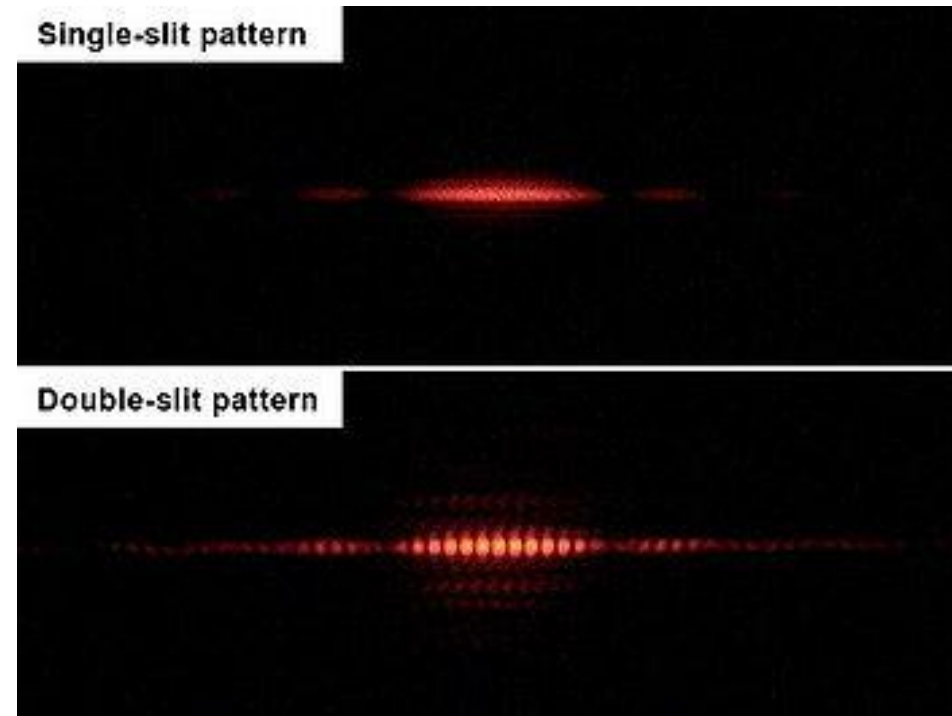
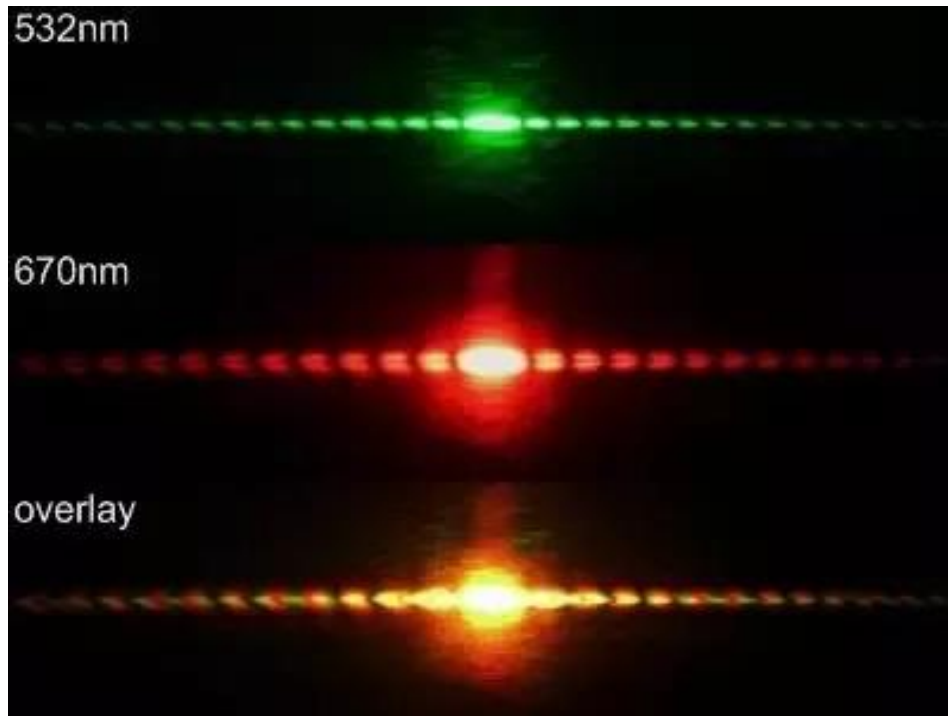
Particle-Model Prediction



Wave-Model Prediction

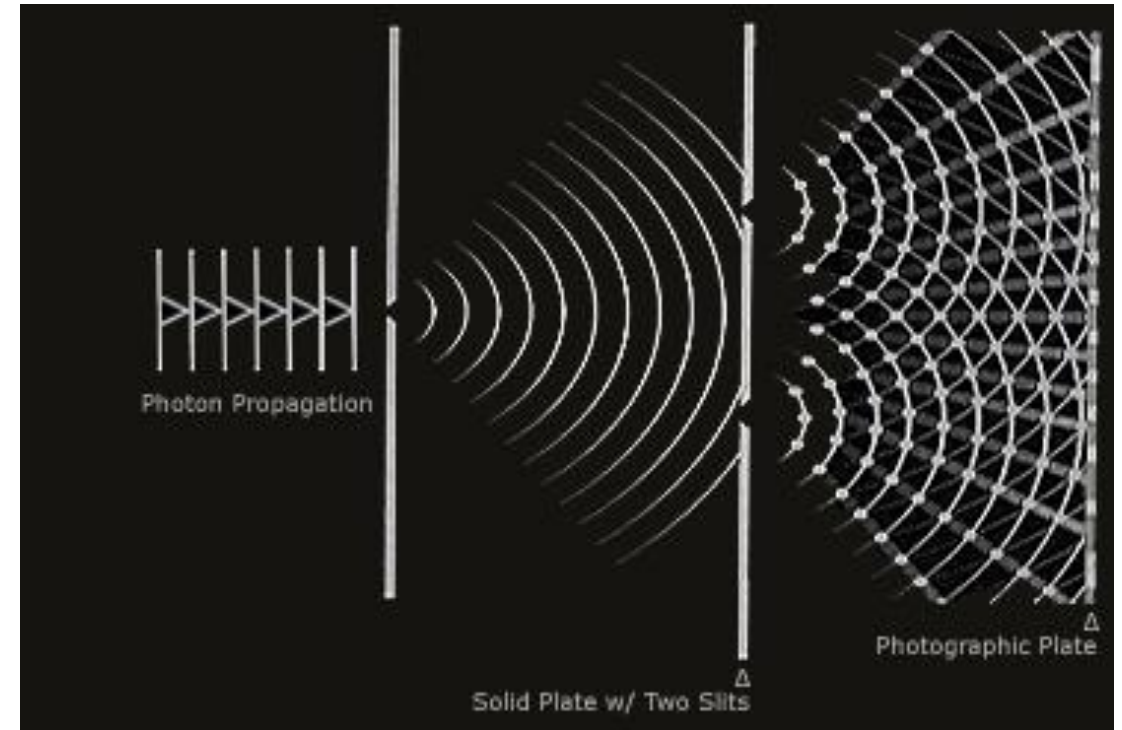
YOUNG'S DOUBLE-SLIT EXPERIMENT

- So, which did he observe?
- Wave-Model solution!

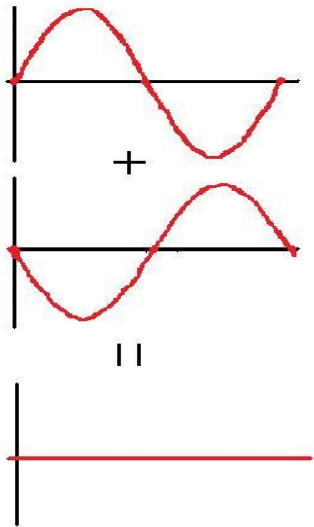


YOUNG'S DOUBLE-SLIT EXPERIMENT

- Why does this occur?
- Diffracted wavefronts from each slit overlap at various points in space; at some points they reinforce each other (constructive interference), whereas at other points they cancel each other out (destructive interference).
- This is called superposition.

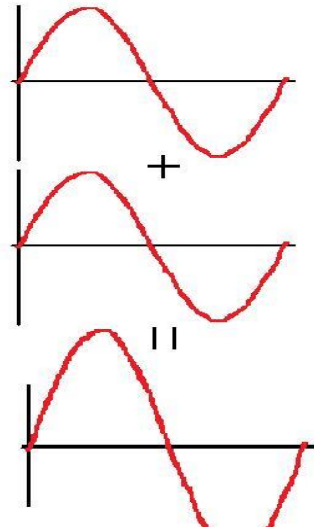


SUPERPOSITION



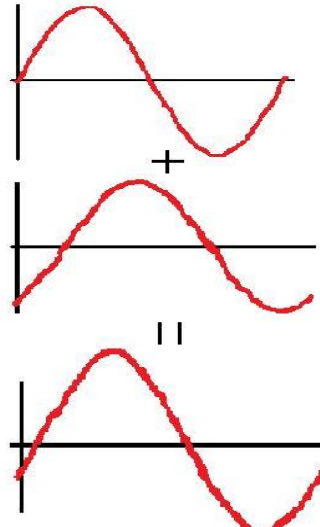
The two waves collide and cancel each other.

Destructive
Interference

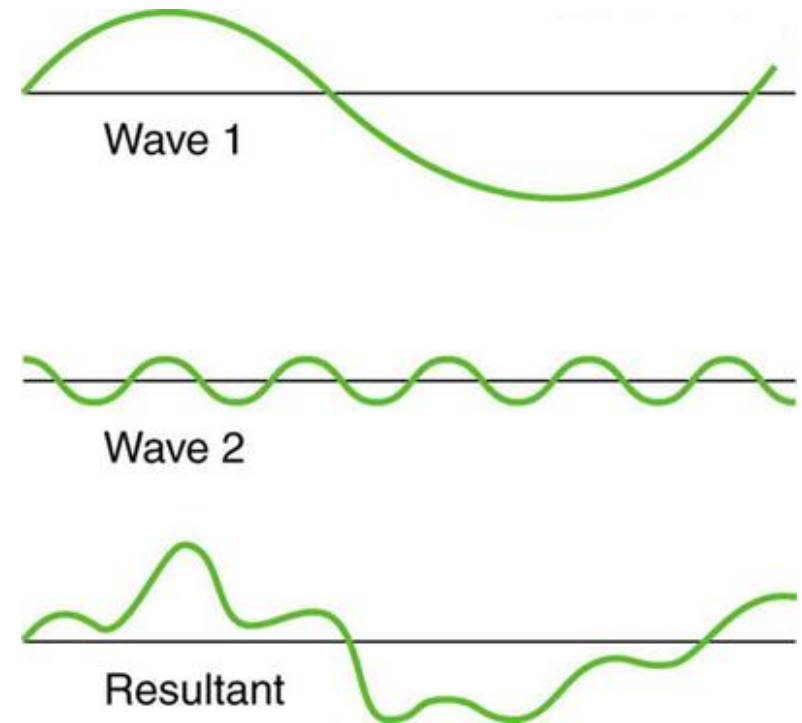


Here the two sine waves are "in phase" with one another and the resultant has a larger amplitude.

Constructive
Interference



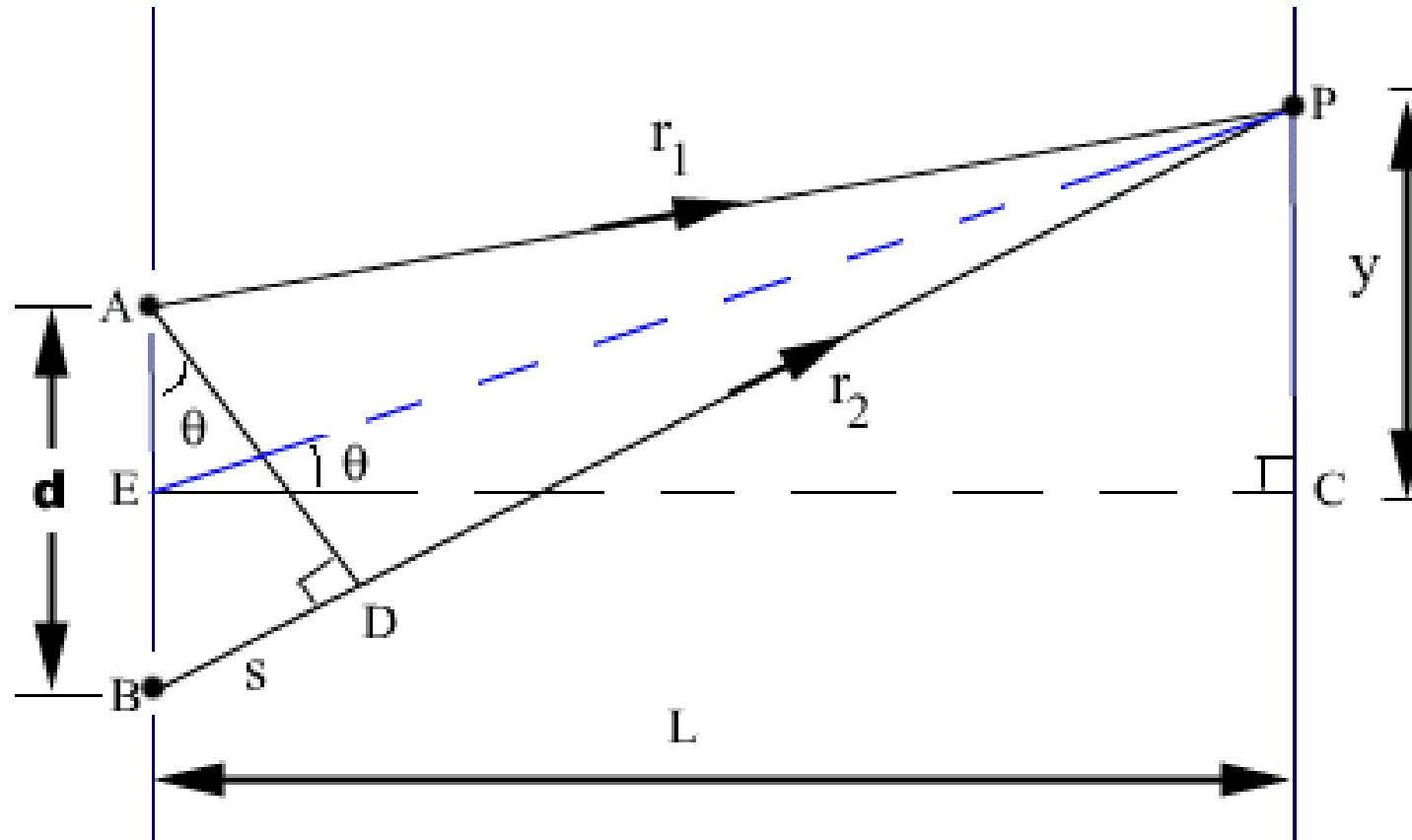
This is a third example of superposition.

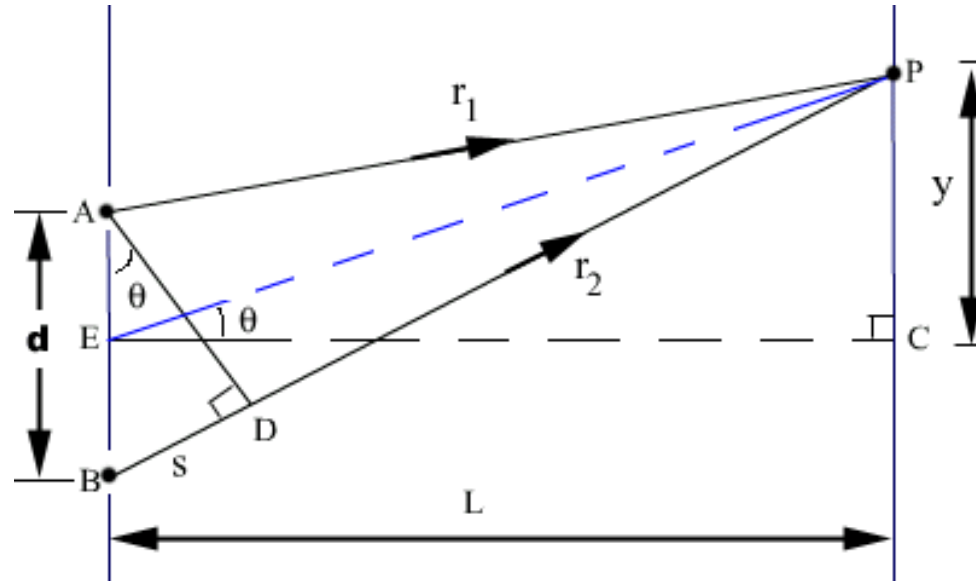


SUPERPOSITION

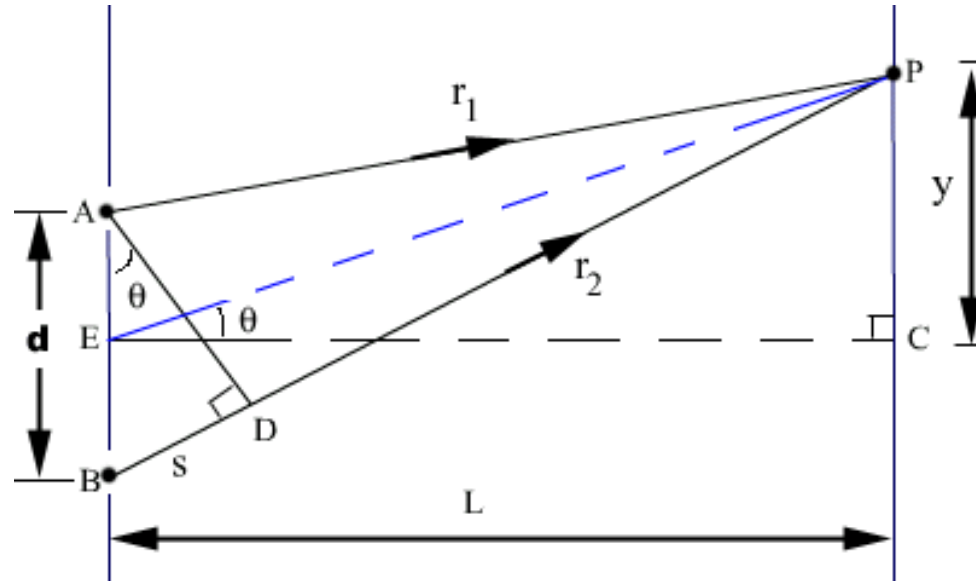


DOUBLE-SLIT EXPERIMENT



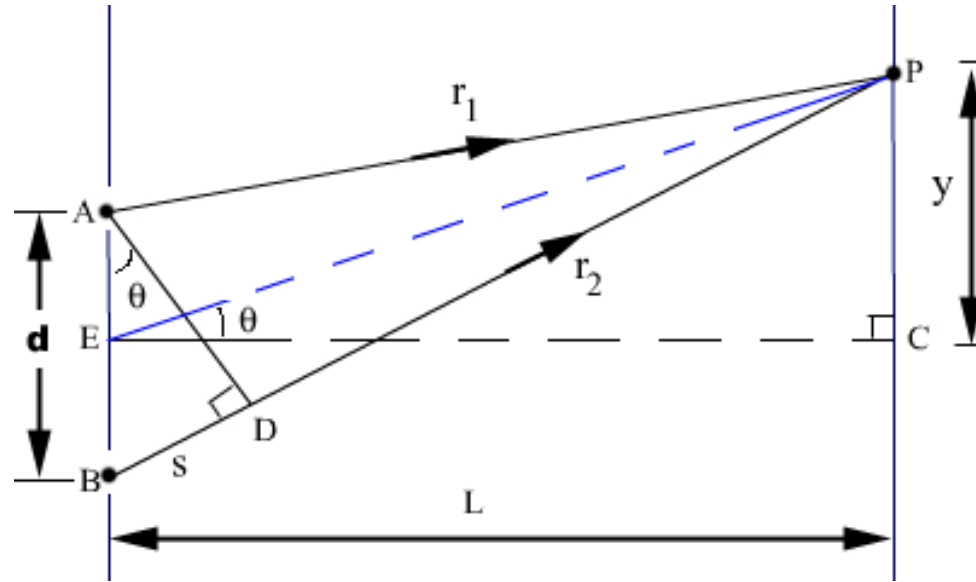


- $S = \text{path difference} = r_2 - r_1$ (magnitude only).
- Path difference depends on the distance between the apertures: d .
- By constructing the right triangle ADB, we can derive the expression $S = d \sin \theta$, where θ is the 'angular distance' between point E (halfway between apertures) and point P.



- By constructing the right triangle ADB, we can derive the expression $S = d \sin \theta$.
- Constructive interference will occur if the path difference is a whole-number of wavelengths: $m\lambda$. These will be the light spots.

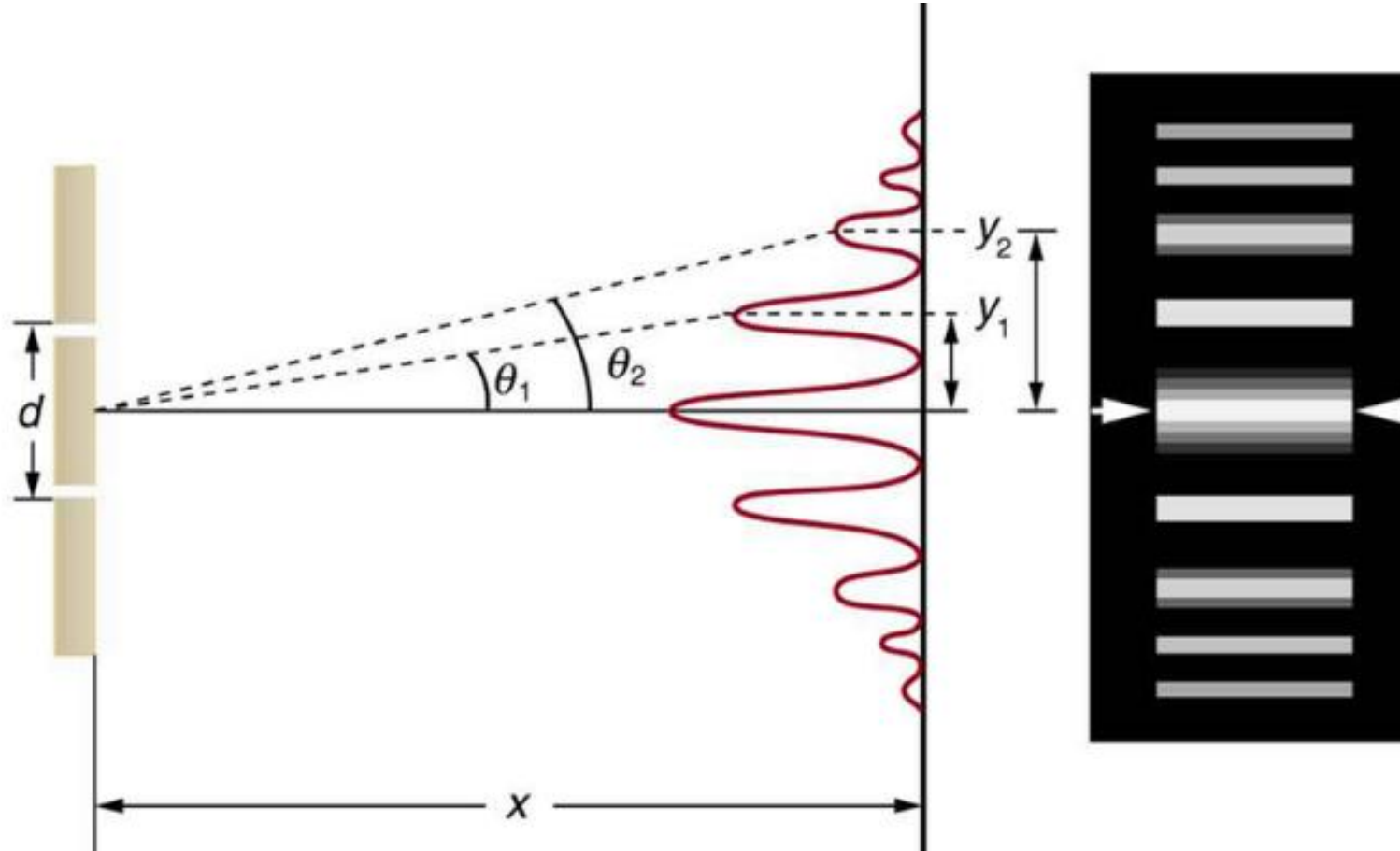
$$d \sin \theta = m\lambda, \text{ where 'm' is an integer}$$

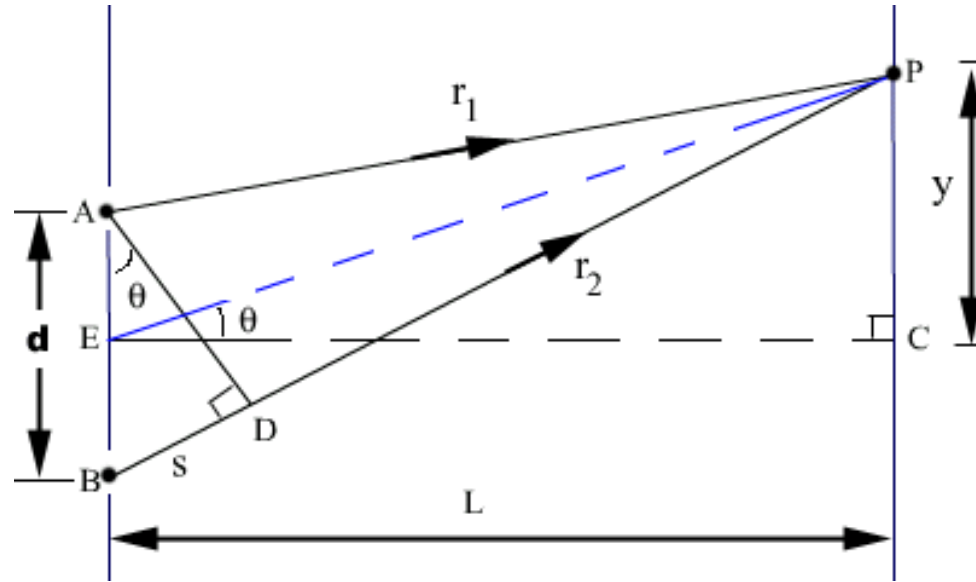


- Destructive interference will occur if the path difference is **not** a whole-number of wavelengths. The point where the waves exactly cancel out will be when they are one-half wavelength apart: $(m + [1/2])\lambda$. These will be the dark spots.

$$d \sin \theta = (m + [1/2]) \lambda, \text{ where 'm' is an integer}$$

DOUBLE-SLIT EXPERIMENT





(Destructive): $d \sin \theta = (m + [1/2]) \lambda$

(Constructive): $d \sin \theta = m\lambda$

For small angles:

$$\sin \theta \approx \tan \theta = \frac{\text{opposite}}{\text{adjacent}} = \frac{y}{L}$$

$$\text{(Destructive): } d \sin \theta = (m + [1/2]) \lambda$$

$$\text{(Constructive): } d \sin \theta = m\lambda$$

For small angles:

$$\sin \theta \approx \tan \theta = \frac{\text{opposite}}{\text{adjacent}} = \frac{y}{L}$$

Therefore (sub. and solve for y):

$$\text{(Destructive): } \frac{d y}{L} = (m + [1/2]) \lambda$$

$$\text{(Constructive): } \frac{d y}{L} = m\lambda$$

Therefore (sub. and solve for y):

$$\text{(Destructive): } \frac{d}{L} y = (m + [1/2]) \lambda$$

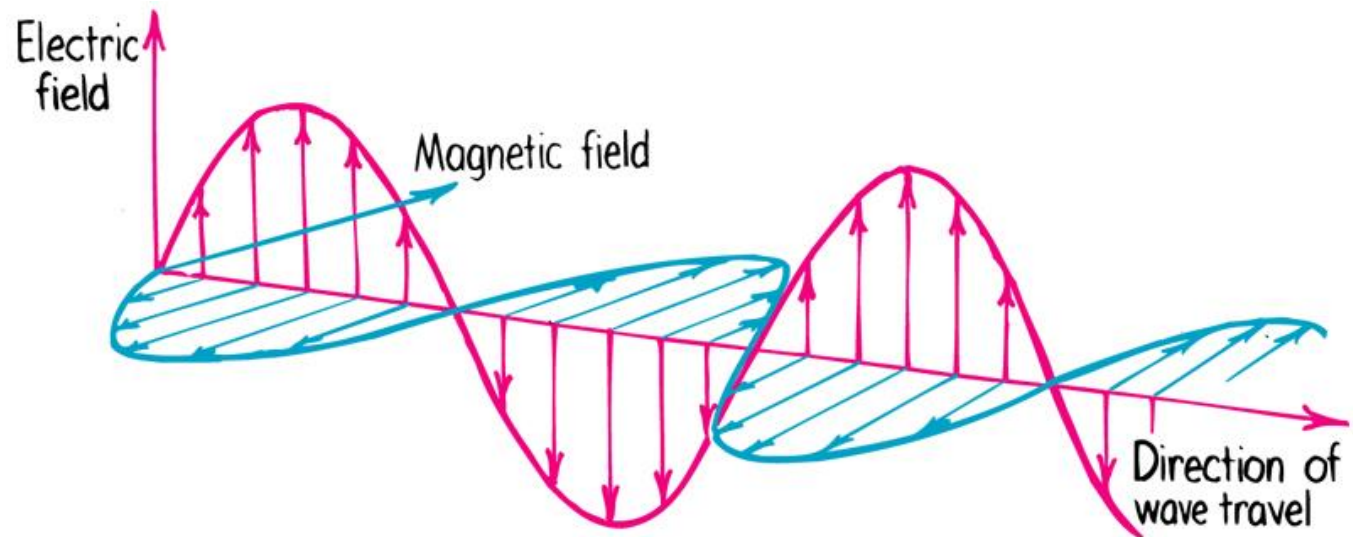
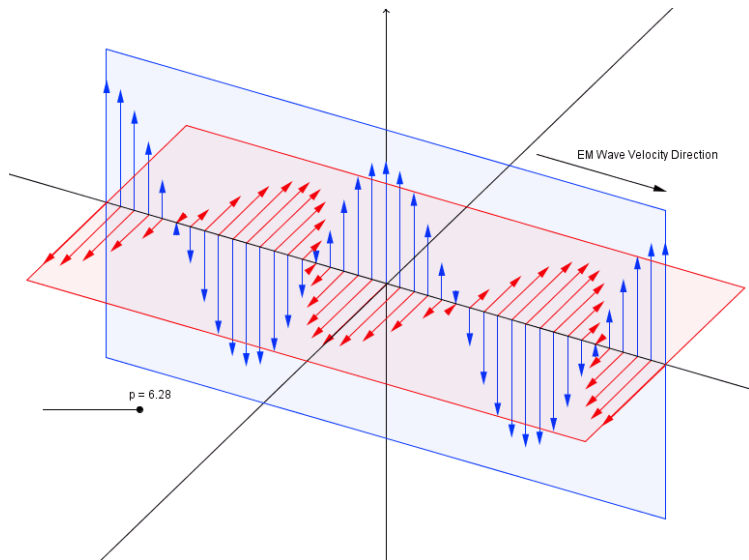
$$\text{(Constructive): } \frac{d}{L} y = m\lambda$$

$$y = \frac{L(m + [1/2]) \lambda}{d} \quad \text{(Destructive – Position of Dark Spots)}$$

$$y = \frac{Lm\lambda}{d} \quad \text{(Constructive – Position of Light Spots)}$$

ELECTROMAGNETIC RADIATION

- Electromagnetic Radiation is comprised of a series of self-propagating, alternating electric and magnetic fields. Changing electric fields induce changing magnetic fields which induce more electric fields... and so on. Each induced field is at right-angles to the field that induced it.



ELECTROMAGNETIC RADIATION

- The Wave Equation:

$$c = f \lambda$$

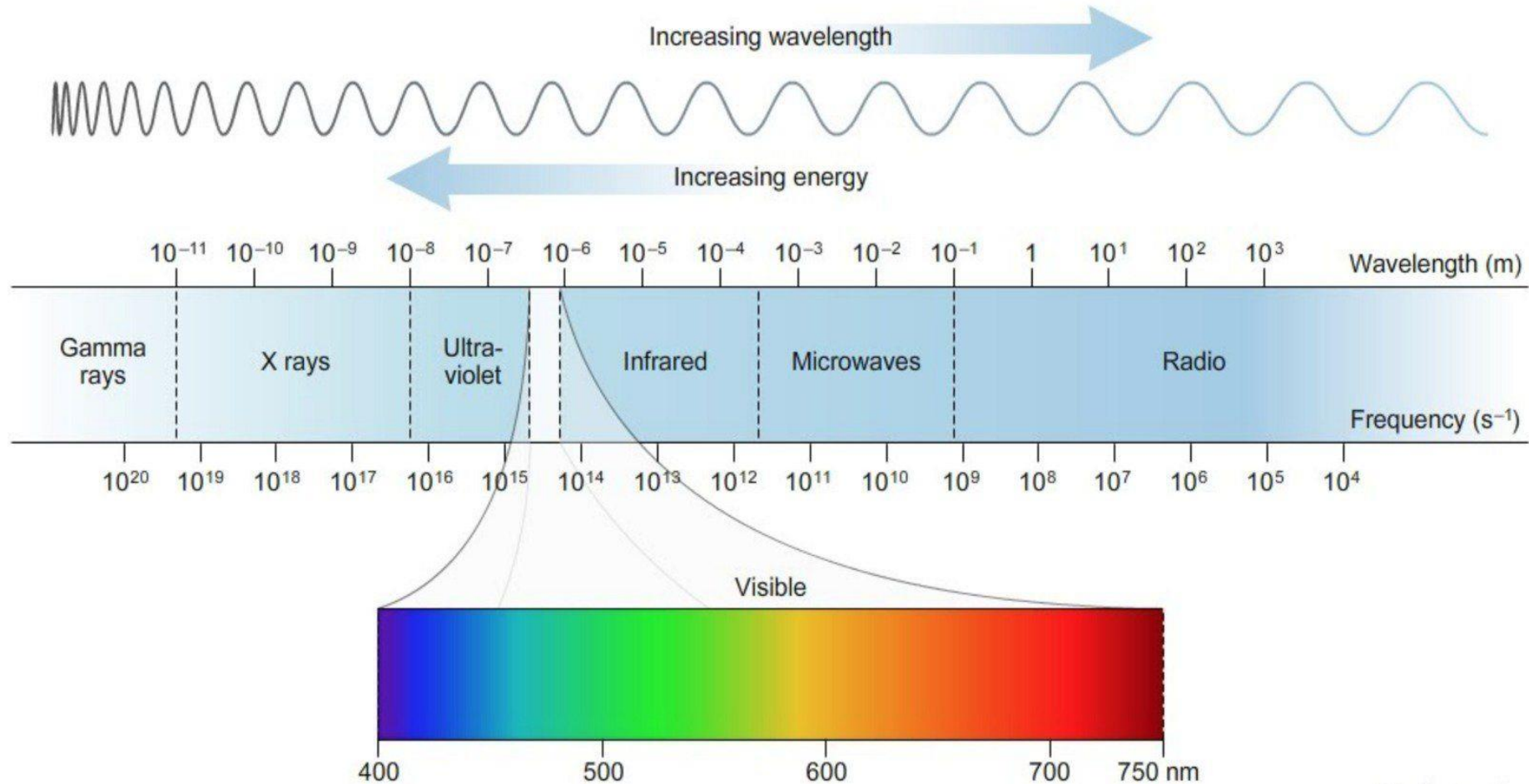
- The speed of light through empty space (c) was theoretically calculated from Maxwell's Equations, and experimentally confirmed by Hippolyte Fizeau in 1849.
- The prediction...

$$c = \frac{1}{\sqrt{\epsilon_0 \mu_0}} = \frac{1}{\sqrt{(8.85 \times 10^{-12})(4\pi \times 10^{-7})}} = 299\,863\,380 \text{ m/s}$$

$$c \approx 3.00 \times 10^8 \text{ m/s}$$

We've seen these before!

ELECTROMAGNETIC RADIATION



ELECTROMAGNETIC RADIATION

- Why is EMR useful? Well, different frequencies have different applications (Infrared, UV, X-Ray, Microwave, Radio Wave etc.). We will look at how some of these are used.



Infrared



X-Ray



Microwave

ELECTROMAGNETIC RADIATION



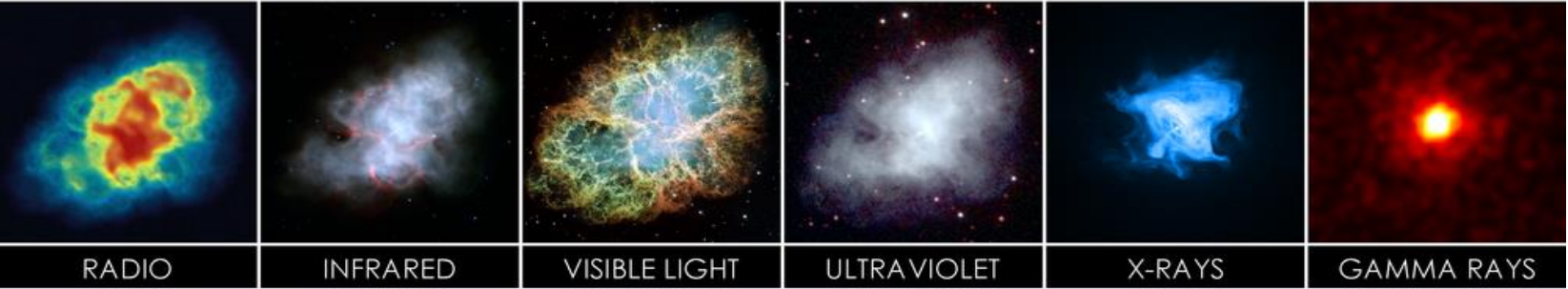
Ultraviolet



Sunscreen (Visible & UV)

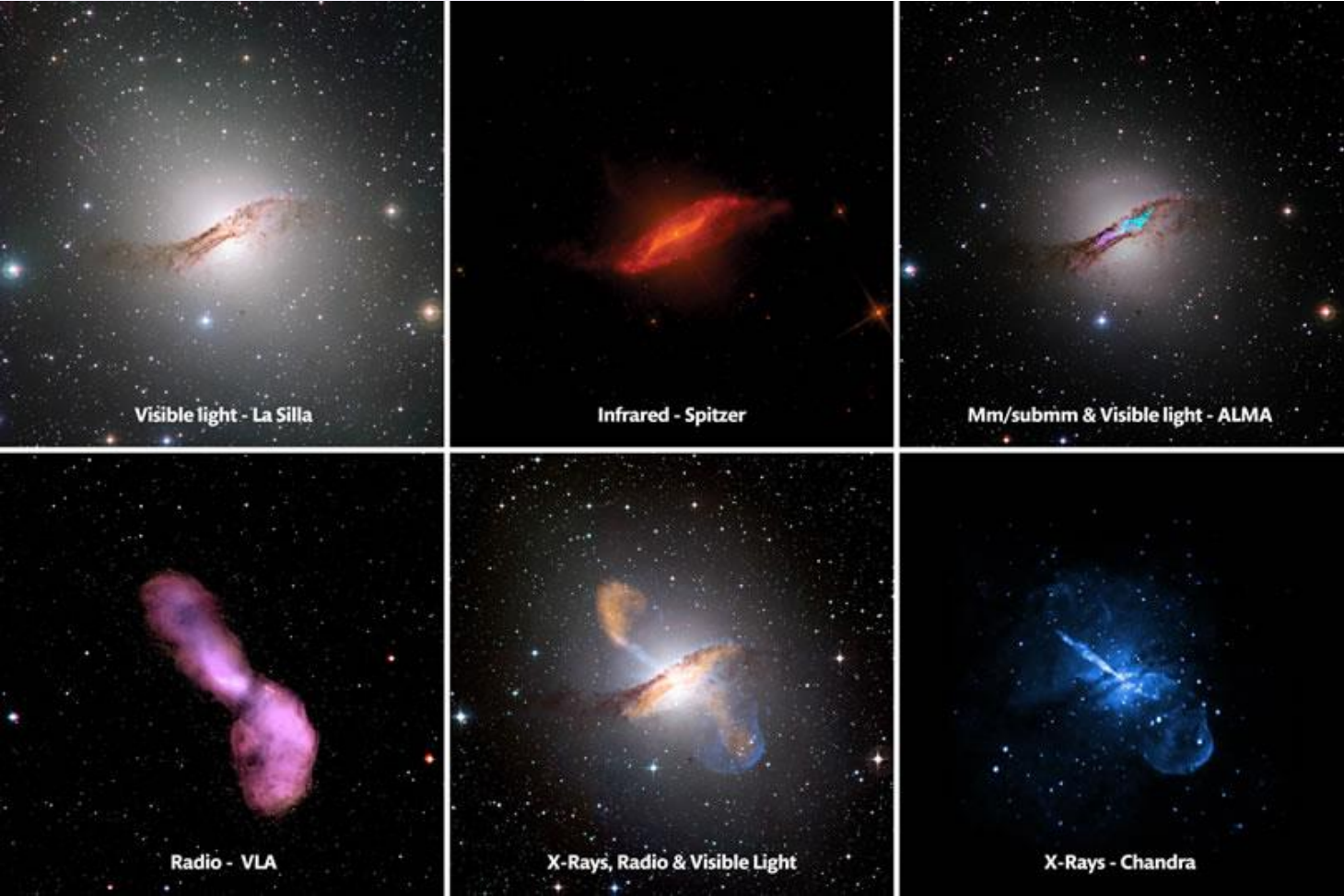
*Sunscreen can function by absorbing UV before it reaches the skin (pictured), reflecting UV, or both.

CRAB NEBULA



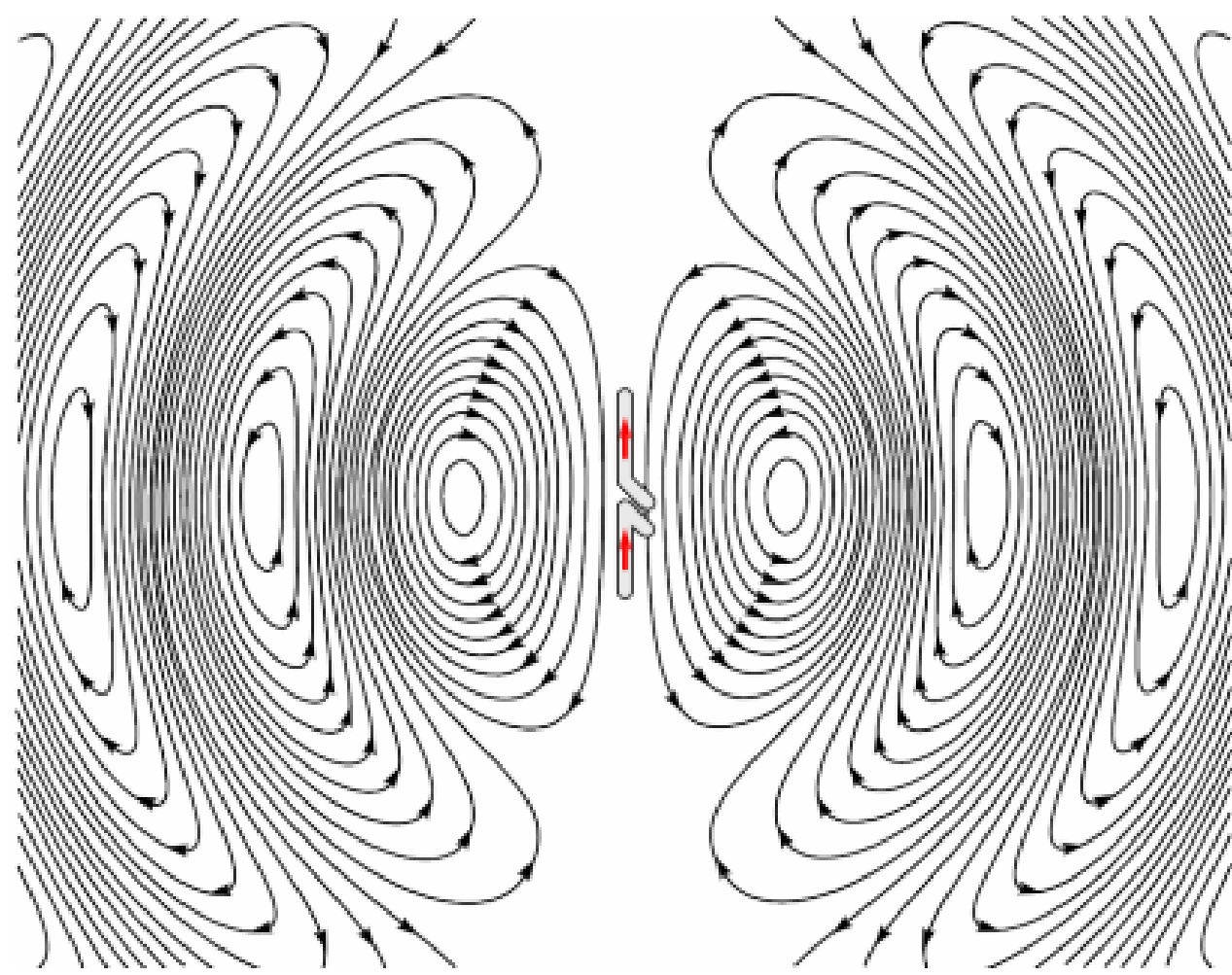
The crab nebula in radio, infrared, visible, ultraviolet, x-ray and gamma-ray wavelengths.

Sources: Radio: NRAO/AUI and M. Bietenholz, J.M. Uson, T.J. Cornwell; Infrared: NASA/JPL-Caltech/R. G. and A.Loll (Arizona State University); Ultraviolet: NASA/Swift/E. Hoversten, PSU, X-ray: NASA/CXC/SA

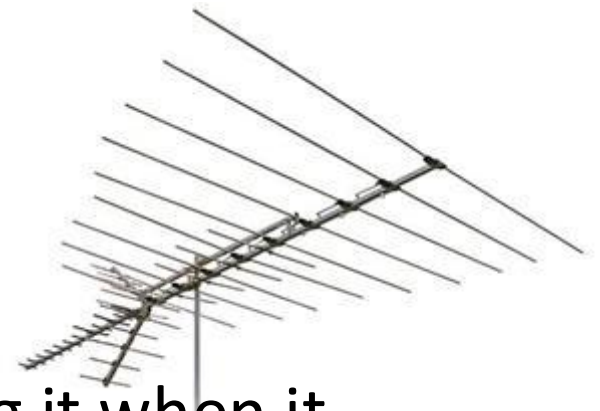


RADIO WAVES

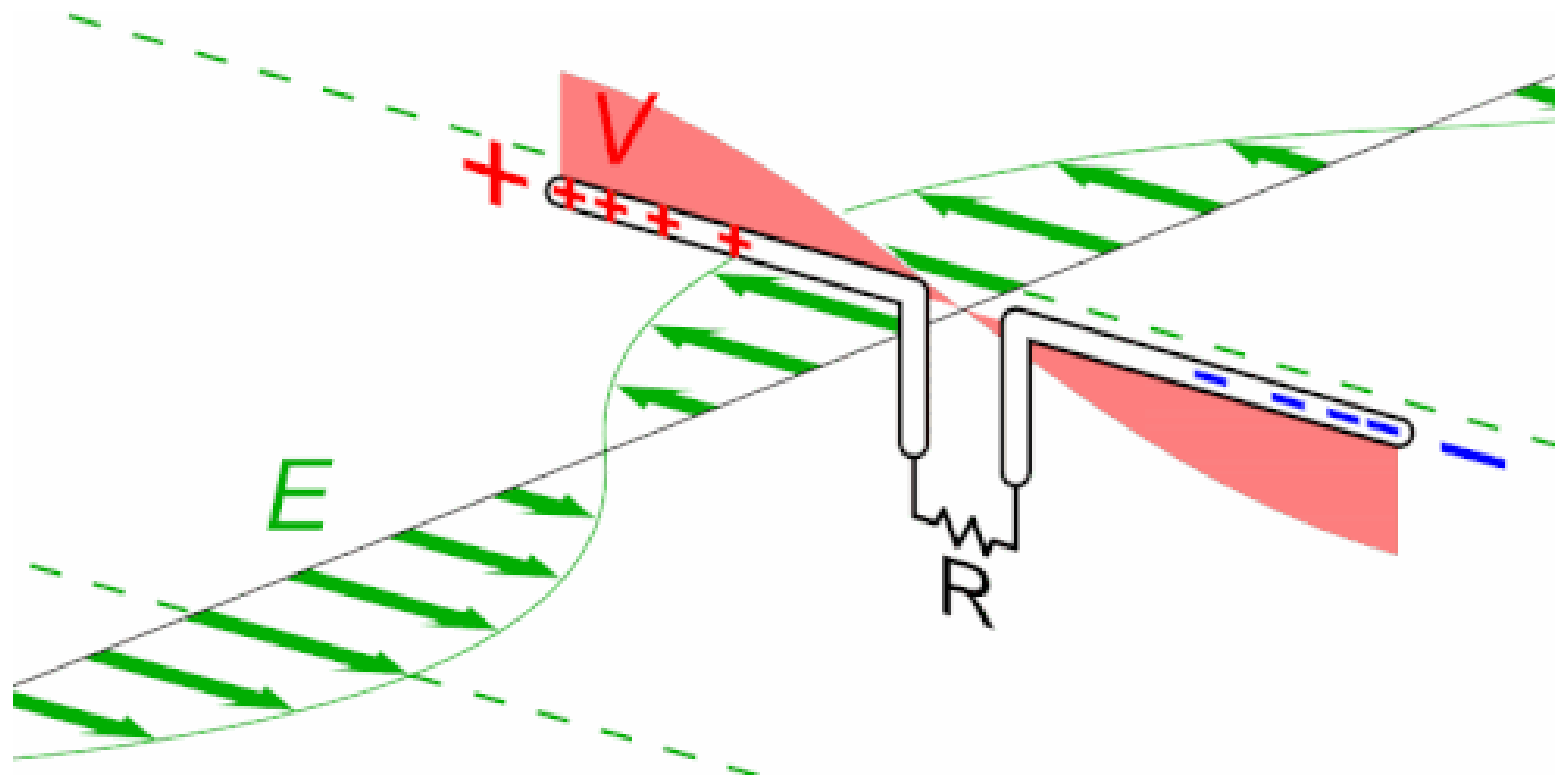
- Radio Waves are used for telecommunication (radio, TV etc.). But how do they work?
- There are *lots* of different things to consider when designing broadcast & reception antennae. We will consider only simple antennae.
- A broadcast antenna has an alternating current running through it; the changing electric field induces a changing magnetic field which induces a changing electric field... etc.



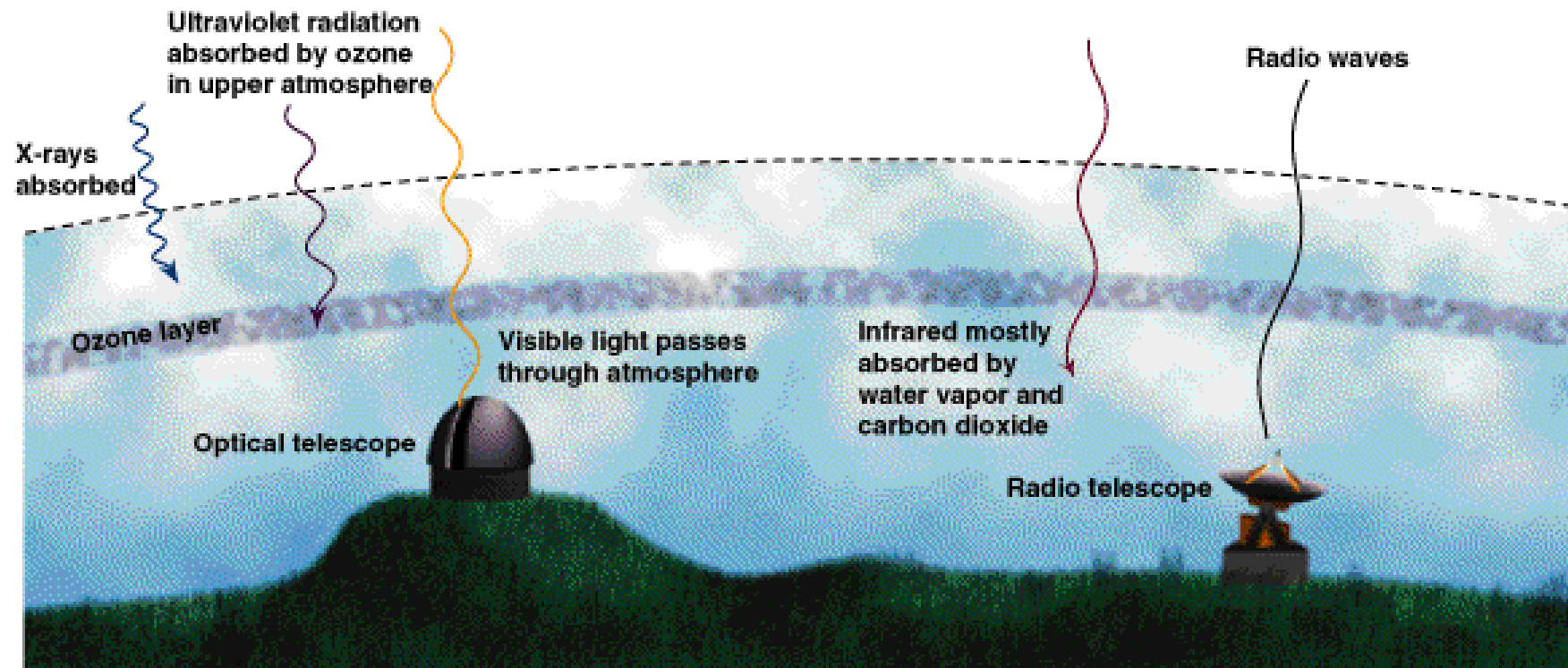
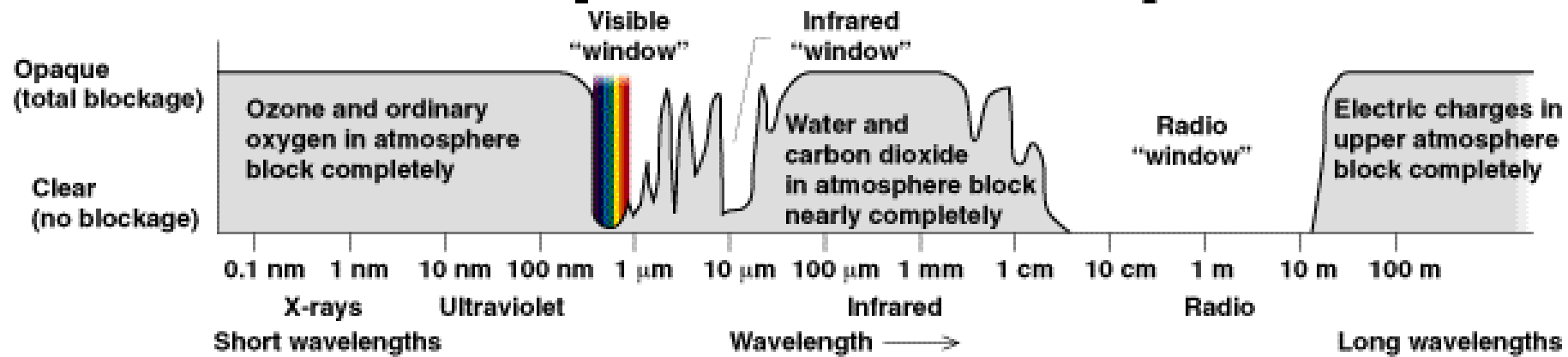
ANTENNAE



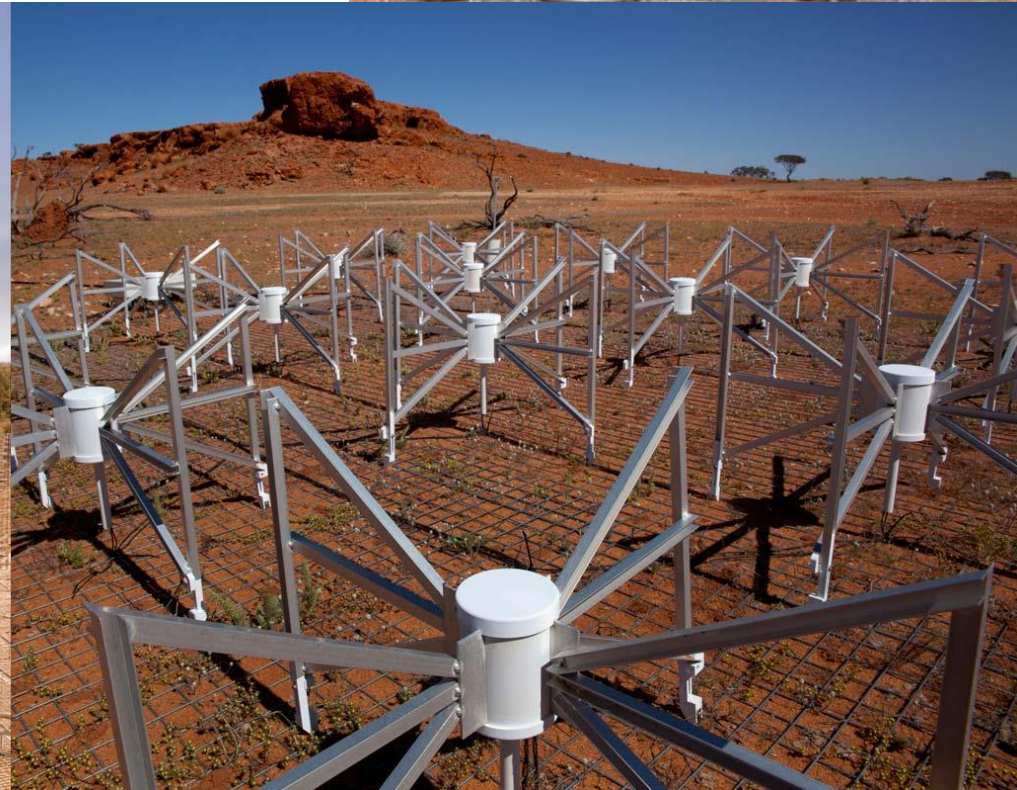
- The receiving antenna will have a current induced along it when it receives the broadcast signal.
- Orientation is important – if transmitter and receiver are oriented differently, induction will be limited and the receiver can't/won't receive.
- Optimal orientation minimises signal “noise” by optimising reception.
- Different lengths of wire are tuned to different frequencies (length from tip-to-tip is a ‘half-wavelength’). Higher frequencies = smaller antennae.



Atmospheric Absorption



SQUARE KILOMETER ARRAY & MURCHISON WIDEFIELD ARRAY



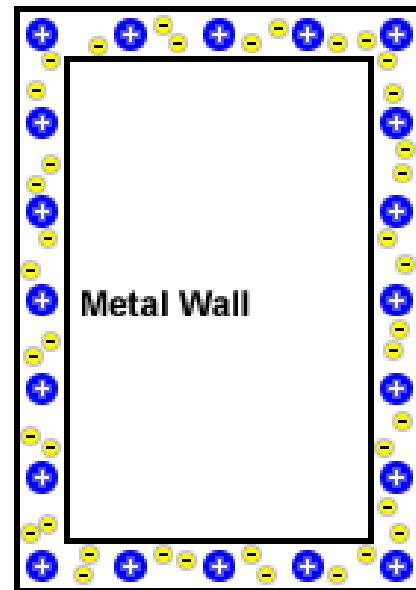
FARADAY CAGE

- A **Faraday cage** or **Faraday shield** is an enclosure used to block electromagnetic fields. A Faraday shield may be formed by a continuous covering of conductive material, or in the case of a Faraday cage, by a mesh of such materials. Faraday cages are named after the English scientist Michael Faraday, who invented them in 1836.
- A Faraday cage operates because an external electrical field causes the electric charges within the cage's conducting material to be distributed so that they cancel the field's effect in the cage's interior. This phenomenon is used to protect sensitive electronic equipment. They are also used to protect people and equipment against actual electric currents such as lightning strikes and electrostatic discharges, since the enclosing cage conducts current around the outside of the enclosed space and none passes through the interior.

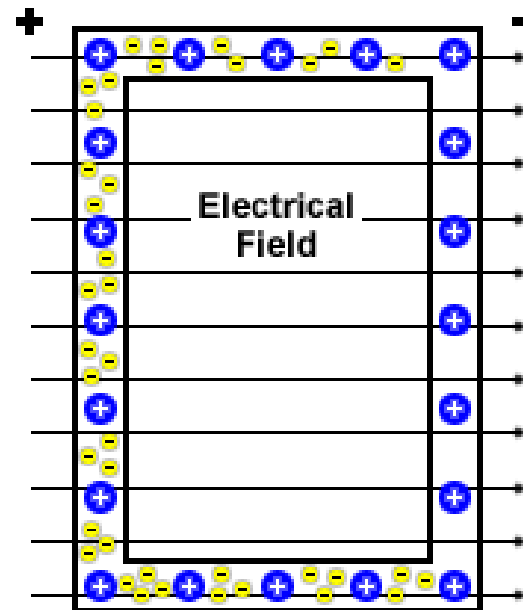
FARADAY CAGE

- Faraday cages cannot block stable or slowly varying magnetic fields, such as the Earth's magnetic field (a compass will still work inside). To a large degree, though, they shield the interior from external electromagnetic radiation if the conductor is thick enough and any holes are significantly smaller than the wavelength of the radiation.
- A common misconception is that a Faraday cage provides full blockage or attenuation; this is not true. The reception or transmission of radio waves, a form of electromagnetic radiation, to or from an antenna within a Faraday cage is heavily attenuated or blocked by the cage; however, a Faraday cage has varied attenuation depending on wave form, frequency or distance from receiver/transmitter, and receiver/transmitter power. Near-field high-powered frequency transmissions like HF RFID are more likely to penetrate. Solid cages generally attenuate fields over a broader range of frequencies than mesh cages.

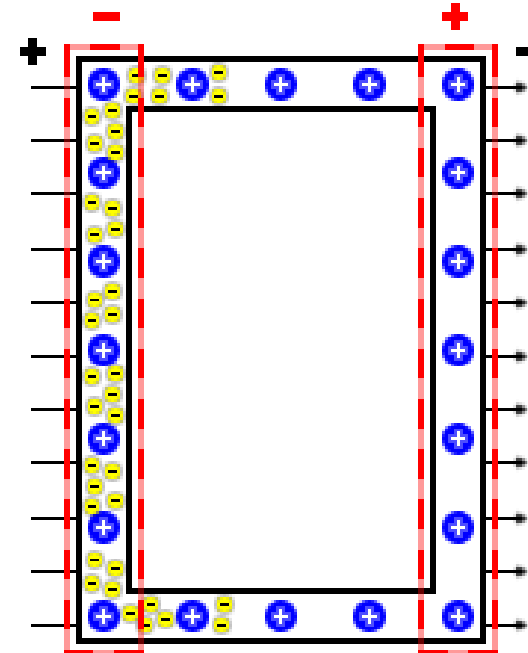
Faraday Cage



Faraday Cage in the absence of an electrical field.

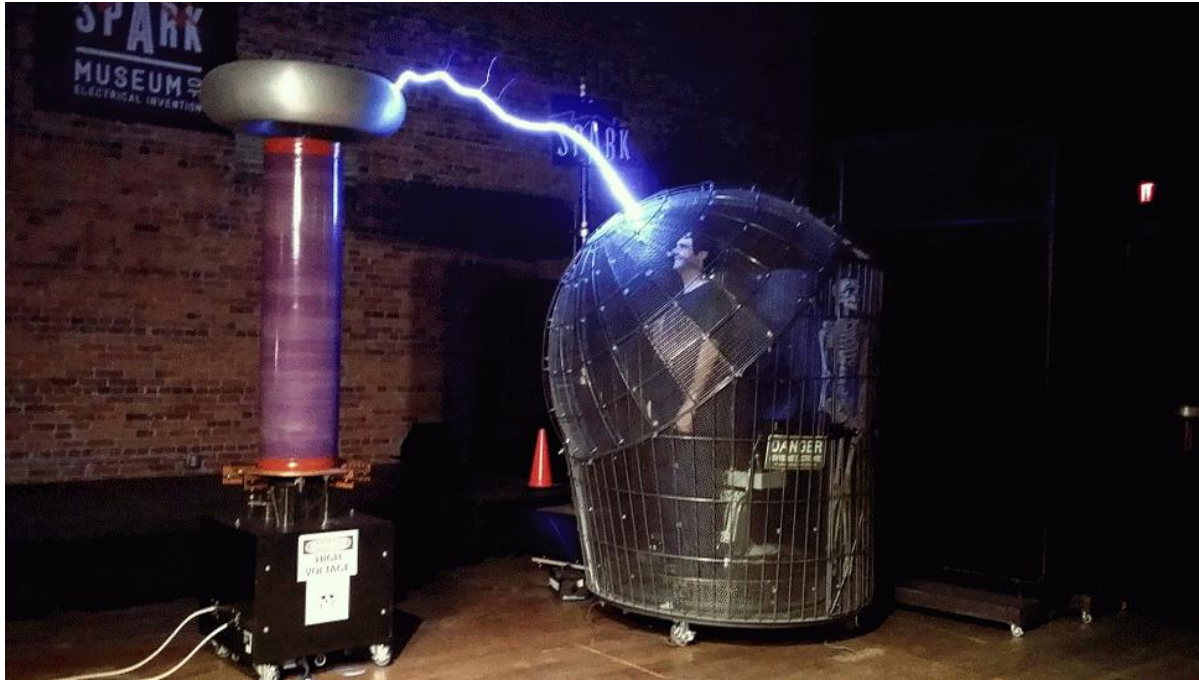


The charged particles in the wall of the Faraday cage respond to an applied electrical field.



Electrical fields generated inside the wall cancel out the applied field, neutralizing the interior of the cage.

FARADAY CAGE



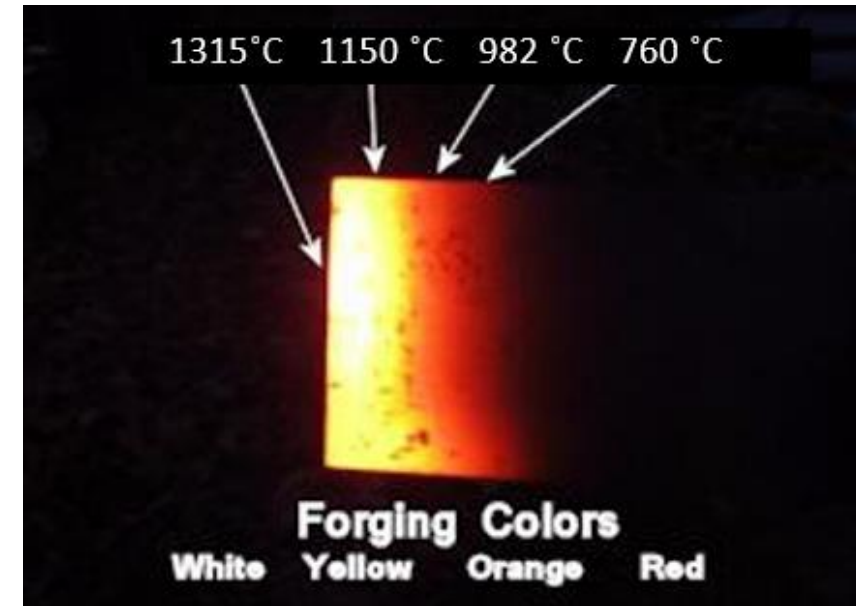
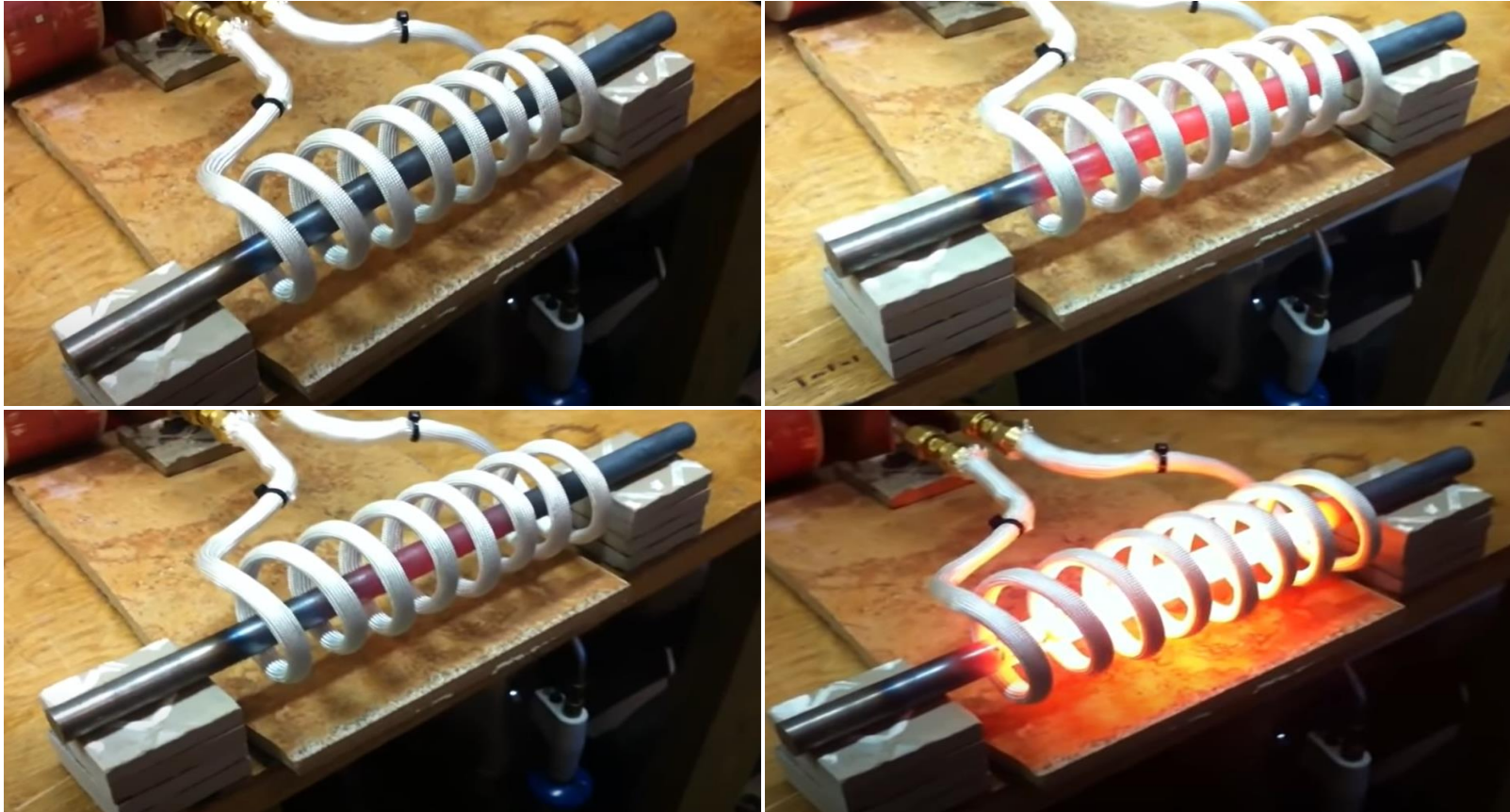
FARADAY CAGE



BLACKBODY RADIATION

- All objects that have a temperature above absolute zero (-273.15°C) emit electromagnetic radiation. At room temperature, most of this radiation is in the infrared part of the spectrum (and is thus invisible to the naked eye).
- At higher temperatures, the amount of radiation emitted is greater, and the peak-frequency increases (wavelength decreases), meaning that more of the emitted radiation is in the visible spectrum.

BLACKBODY RADIATION



BLACKBODY RADIATION

- For an ideal blackbody radiator, the emission spectrum will depend solely upon temperature. In reality, the type and shape of material also play a role.
- Why is it called a blackbody radiator? A blackbody is an ideal absorber that absorbs all incident radiation (by contrast, a *whitebody* would be a perfect reflector, having a high albedo). It is also in thermal equilibrium with its surroundings, so it absorbs and emits/radiates energy at the same rate.

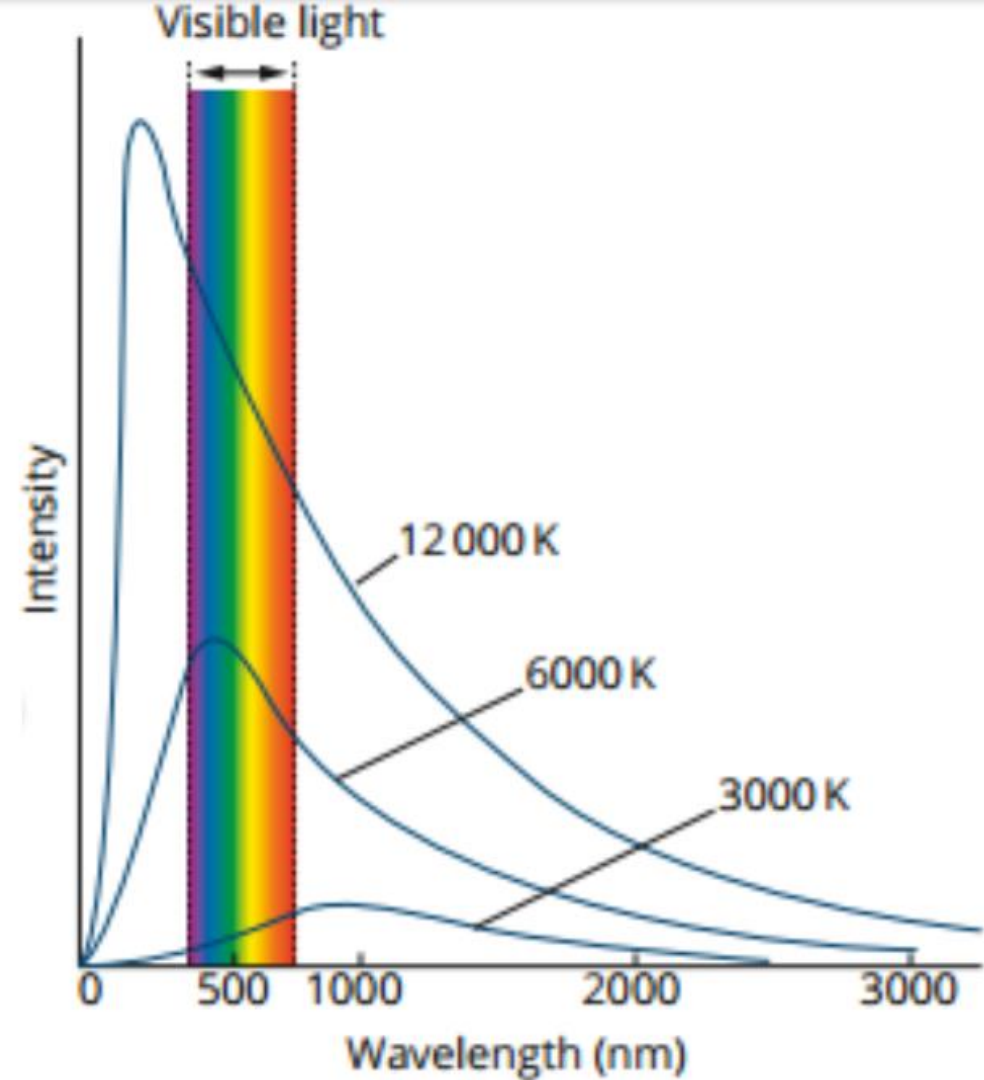


FIGURE 7.4.3 Objects at temperatures of 3000 K, 6000 K and 12 000 K have different emission spectra. As the temperature increases, the amount of energy radiated (shown by the area under these graphs) increases, and the peak wavelength decreases (it becomes more blue).

BLACKBODY RADIATION

- In the laboratory, blackbody radiation is approximated by the radiation from a small hole in a large cavity in an entirely opaque body that is maintained at a constant temperature.
- Any light entering the hole would have to reflect off the walls of the cavity multiple times before it escaped, in which process it is nearly certain to be absorbed. Absorption occurs regardless of the wavelength of the radiation entering (as long as it is small compared to the hole).
- The hole, then, is a close approximation of a theoretical black body and, if the cavity is heated, the spectrum of the hole's radiation (i.e., the amount of light emitted from the hole at each wavelength) will be continuous, and will depend only on the temperature.

BLACKBODY RADIATION

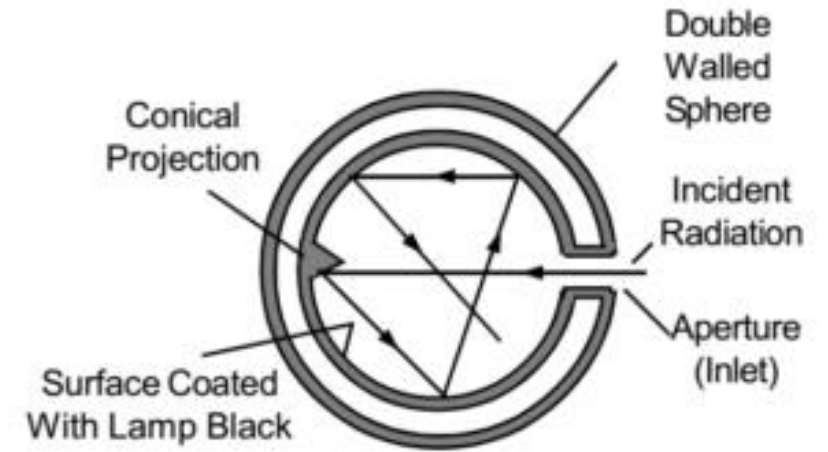
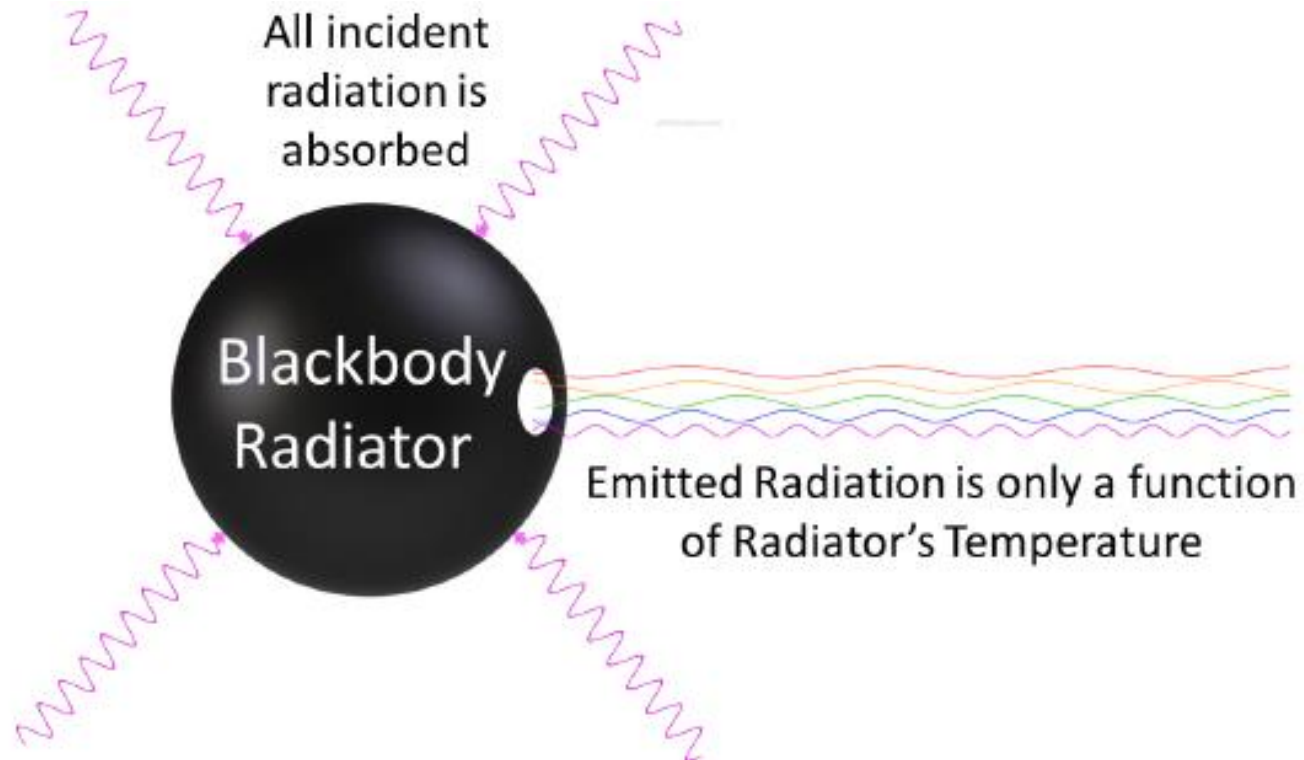


Figure 40.1 A physical model of a black body.

BLACKBODY RADIATION

- For an ideal blackbody radiator, there is a mathematical relationship between the peak wavelength emitted and the temperature, called Wien's Displacement Law:

$$\lambda_{peak} = \frac{b}{T}$$

Where 'b' is Wien's Displacement Constant

($b = 2.897771955 \times 10^{-3} \text{ m K}$)

'T' is temperature in Kelvin (K)

BLACKBODY RADIATION

- So, why all this fuss over blackbody radiators? Well, an understanding between emitted wavelength and temperature allows us to approximate the temperature of things that otherwise could not; stars, galaxies, even space itself.

BLACKBODY RADIATION

PHYSICSFILE

Measuring the temperatures of stars

With an understanding of the radiation spectra of different objects, a great deal of information is gained about a star by assessing the colour of the light it emits. Colour is so important to astronomers that they refer to it in their classification system for stars, using names such as red giant, white dwarf, blue supergiant.

A graph that plots the absolute magnitude (luminosity relative to the Sun) of a star against its surface temperature is called a Hertzsprung–Russell diagram (Figure 7.4.4). Low-mass stars are cool and red. Medium-mass stars such as the Sun (circled in Figure 7.4.4) are yellow. High-mass stars are hot and blue.

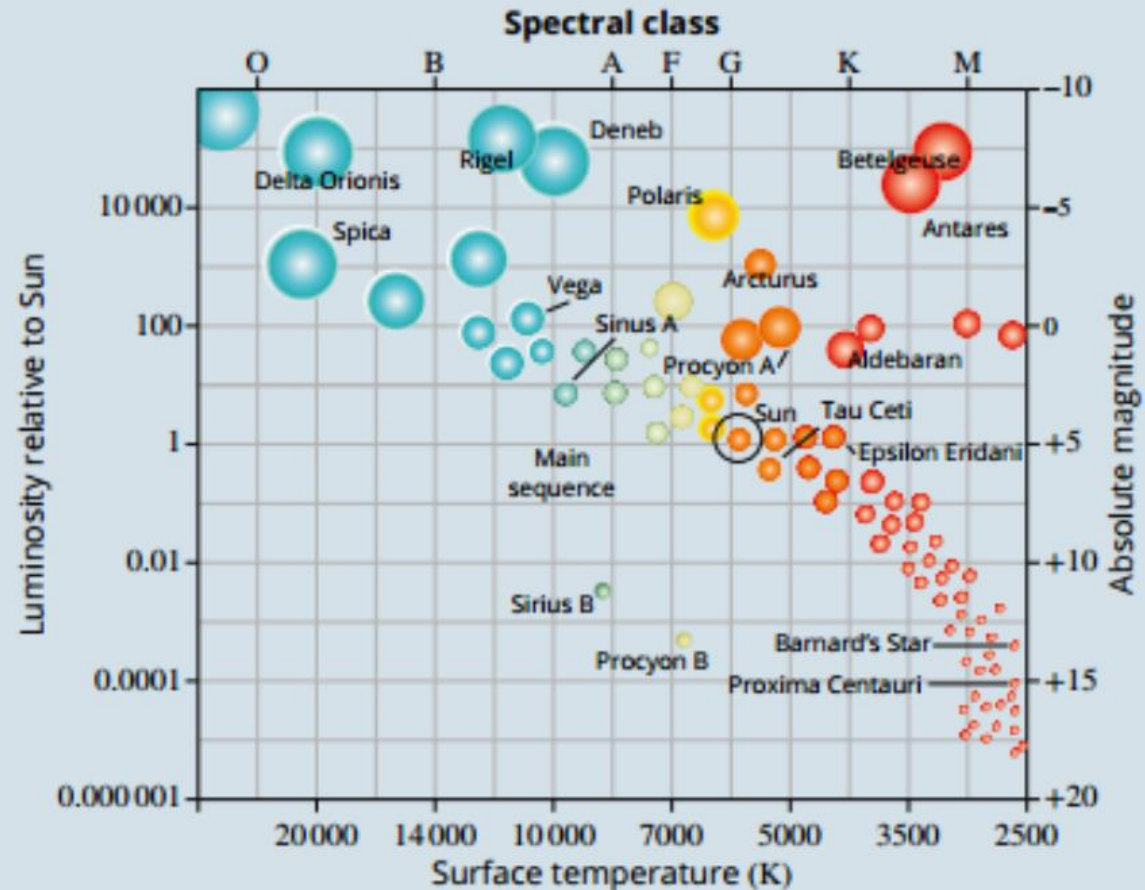
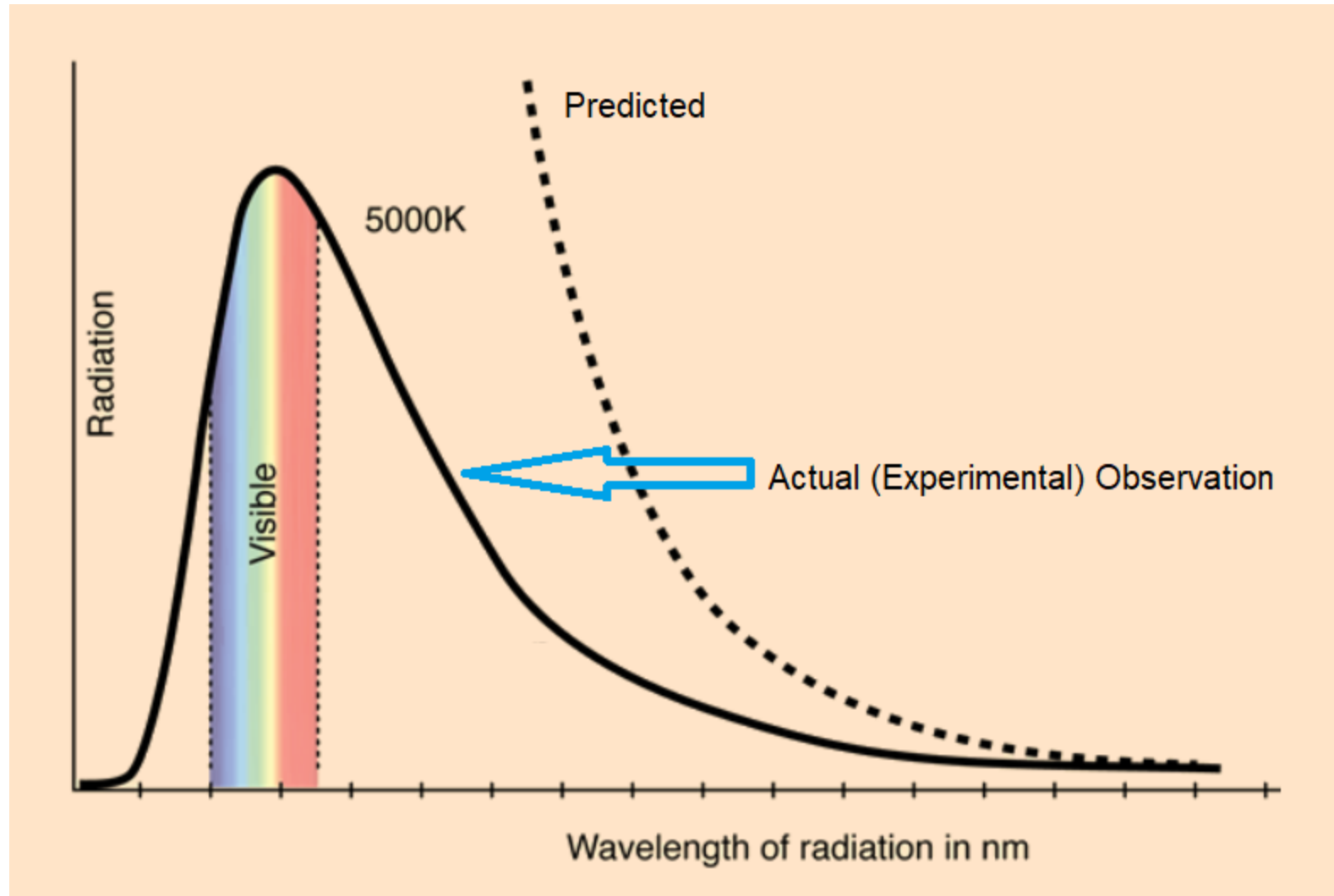


FIGURE 7.4.4 Stellar classification of stars using a Hertzsprung–Russell diagram.

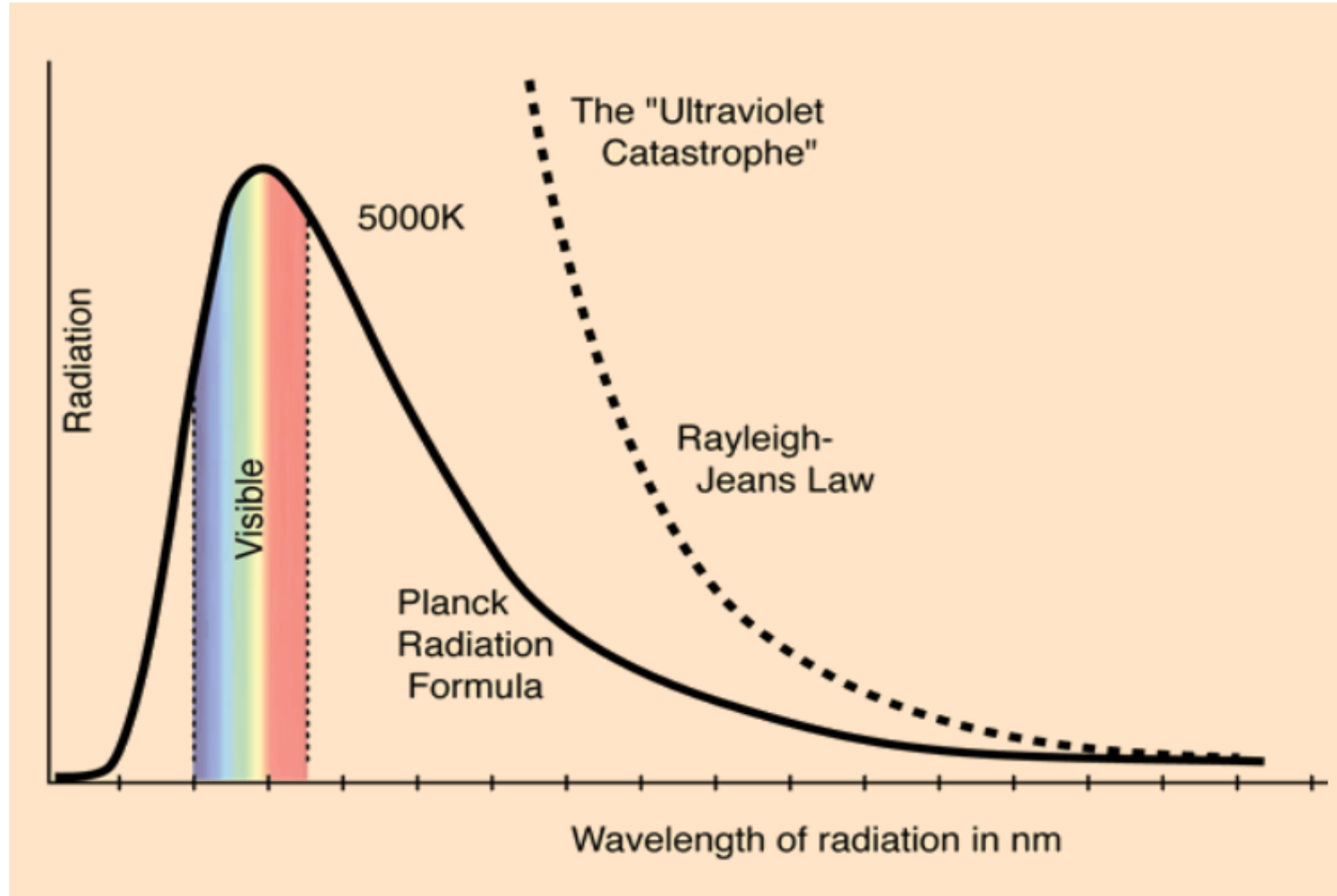
BLACKBODY RADIATION

- The science of blackbody radiators has been well understood for well over a century. The mathematics of blackbody radiators, however, ran into a slight hiccup around 1900. The problem? It didn't work!
- Well, that's not strictly true. The law used to predict the intensity & peak wavelength was called the Rayleigh-Jeans Law, and it worked pretty well for long wavelengths (low frequencies). However, for short wavelengths (high frequencies), the prediction varies wildly from reality. This clash between classical theory and reality became known as the *Ultraviolet Catastrophe*.

BLACKBODY RADIATION



BLACKBODY RADIATION



BLACKBODY RADIATION

- Essentially, the theory predicted that almost all objects would spontaneously emit an arbitrarily high amount of energy until it was near absolute zero. Since this does not happen (I mean, just look around!), this is obviously wrong.
- Max Planck resolved the issue by assuming that molecules vibrate at discrete frequencies, and that they gain or lose energy by absorbing or emitting radiation in discrete packets called *photons* (later confirmed by Bose & Einstein).
- He even developed an equation to determine the energy of these photons.

PLANCK'S EQUATION

PLANCK'S EQUATION

As discussed above, Max Planck proposed that light existed in discrete packets of energy. He called the discrete packets of energy 'quanta', and developed an equation for the energy, E , of each **quantum**:

i $E = hf$

where E is the energy of a quantum of light (J)

f is the frequency of the electromagnetic radiation (Hz)

h is the constant 6.63×10^{-34} Js, now known as Planck's constant

Electromagnetic radiation is often described according to its wavelength, so Planck's equation can be combined with the wave equation for light $c = f\lambda$ as follows:

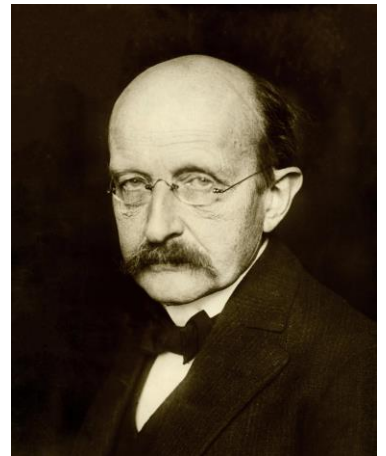
$$E = hf \text{ and } f = \frac{c}{\lambda}$$

so

i $E = \frac{hc}{\lambda}$

PLANCK'S EQUATION

- This 'quantum' interpretation worked beautifully in resolving the apparent paradox of the Ultraviolet Catastrophe; however, those little 'packets' of light – the photons – were essentially light *particles*. Up until this point, the wave theory for light was generally accepted (it easily explains just about every property of light); on this occasion, however, the wave model for light was unable to explain the experimental observations being made, but a particle model was successful in doing so. Thus, Quantum Theory was born!



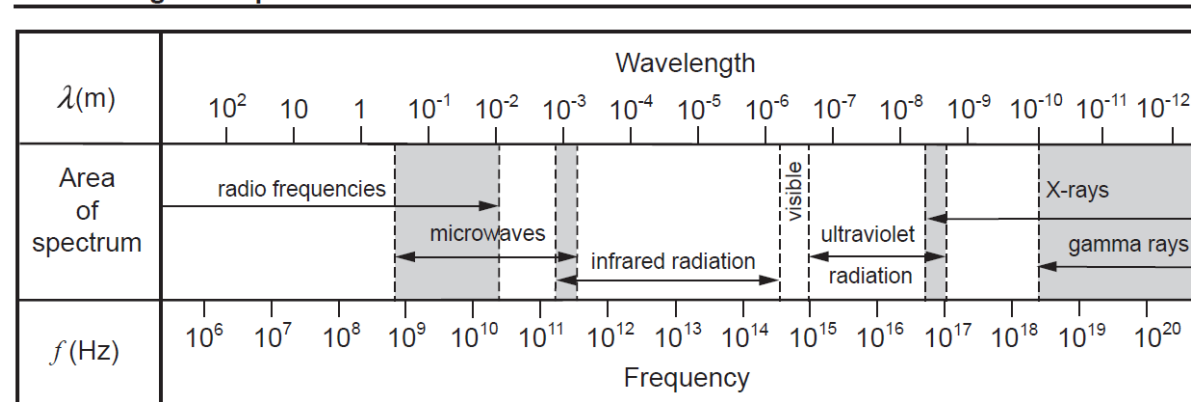
PLANCK'S EQUATION

- Example: Find the energy for a UV photon ($f = 2.00 \times 10^{15}$ Hz):
- Answer: $E = hf = (6.63 \times 10^{-34})(2.00 \times 10^{15}) = \underline{1.33 \times 10^{-18} \text{ J.}}$
- Because the energies involved are so small, it is common to convert these energies to electron-volts ($1 \text{ eV} = 1.60 \times 10^{-19} \text{ J}$). Therefore:
- $1.33 \times 10^{-18} \text{ J} / 1.60 \times 10^{-19} = 8.3125 = \underline{8.31 \text{ eV}}$

PLANCK'S EQUATION

- Example: A photon has an energy of 1.24 MeV. What part of the spectrum is it from?
- Answer: $(1.24 \times 10^6)(1.60 \times 10^{-19}) = 1.984 \times 10^{-13} \text{ J}$
- $f = E/h = 1.984 \times 10^{-13} / 6.63 \times 10^{-34} = 2.99 \times 10^{20} \text{ Hz} = \text{Gamma Ray/X-Ray}$

Electromagnetic spectrum



Note: shaded areas represent regions of overlap.

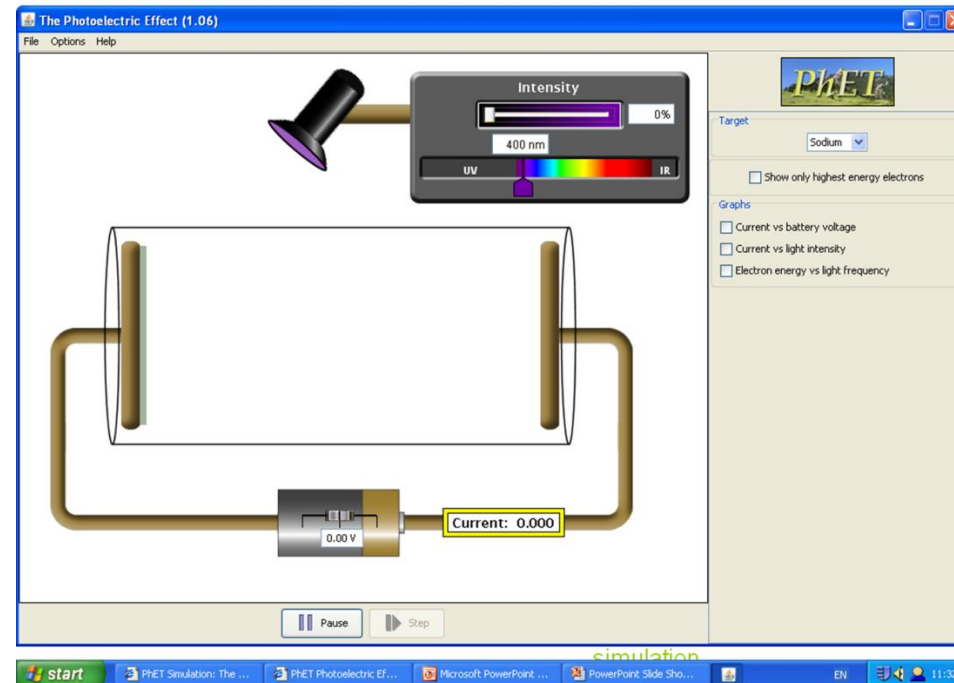
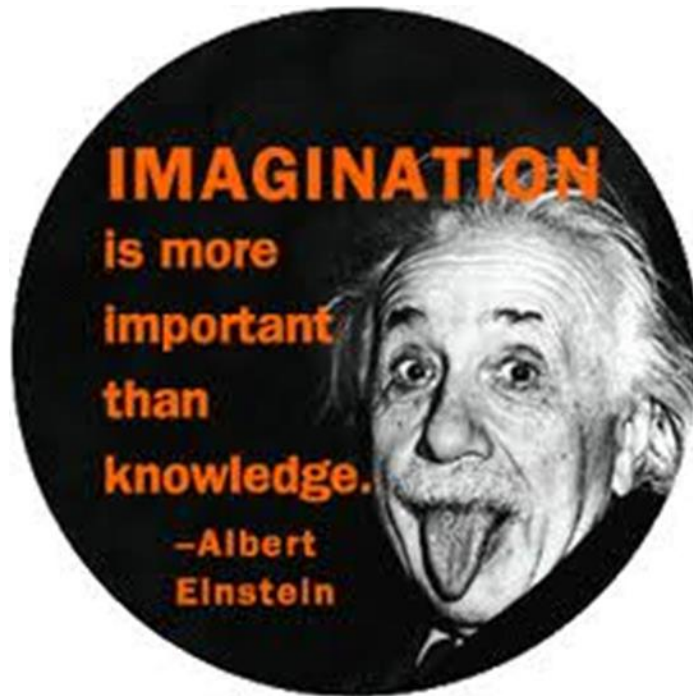
THE PHOTOELECTRIC EFFECT

- Another phenomenon that could not be explained via the wave model for light was the Photoelectric Effect.

Scientists noticed that when some types of electromagnetic radiation are incident on a piece of metal, the metal becomes positively charged. This positive charge is due to electrons being ejected from the surface of the metal. The electrons became known as **photoelectrons** because they were released due to light or other forms of electromagnetic radiation. The phenomenon is known as the **photoelectric effect**.

A common apparatus used to observe the photoelectric effect consists of a clean metal surface (the cathode) illuminated with light from an external source. If the light causes photoelectrons to be emitted, they are detected at the anode. The flow of electrons is called the **photocurrent** and is registered by a sensitive ammeter.

THE PHOTOELECTRIC EFFECT



The Photoelectric Effect

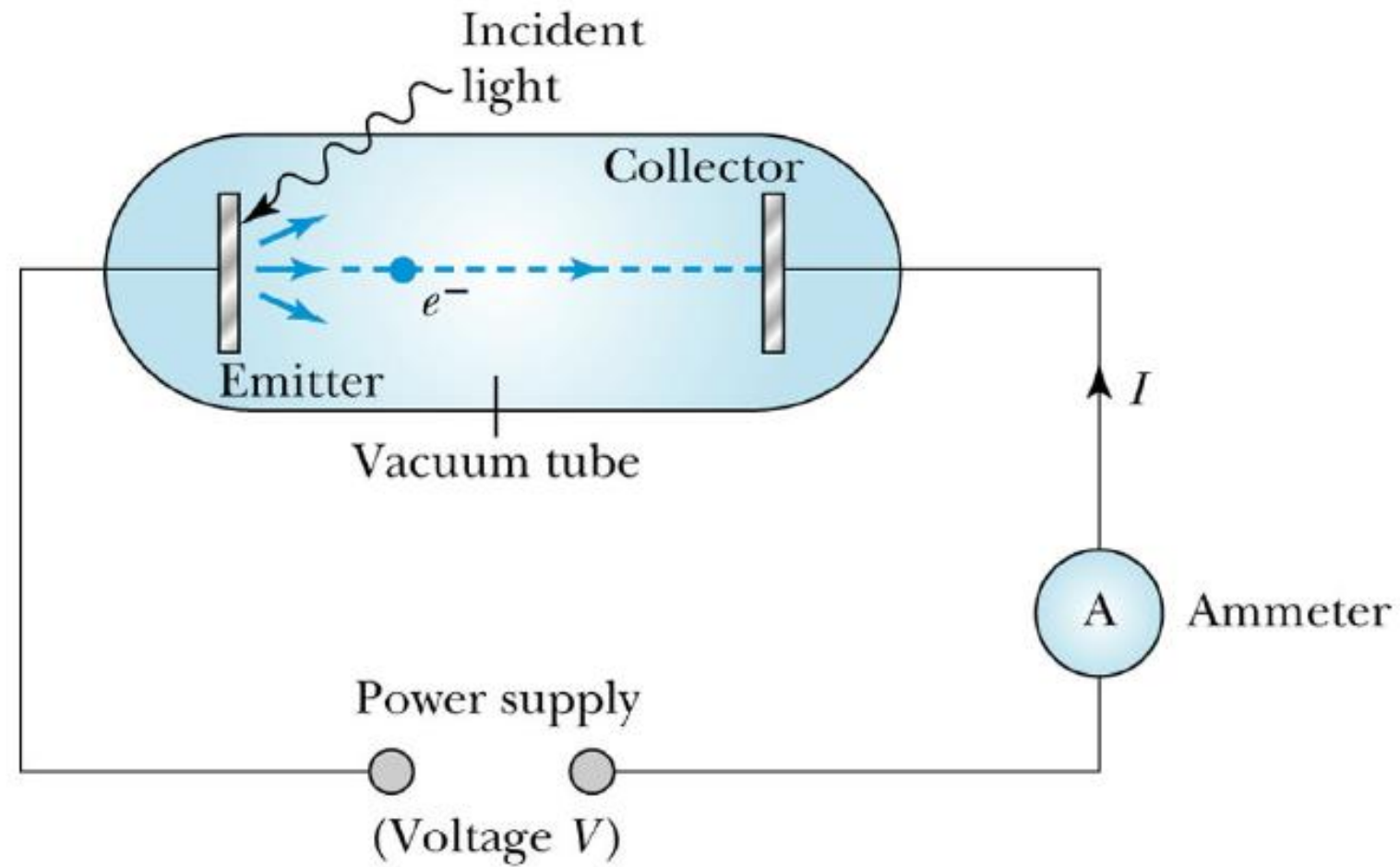
(sort of how a solar panel works)



We need to understand:

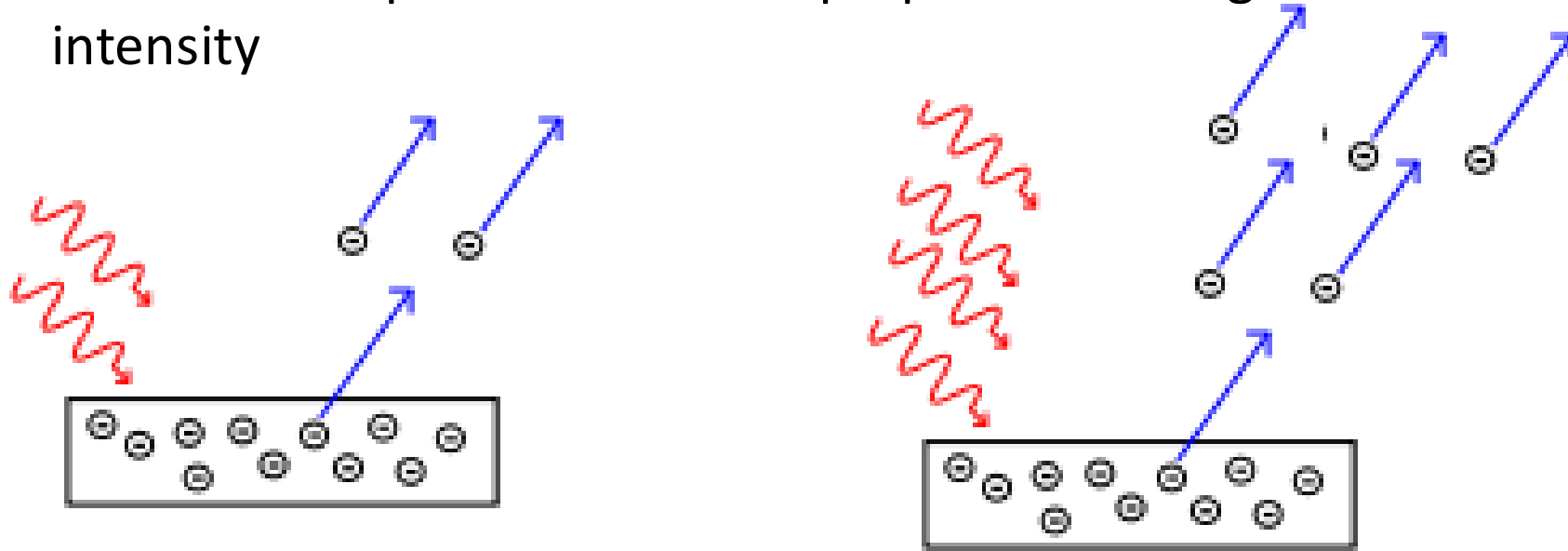
- Basic nature of the photoelectric effect
- Plank's equation for the energy of a photon ($E_{\text{ph}} = hf$)
- Frequency threshold
- Work function (Φ)
- Stopping voltage (stopping potential)
- Equation for an ejected photoelectron: $KE_{\text{max}} = hf - \phi$

Experimental setup to show
the photoelectric effect



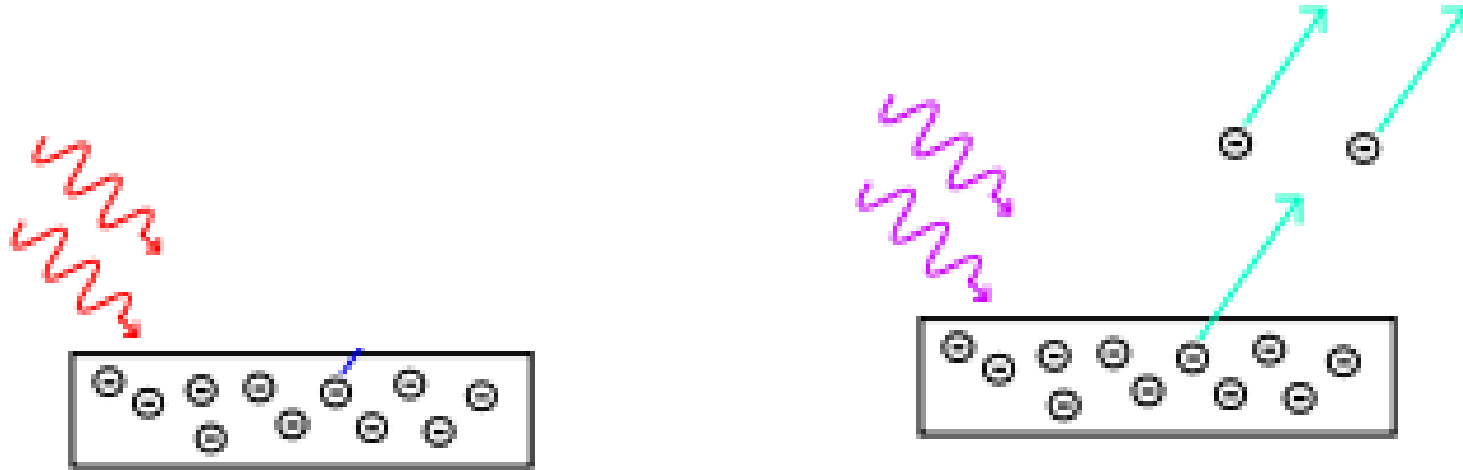
Laws of photoelectric effect

1. Number of photoelectrons is proportional to light intensity



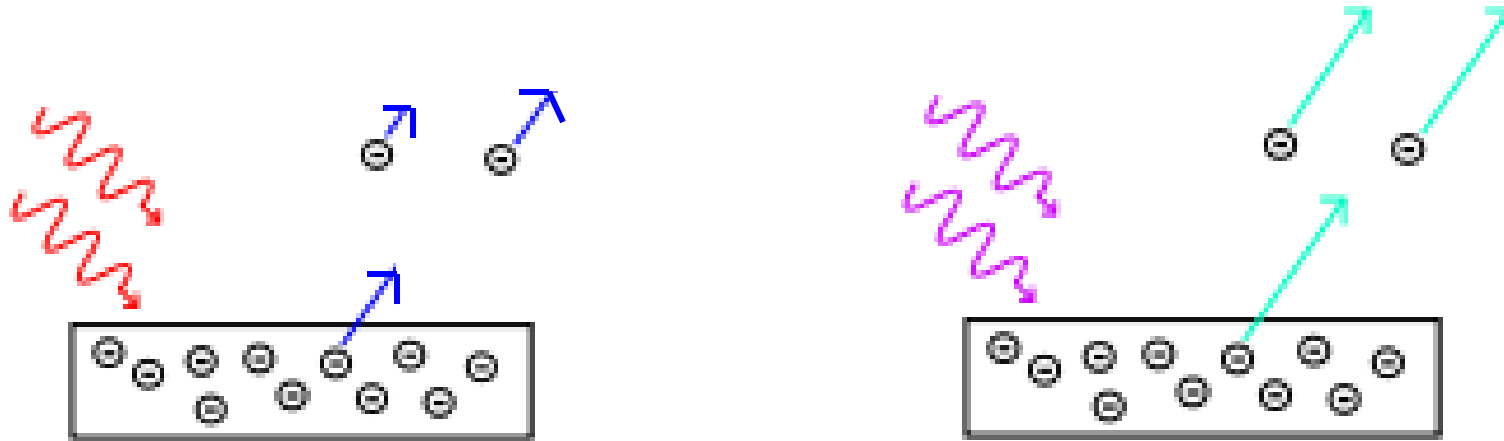
Laws of photoelectric effect

2. Each metal has a different threshold frequency below which no photoelectrons are produced irrespective of light intensity.



Laws of photoelectric effect

3. A range of energies are produced but the maximum value depends on colour (frequency)



Wave theory can't explain why colour is the most important factor in the photoelectric effect. It predicts that intensity should be the main factor.



1905.

№ 6.

ANNALEN DER PHYSIK.

BEGRÜNDET UND FORTGEFÜHRT DURCH

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VIERTE FOLGE.

BAND 17. HEFT 1.

DER GANZEN REIHE 322. BANDES 1. HEFT.

KURATORIUM:

F. KOHLRAUSCH, M. PLANCK, G. QUINCKE,
W. C. RÖNTGEN, E. WARBURG.

UNTER MITWIRKUNG

DER DEUTSCHEN PHYSIKALISCHEN GESELLSCHAFT

UND INSBESONDERE VON

M. PLANCK

HERAUSGEGEBEN VON

PAUL DRUDE.

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LEIPZIG, 1905.

VERLAG VON JOHANN AMBROSIVS BARTH.
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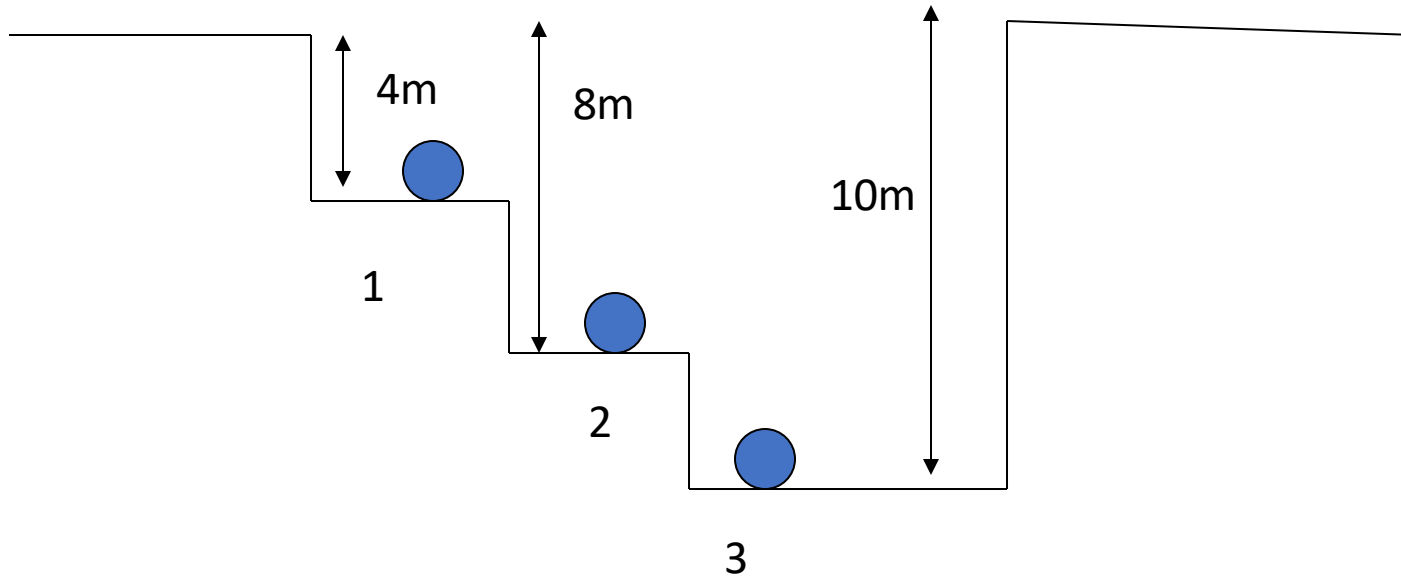
Albert Einstein

1879-1955

Nobel Prize in Physics 1921

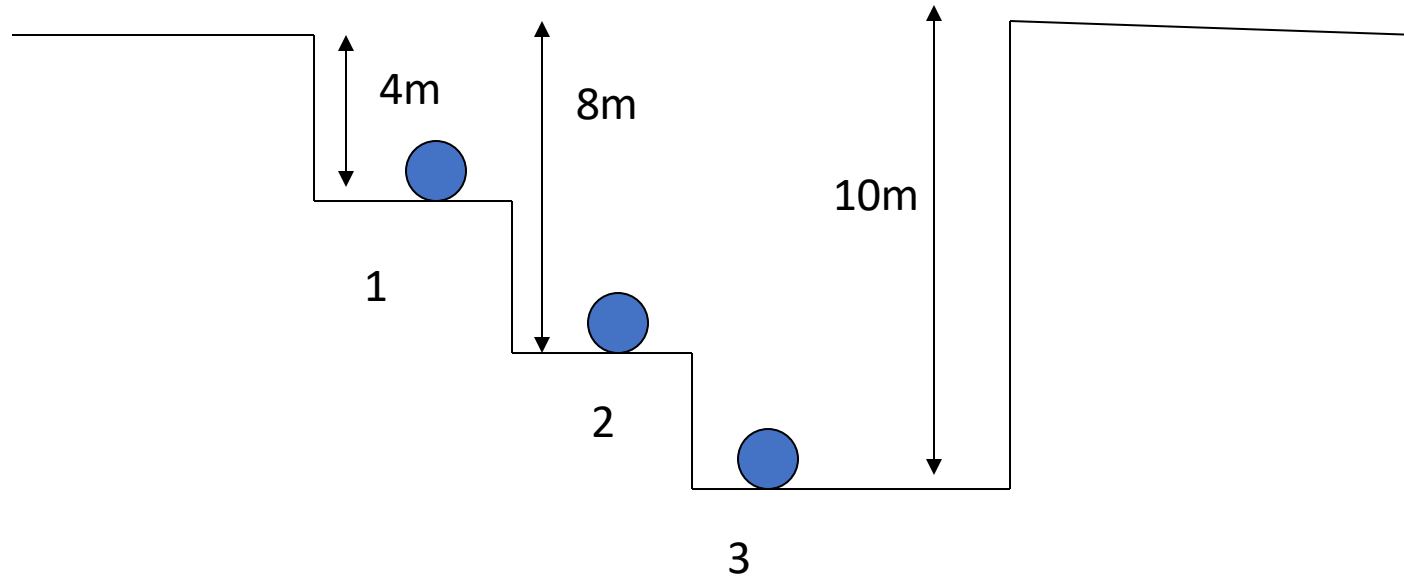


All balls are 2 kg.



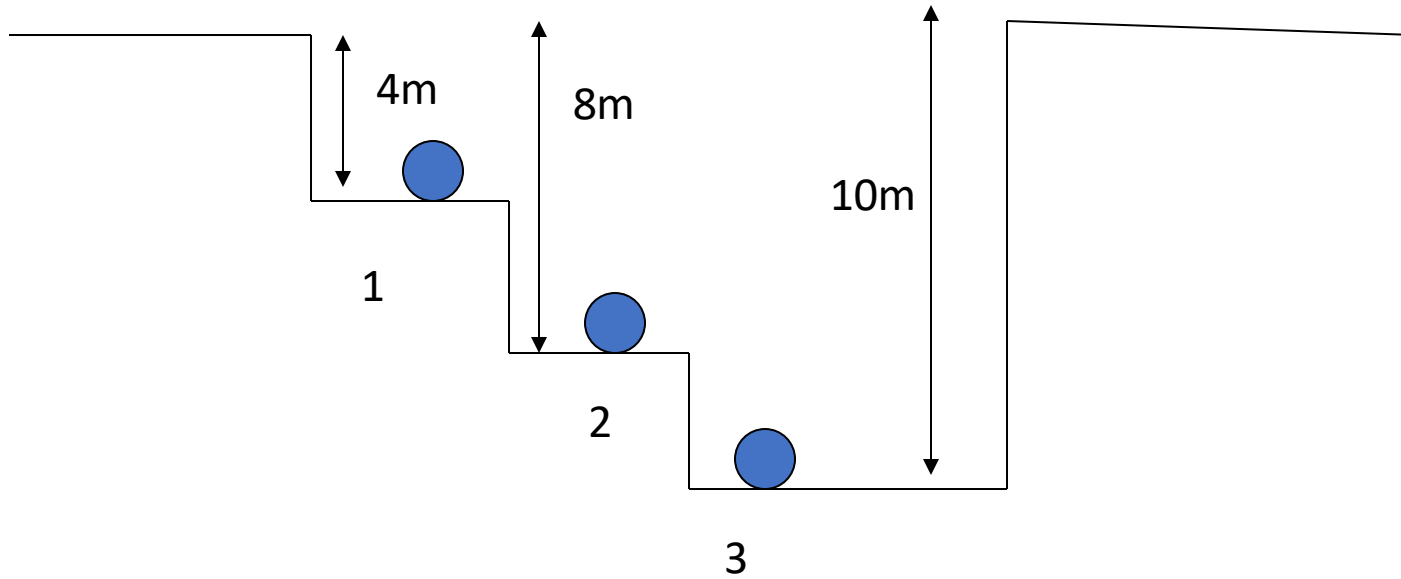
Give ball one **50 J** what happens?

All balls are 2 kg.



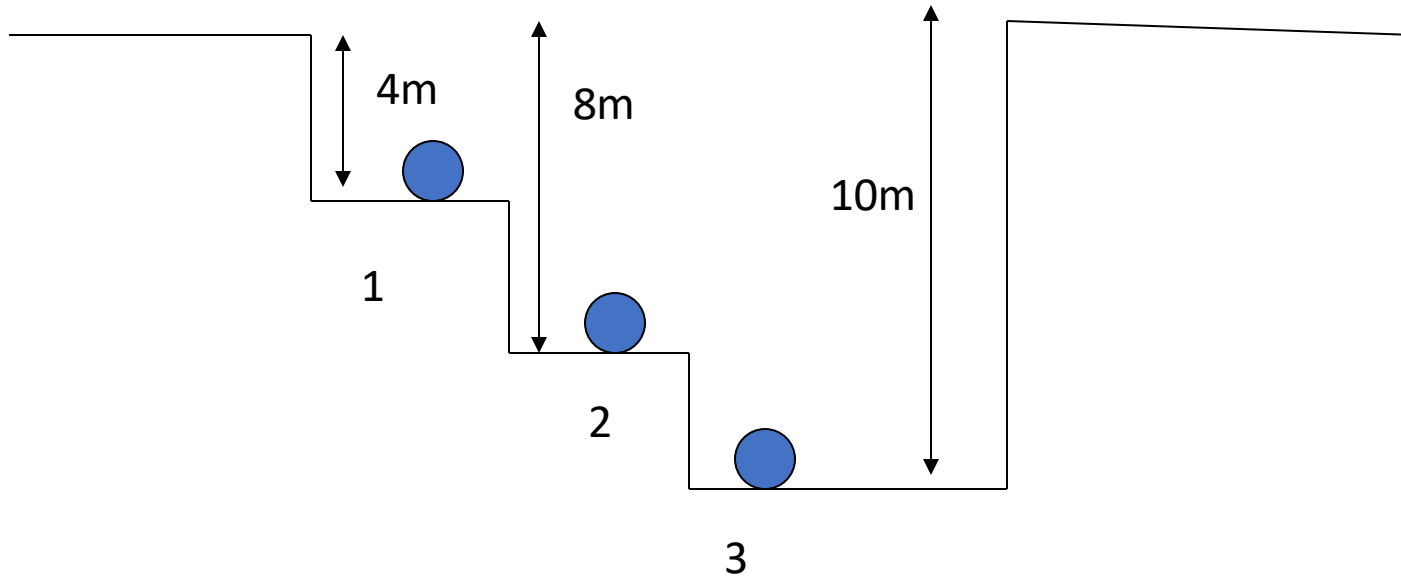
Give ball one **80 J** what happens?

All balls are 2 kg.



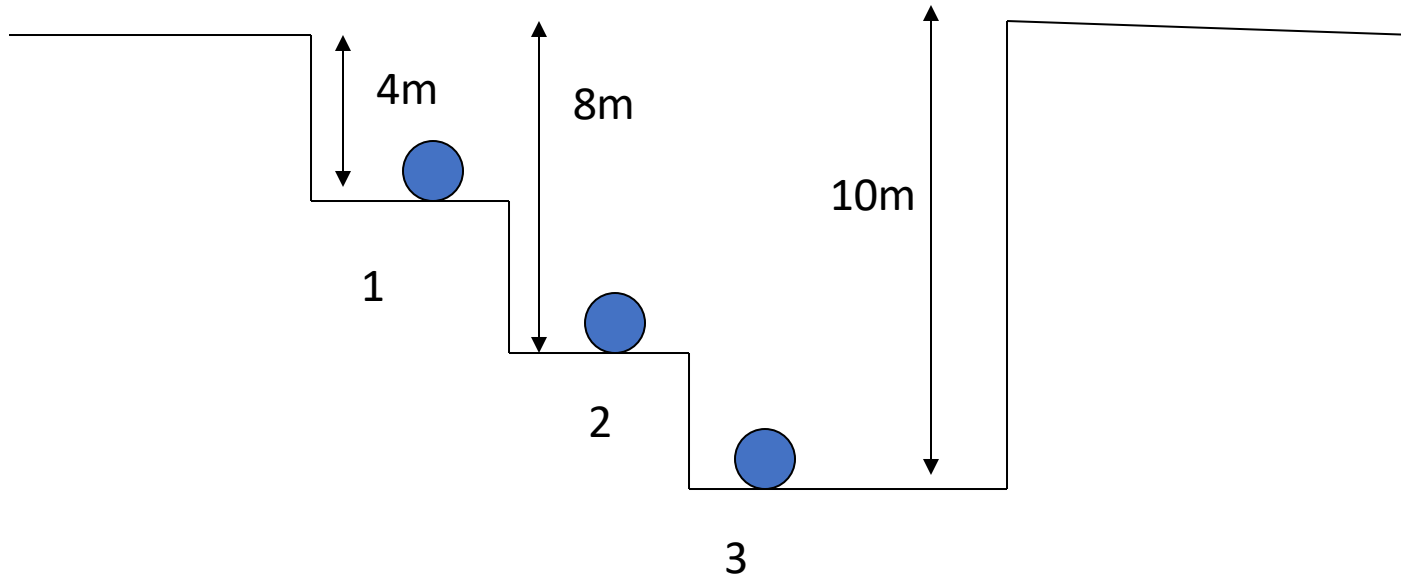
Give ball one **100 J** what happens?

All balls are 2 kg.



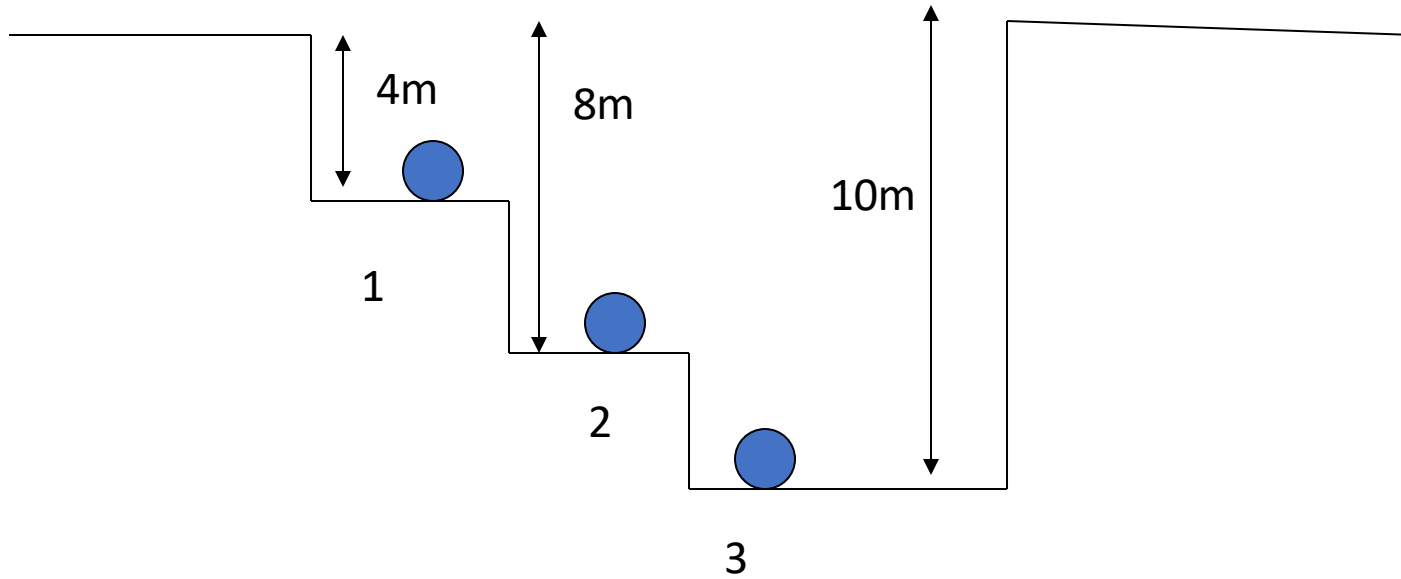
Give ball two **100 J** what happens?

All balls are 2 kg.



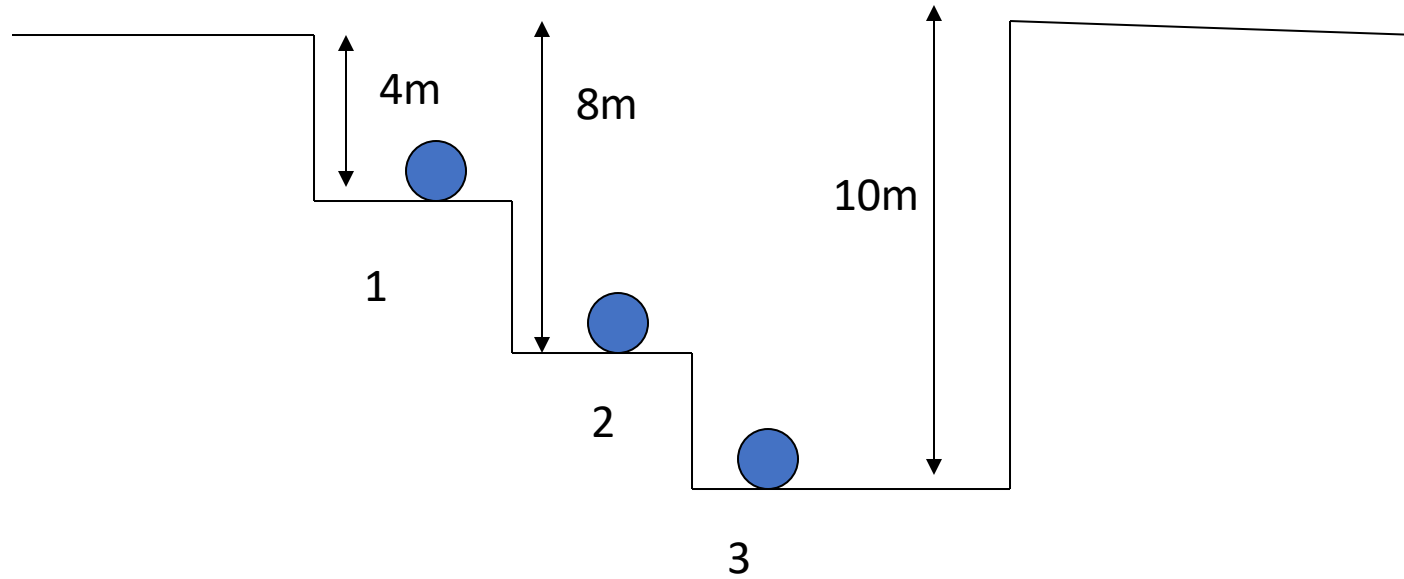
Give ball two **200 J** what happens?

All balls are 2 kg.

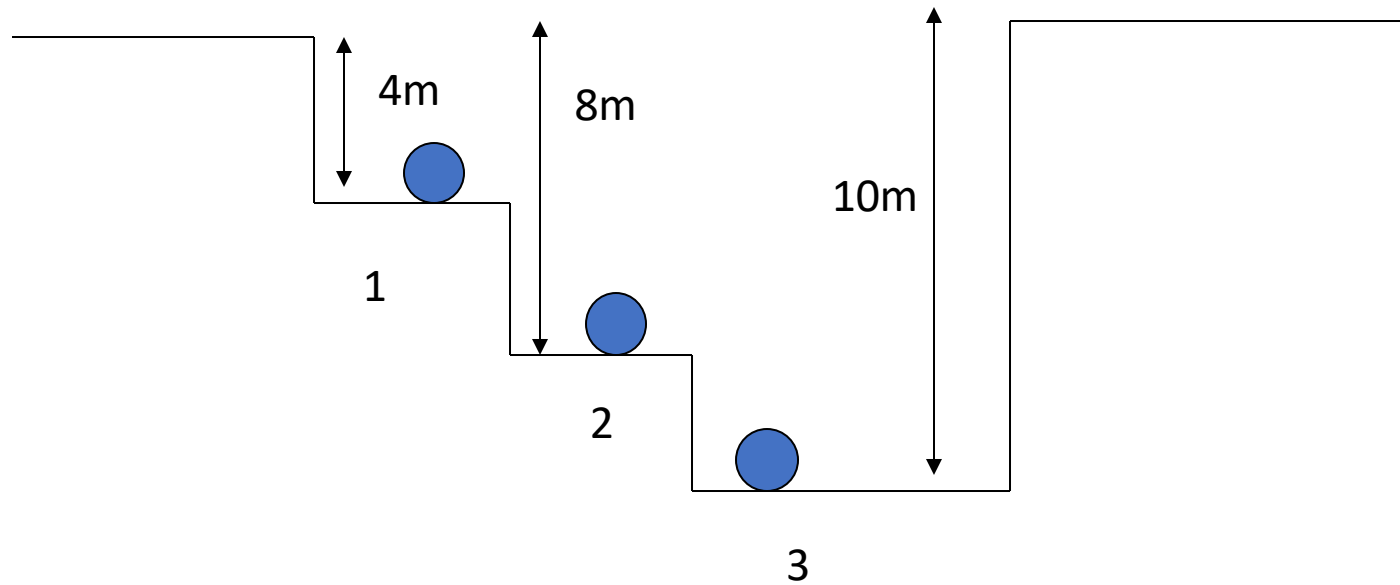


Give ball three **100 J** what happens?

All balls are 2 kg.



Give ball three **200 J** what happens?



Summary:

If energy is supplied in 'packets', then there is a minimum amount of energy for any ball to escape. Nothing will escape below this threshold.

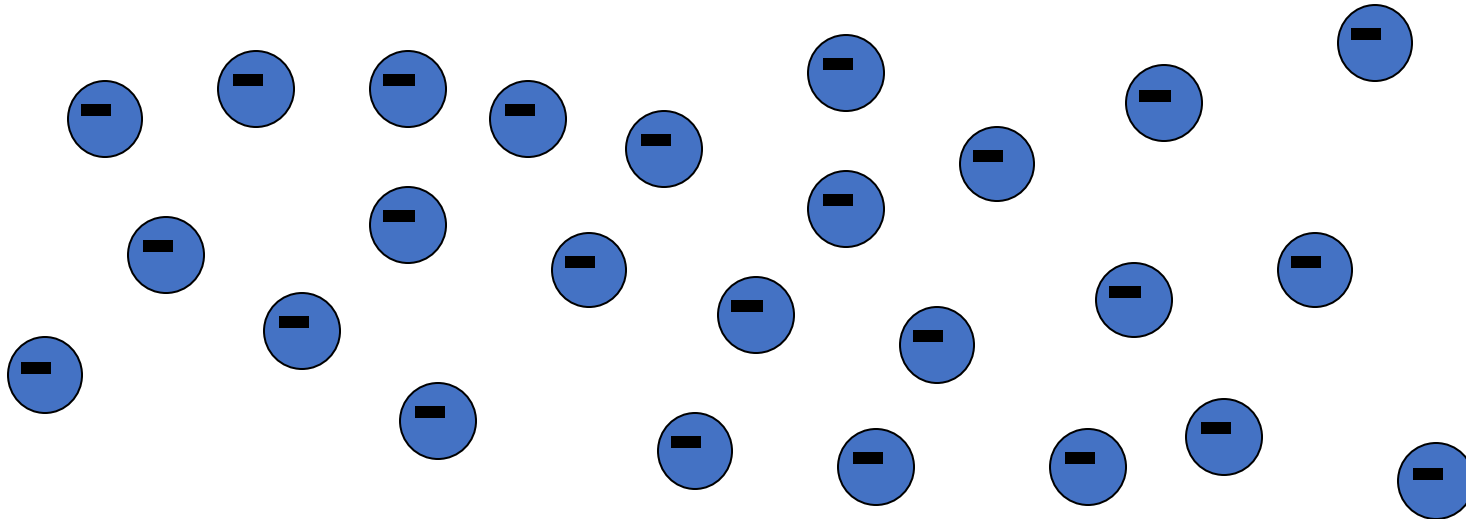
Remaining energy presents as a range of KE

If the energy supplied just equals threshold energy, then $KE = 0$

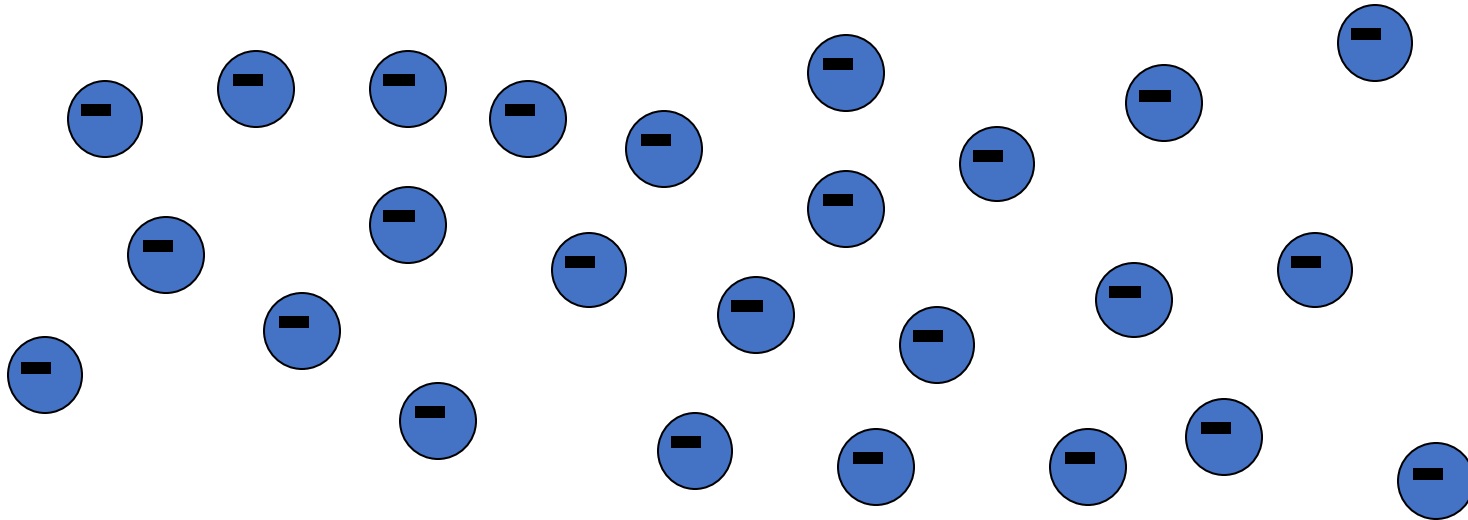


Einstein explained the photoelectric effect by assuming that light is made up of photons **each with an energy given by Planck's equation** $E_{ph} = hf$

(where Planck's constant is $6.63 \times 10^{-34} \text{ J s}$)



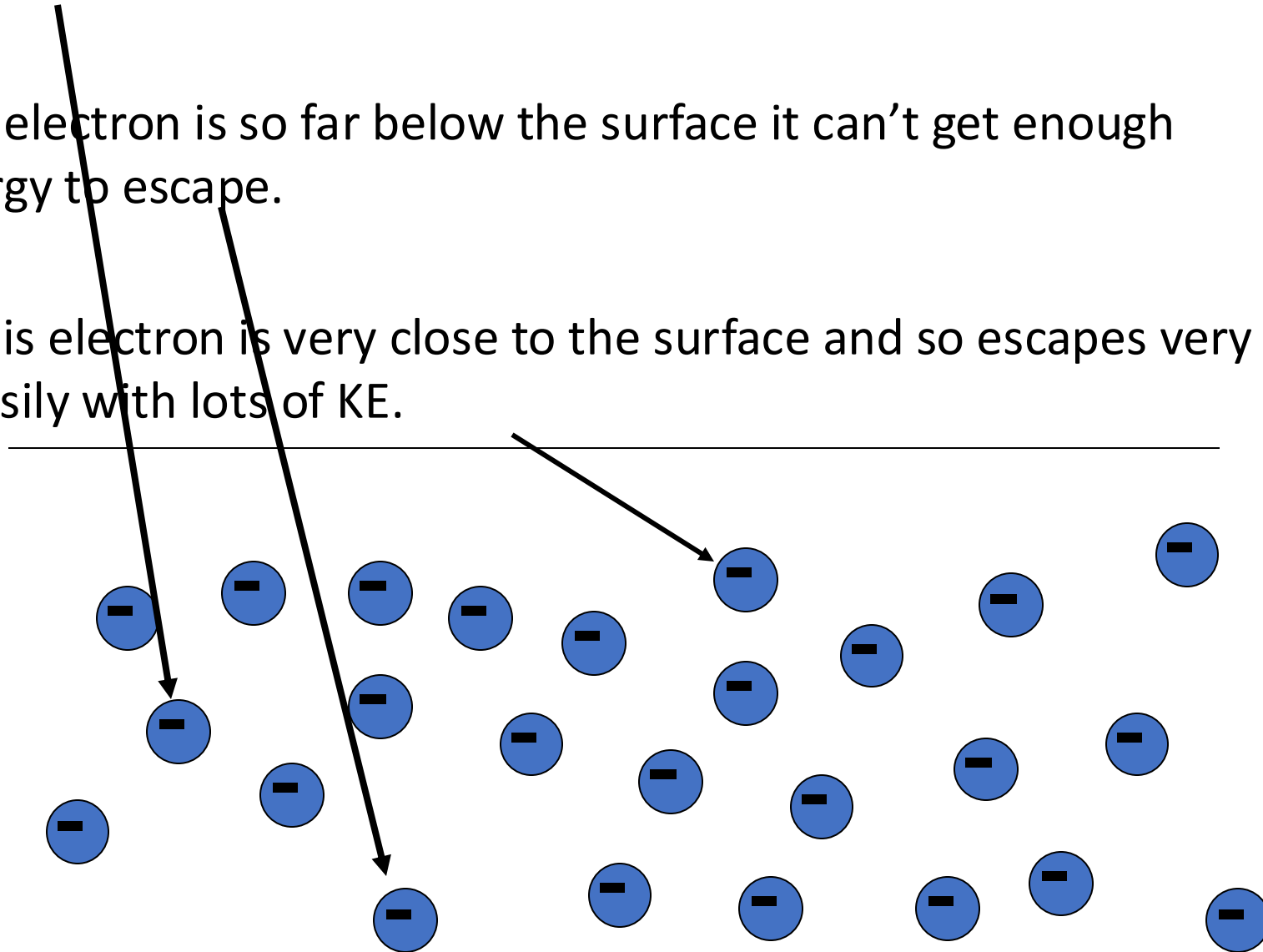
The electrons use some energy to escape from the metal and so escape with a range of kinetic energy



This electron is far below the surface and needs a lot of energy to escape. It emerges with very little KE.

This electron is so far below the surface it can't get enough energy to escape.

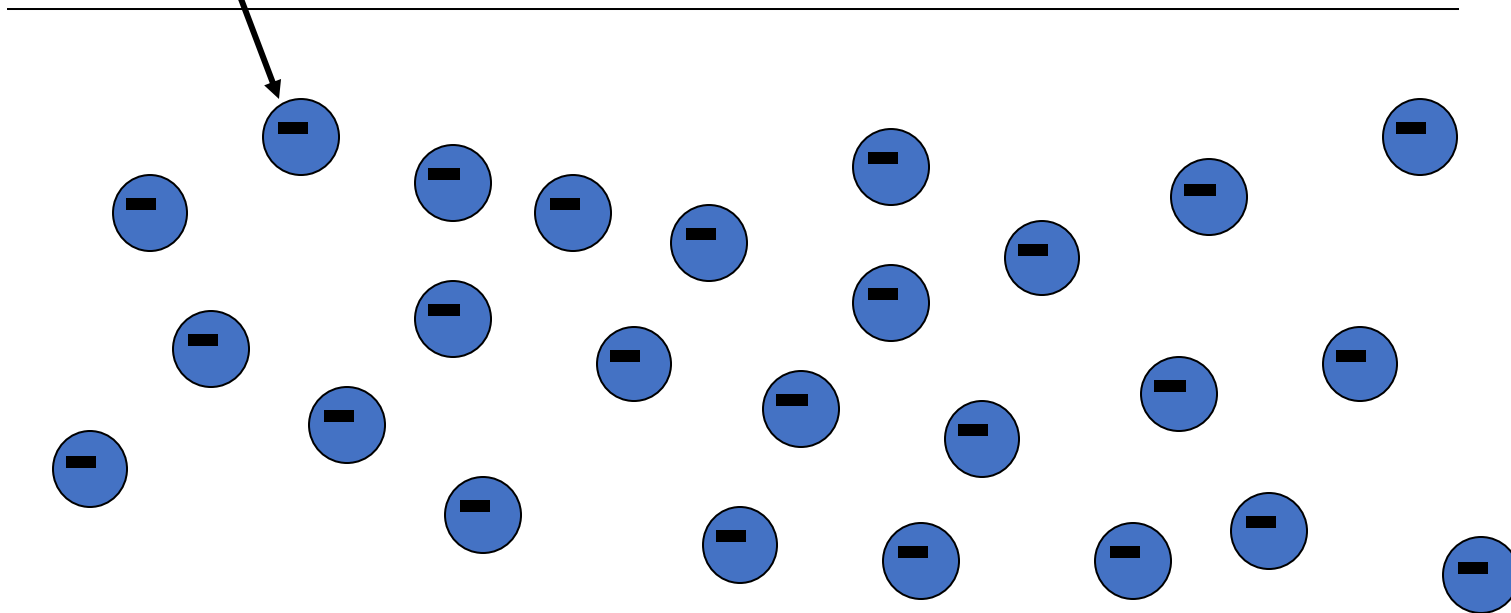
This electron is very close to the surface and so escapes very easily with lots of KE.



● This photon has an energy $E = hf$.

The minimum energy to knock an electron out of the metal – even at the surface of the metal - is called the ***work function***, Φ .

The maximum possible kinetic energy of an ejected photoelectron is the difference between the photon energy and the work function.



Einstein's Photoelectric equation :

$$K.E_{\max} = hf - \Phi$$

Maximum
possible
Kinetic energy
of ejected
electrons (J)

Energy of
photon (J)

Work
function
(J)

OR:

$$hf = \Phi + K_{max}$$

Table 3.3 Work Functions

Element	ϕ (eV)	Element	ϕ (eV)	Element	ϕ (eV)
Ag	4.64	K	2.29	Pd	5.22
Al	4.20	Li	2.93	Pt	5.64
C	5.0	Na	2.36	W	4.63
Cs	1.95	Nd	3.2	Zr	4.05
Cu	4.48	Ni	5.22		
Fe	4.67	Pb	4.25		

eV (electron-Volt: energy gained by an electron in an accelerating potential of 1V)

New unit of energy: $1 \text{ eV} = 1.6 \times 10^{-19} \text{ J}$

Threshold frequency $f_{\min}$

Emission of electrons from the surface can only occur if :

$$hf > \Phi$$

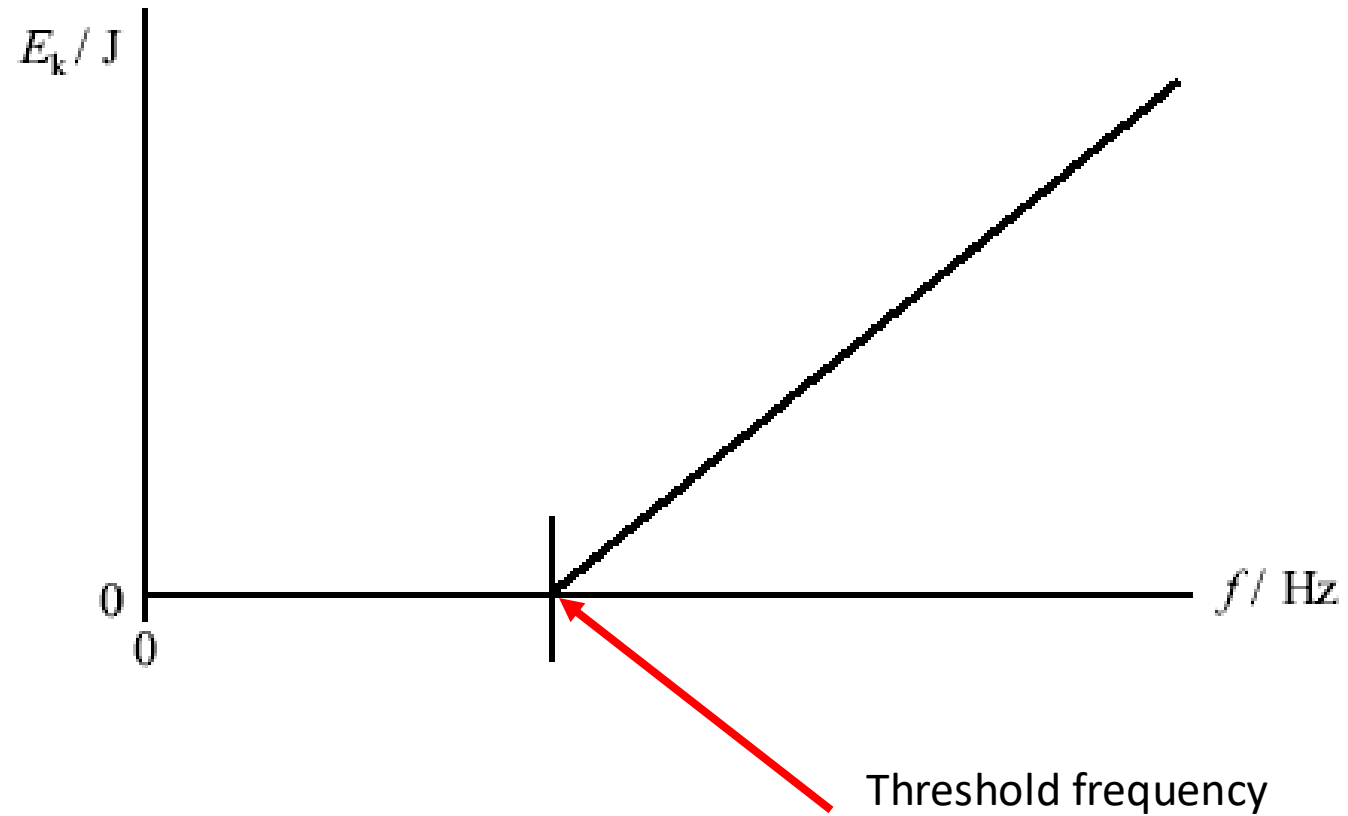
$$f > \frac{\Phi}{h}$$

$$f_{\min} = \frac{\Phi}{h}$$

The threshold frequency is the minimum frequency (of the incident radiation) required for an emission of an electron to occur. If the frequency is below this threshold, no electrons are ejected.

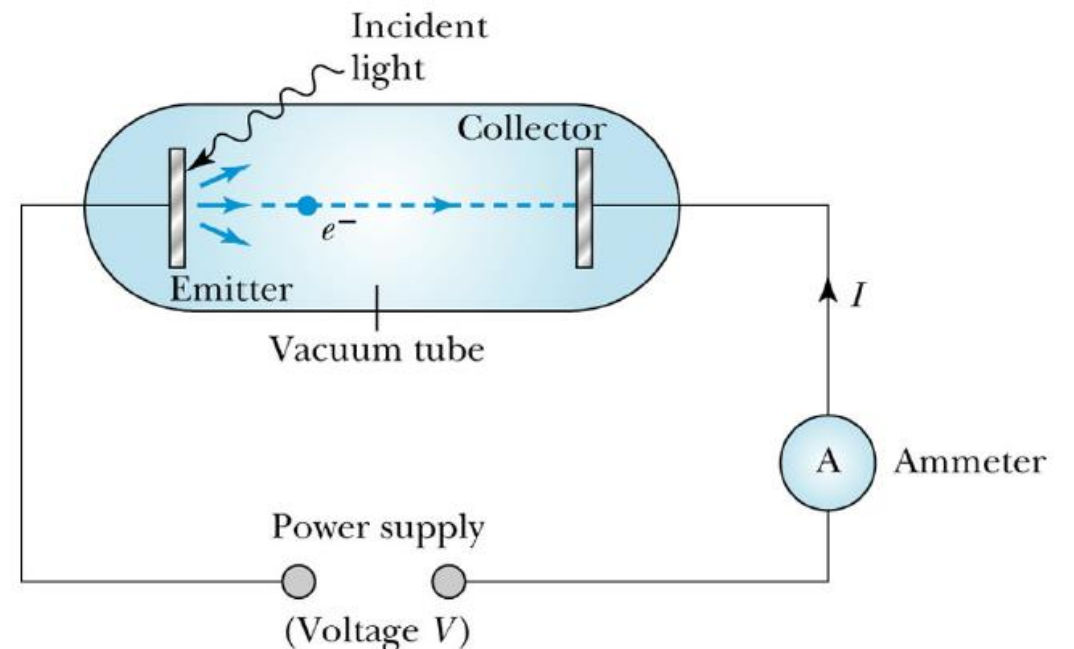
KE of the electrons = 0 when the frequency of the incident radiation is $f_{\min}$

KE_{max} versus frequency graph



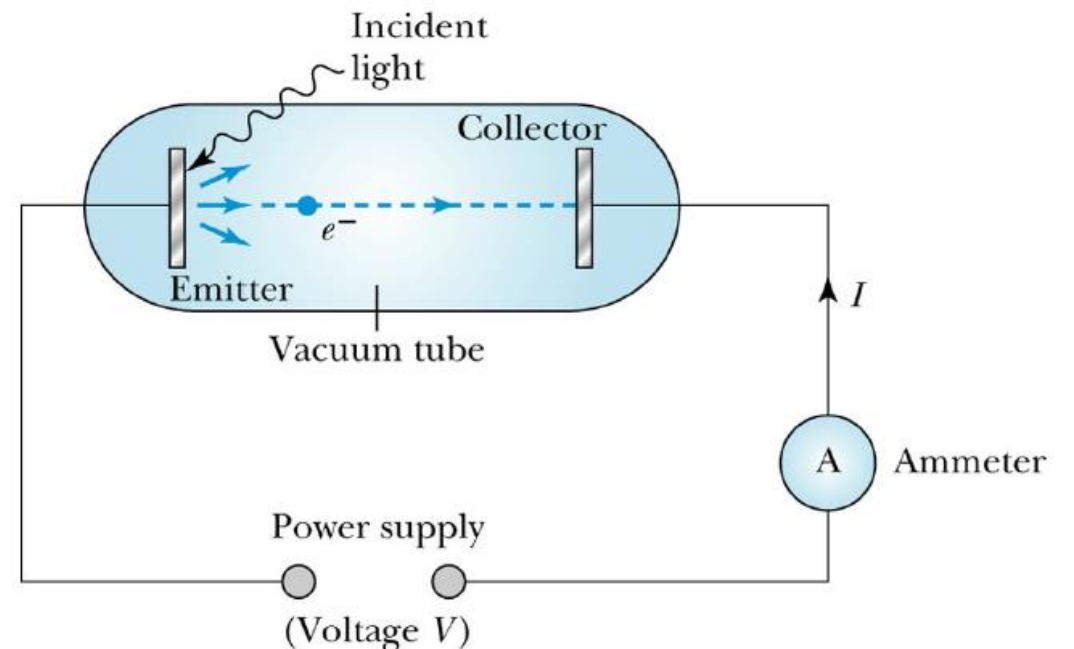
STOPPING VOLTAGE/BIAS

- Consider the setup for the Photoelectric Effect (below). Now, consider that the 'collector' plate was at a *positive* potential compared to the 'emitter' plate. The electrons would *accelerate* towards the collector plate when ejected from the emitter plate.
- What if the collector plate was at a *negative* potential instead?



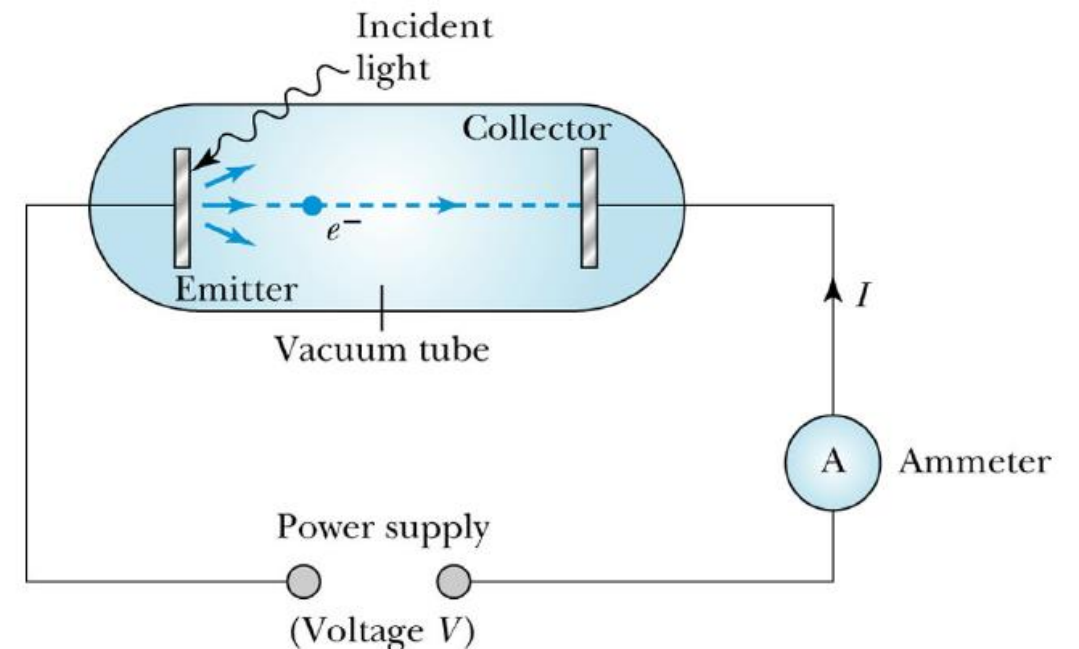
STOPPING VOLTAGE/BIAS

- The electrons, when ejected from the emitter, would begin to slow down as they approached the collector plate. If the potential difference was sufficient, then it would be possible to ensure that no electrons reach the collector plate at all. The minimum voltage required to do this is called the *stopping voltage*.



STOPPING VOLTAGE/BIAS

- In order to prevent *all* electrons from reaching the collector plate, we must do work on the electrons. The work done must be enough to ensure that even the most energetically ejected electrons have insufficient kinetic energy to reach the plate (by making the total energy go to 0).



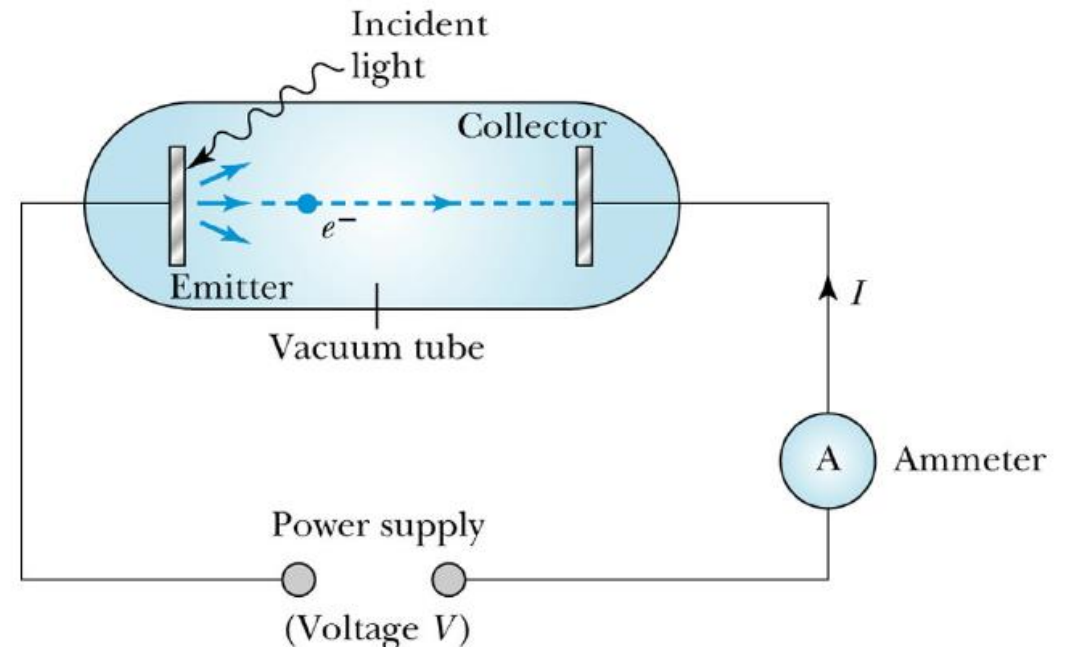
STOPPING VOLTAGE/BIAS

- The work done on a charged particle in an electric field (i.e. passing through a potential difference): $W = qV$.
- Therefore:

$$E_{K(Max)} = q_e V_S$$

Where q_e = electron charge &

V_S = Stopping Voltage.



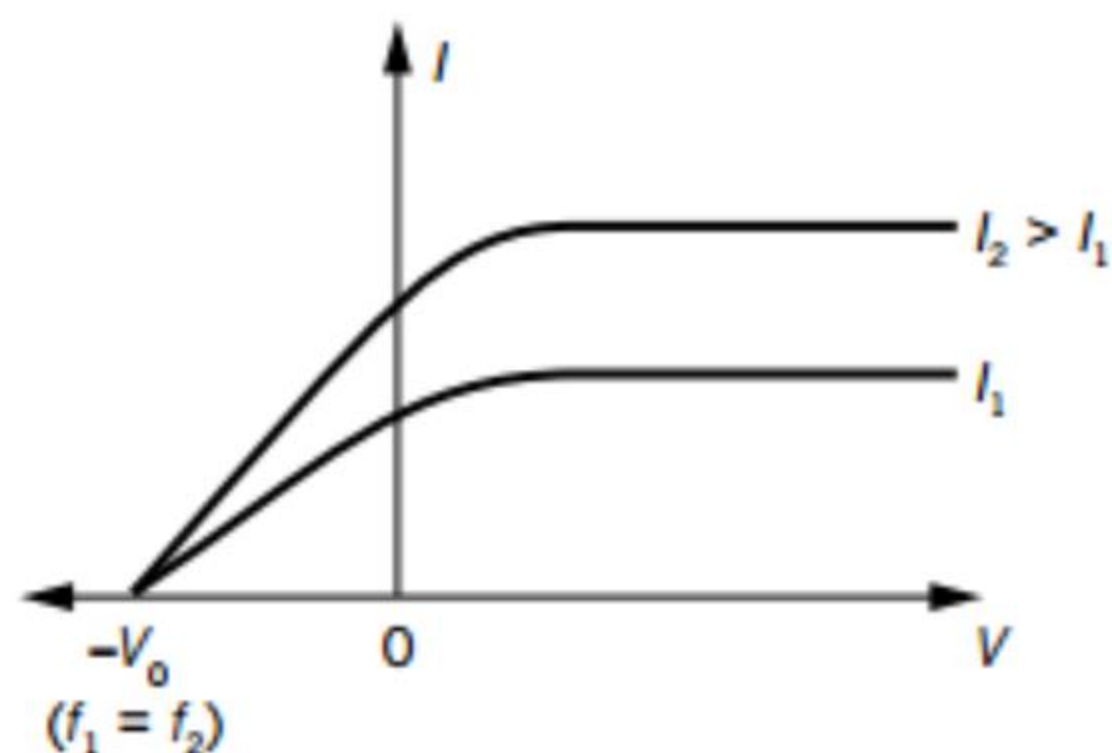


FIGURE 7.4.10 Photocurrent (I) plotted as a function of the voltage (V) applied between the cathode and the anode for different light intensities. For brighter light ($I_2 > I_1$) of the same frequency ($f_1 = f_2$), there is a higher photocurrent, but the same stopping voltage, V_0 .

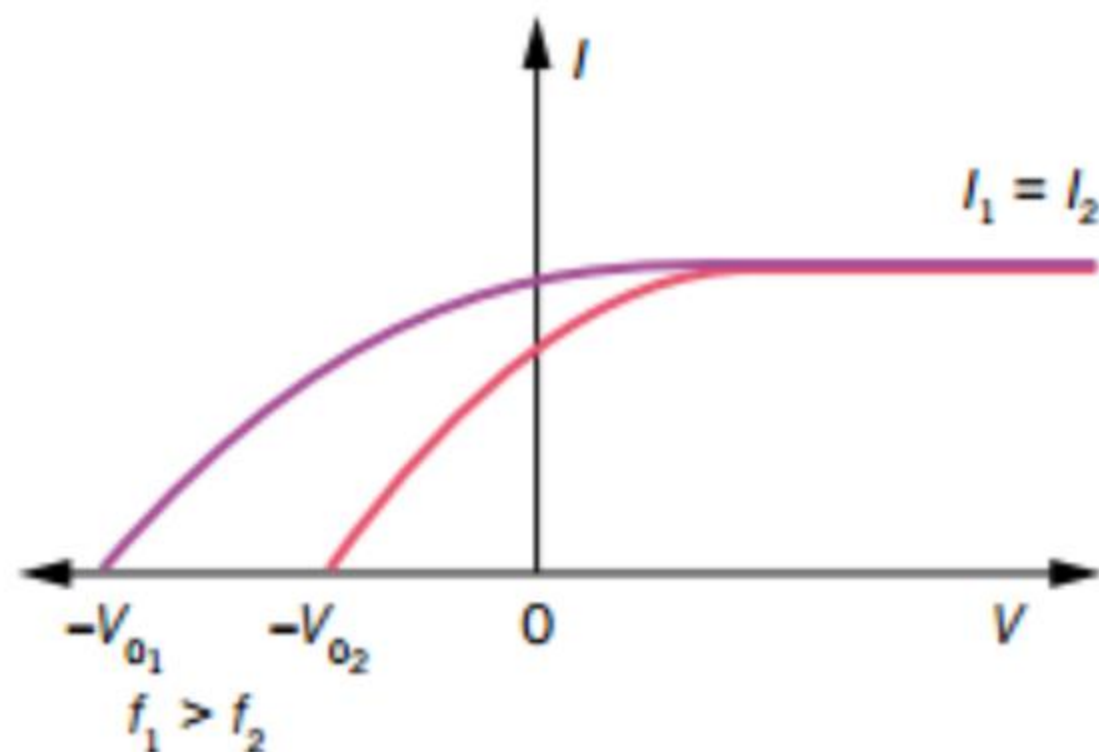


FIGURE 7.4.11 Photocurrent (I) plotted as a function of the voltage (V) applied between the cathode and the anode for different frequencies ($f_1 > f_2$) of incident light with the same intensity ($I_1 = I_2$). Both frequencies produce the same maximum photocurrent; however, light with the higher frequency requires a larger stopping voltage.

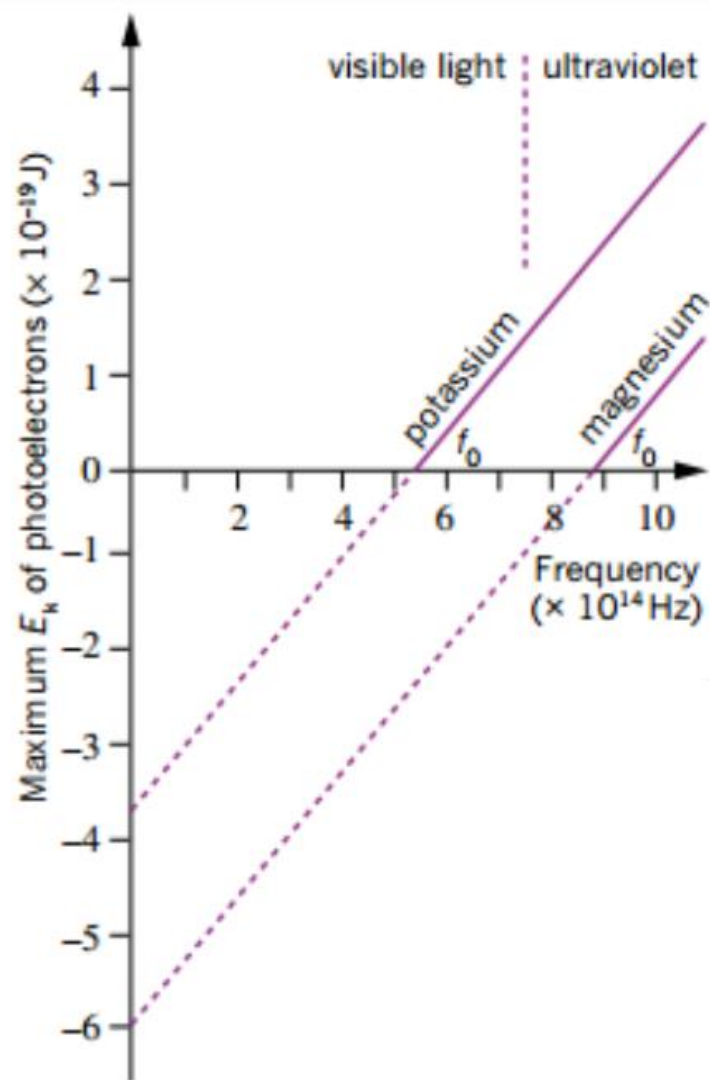
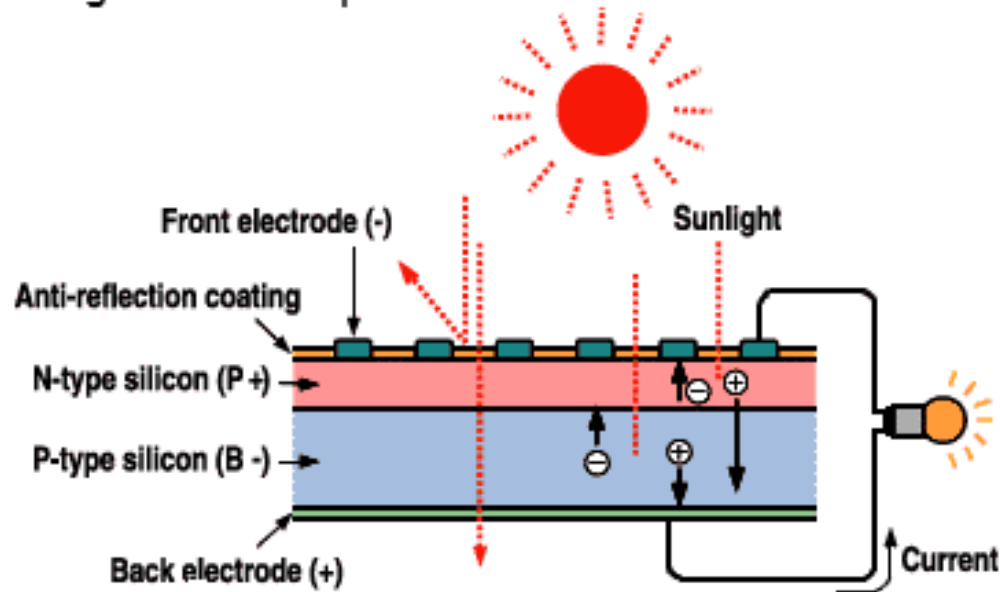


FIGURE 7.4.12 Magnesium has a high threshold frequency that is in the ultraviolet region. The x-intercept gives the threshold frequency, f_0 . For potassium, the threshold frequency is in the visible region. The gradient of the graph for each metal is Planck's constant, h . The magnitude of the y-intercept gives the work function, ϕ .

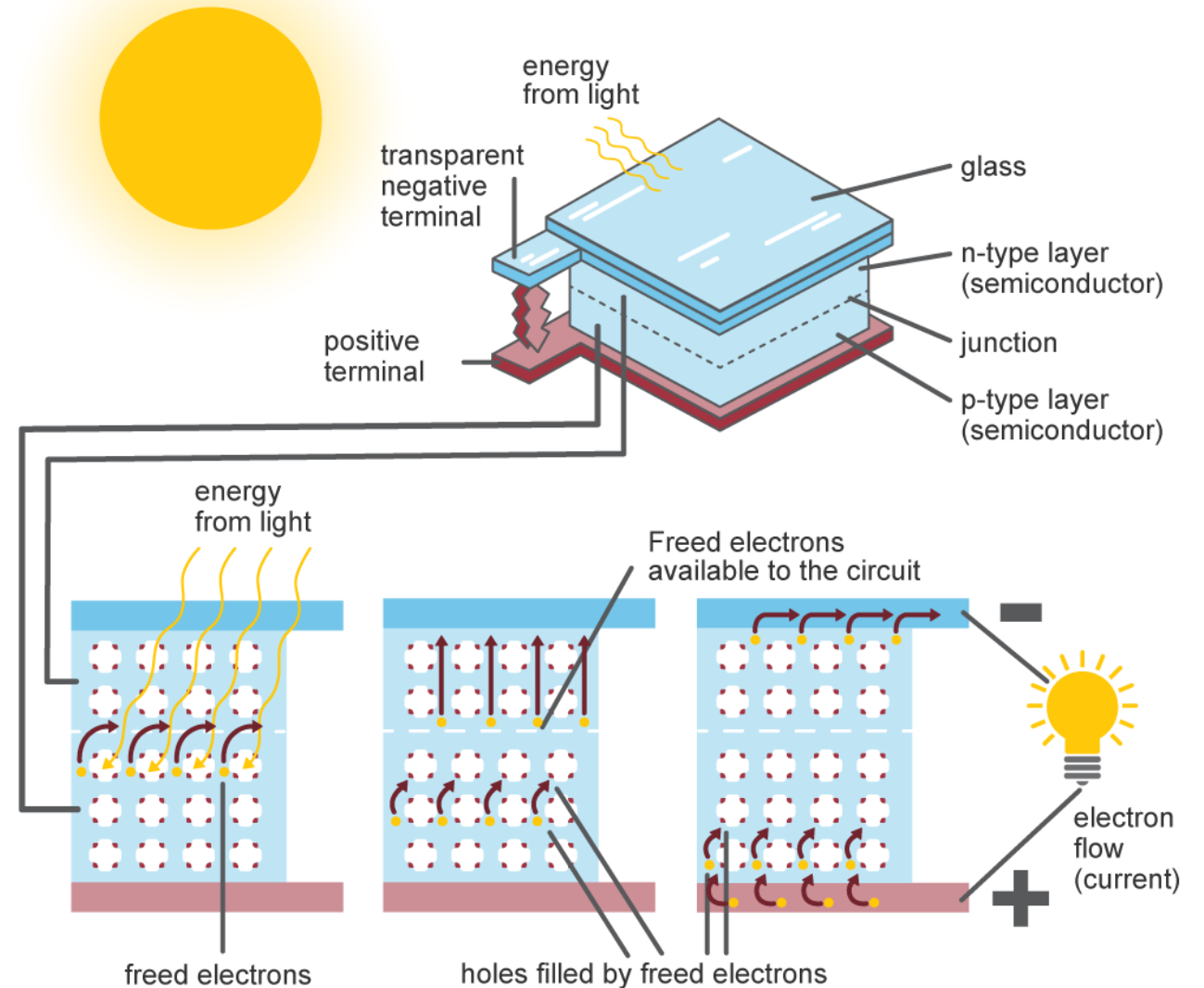
SOLAR PANELS

- So, how is this relevant to solar panels?

Diagram 1. The photovoltaic effect



Inside a photovoltaic cell



Source: U.S. Energy Information Administration

Summary

- Light comes in packets called photons.
- A photon's energy is given by $E = hf$
- Photons can knock electrons from a metal surface – photoelectron effect
- Each metal has a minimum photon energy to for an electron to be ejected (the work function, Φ)
- Thus, there is a minimum frequency for an electron to be ejected (**threshold frequency**)
- Photons *below* the threshold will never eject e^- s
- Photons *above* the threshold frequency can eject e^- s, and the extra energy ($hf - \Phi$) will be present as KE in the e^- s

Questions

1. Calculate the wavelength of light, in nm, illuminating a barium surface if the threshold frequency of barium is 6.07×10^{14} Hz and the kinetic energy of the electron emitted is 1.31×10^{-18} J.

2. If barium has a threshold frequency of 6.07×10^{14} Hz, calculate the kinetic energy of the electron emitted when barium is illuminated with a wavelength of 385 nm.

1. Calculate the wavelength of light, in nm, illuminating a barium surface if the threshold frequency of barium is 6.07×10^{14} Hz and the kinetic energy of the electron emitted is 1.31×10^{-18} J.

Ans. 116 nm

2. If barium has a threshold frequency of 6.07×10^{14} Hz, calculate the kinetic energy of the electron emitted when barium is illuminated with a wavelength of 385 nm.

Ans. 1.14×10^{-19} J.

3. Calculate the threshold wavelength, in nm, of a metal if that metal emits an electron with a kinetic energy of 6.32×10^{-19} J when illuminated with light that has a frequency of 1.97×10^{15} Hz.

4. Calculate the frequency of light illuminating a copper surface if the threshold frequency of copper is 1.08×10^{15} Hz and the velocity of the electron emitted is 1.11×10^6 m/s.

3. Calculate the threshold wavelength, in nm, of a metal if that metal emits an electron with a kinetic energy of 6.32×10^{-19} J when illuminated with light that has a frequency of 1.97×10^{15} Hz.

Ans. 295 nm

4. Calculate the frequency of light illuminating a copper surface if the threshold frequency of copper is 1.08×10^{15} Hz and the velocity of the electron emitted is 1.11×10^6 m/s.

Ans. 1.92×10^{15} Hz.

5. An instrument is sensitive to light packets that carry at least 1.03×10^{-16} J of energy. How many photons of light of frequency of 3.02×10^{13} Hz can the instrument detect?

6. A laser emits a pulse of light containing 2.46×10^{-16} J of energy at a frequency of 3.25×10^{14} Hz. How many photons are present in the pulse?

7. Calculate the threshold wavelength, in nm, of a metal if that metal emits an electron with a kinetic energy of 2.09×10^{-19} J when illuminated with light that has a frequency of 1.40×10^{15} Hz.

Extension questions

5. An instrument is sensitive to light packets that carry at least 1.03×10^{-16} J of energy. How many photons of light of frequency of 3.02×10^{13} Hz can the instrument detect?

Ans. 5144 photons.

~~Ans. 5160 photons.~~

6. A laser emits a pulse of light containing 2.46×10^{-16} J of energy at a frequency of 3.25×10^{14} Hz. How many photons are present in the pulse?

Ans: 1140

7. Calculate the threshold wavelength, in nm, of a metal if that metal emits an electron with a kinetic energy of 2.09×10^{-19} J when illuminated with light that has a frequency of 1.40×10^{15} Hz.

Ans. 278 nm

-
- 6 (a) (i) Explain the meaning of the term *work function* of a metal.

.....

.....

.....

- (ii) State what you would need to change in an experiment to investigate the effect of the work function on the photoelectric effect.

.....

(3 marks)

- (b) Experiments based on the photoelectric effect support the particle theory of light. State **one** conclusion drawn from these experiments and explain how it supports the particle theory.

You may be awarded marks for the quality of written communication in your answer.

.....

.....

.....

.....

.....

.....

(2 marks)

- (c) Monochromatic light of wavelength $4.80 \times 10^{-7} \text{ m}$ falls onto a metal surface which has a work function of $1.20 \times 10^{-19} \text{ J}$.

Calculate

- (i) the energy, in J, of a single photon of this light,

.....
.....
.....
.....
.....

- (ii) the maximum kinetic energy, in J, of an electron emitted from the surface.

.....
.....
.....
.....
.....

(5 marks)

- 6 (a) (i) Explain the meaning of the term *work function* of a metal.

the minimum energy ✓
energy required to eject a (photo)electron (from the metal surface) ✓

- (ii) State what you would need to change in an experiment to investigate the effect of the work function on the photoelectric effect.

changing the metal/cathode ✓

(3 marks)

- (b) Experiments based on the photoelectric effect support the particle theory of light. State **one** conclusion drawn from these experiments and explain how it supports the particle theory.

You may be awarded marks for the quality of written communication in your answer.

conclusion : light below a threshold frequency does not release electrons ✓
explanation: photons carry quanta of energy ✓

[or conclusion: electrons are emitted immediately the light hits the metal surface ✓
explanation : photons carry quanta of energy ✓]

(2 marks)

- (c) Monochromatic light of wavelength $4.80 \times 10^{-7} \text{ m}$ falls onto a metal surface which has a work function of $1.20 \times 10^{-19} \text{ J}$.

Calculate

- (i) the energy, in J, of a single photon of this light,

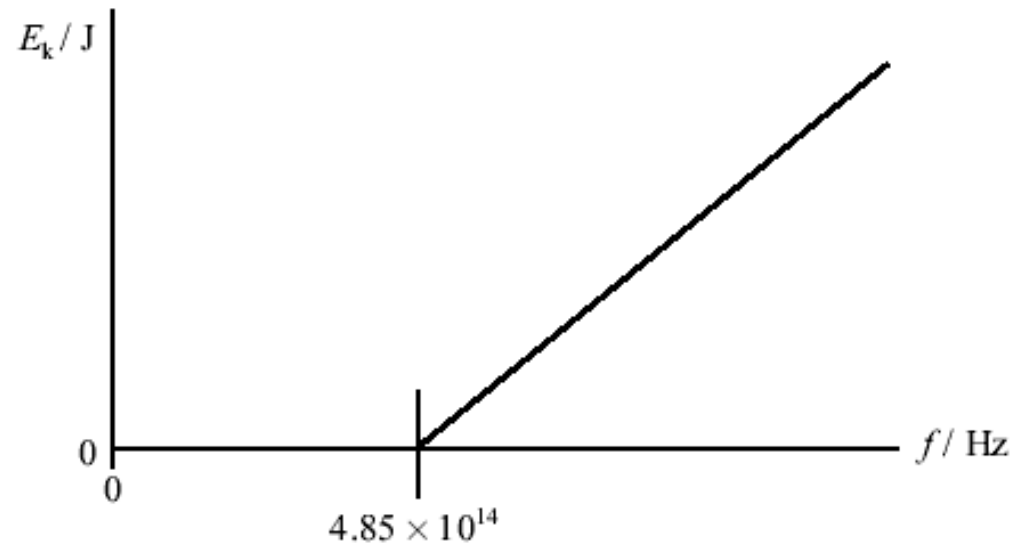
$$\text{(use of } E = hf = \frac{hc}{\lambda} \text{ gives) } E = \frac{6.63 \times 10^{-34} \times 3 \times 10^8}{4.80 \times 10^{-7}} \checkmark = 4.14 \times 10^{-19} \text{ (J)} \checkmark$$

- (ii) the maximum kinetic energy, in J, of an electron emitted from the surface.

$$\begin{aligned} hf &= \phi + E_k \checkmark \\ E_k &= 4.14 \times 10^{-19} - 1.20 \times 10^{-19} \checkmark \\ &= 2.94 \times 10^{-19} \text{ (J)} \checkmark \\ &\text{(allow C.E for value of } E \text{ from (i))} \end{aligned}$$

(5 marks)

- 4 The graph shows how the maximum kinetic energy, E_k , of electrons emitted from the surface of metal A, varies with the frequency, f , of the incident electromagnetic radiation.



- (a) Explain the meaning of the term *threshold frequency*.

.....

.....

(2 marks)

(b) Calculate the work function of metal A,

(i) in J,

.....

.....

.....

(ii) in eV.

.....

.....

(3 marks)

- (c) Metal A is exchanged for metal B. The work function of metal B is half that of metal A. Add a line to the graph opposite to show how the maximum kinetic energy of electrons emitted from metal B varies with frequency. *(2 marks)*

- (d) At a fixed frequency of the incident radiation, the emitted electrons are collected and a current is recorded.

State and explain a change that may be made to increase this current.

You may be awarded marks for the quality of written communication in your answer.

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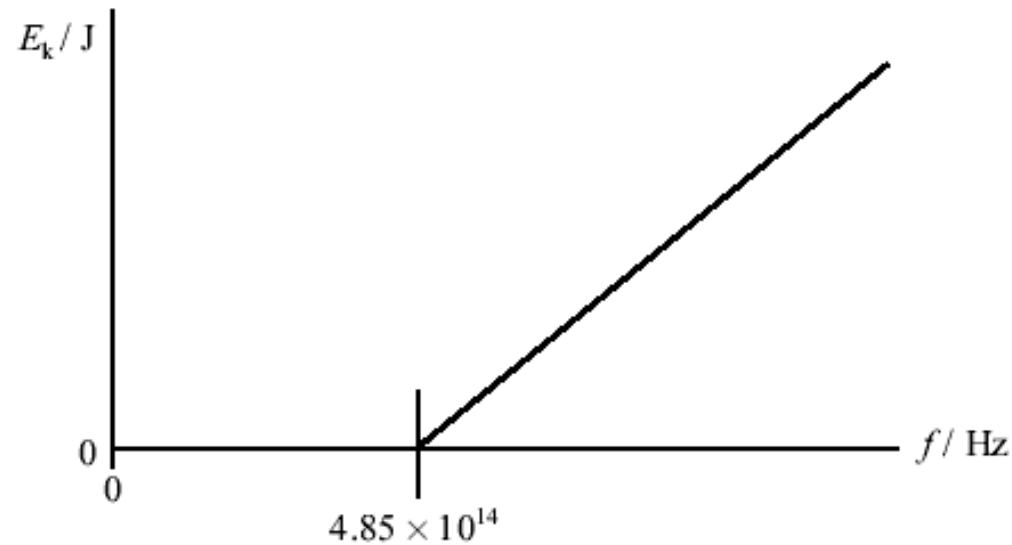
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(3 marks)

- 4 The graph shows how the maximum kinetic energy, E_k , of electrons emitted from the surface of metal A, varies with the frequency, f , of the incident electromagnetic radiation.



- (a) Explain the meaning of the term *threshold frequency*.

(a) the minimum frequency (of radiation) ✓
required to eject photoelectrons ✓

(2 marks)

(b) Calculate the work function of metal A,

(i) in J,

$$\begin{aligned} \text{(b)(i)} \quad (\text{use of } \phi = hf_0 \text{ gives}) \quad \phi &= 6.63 \times 10^{-34} \times 4.85 \times 10^{14} \quad \checkmark \\ &= 3.22 \times 10^{-19} \text{ (J)} \quad \checkmark \end{aligned}$$

.....

(ii) in eV.

$$\text{(b)(ii)} \quad \phi \left(= \frac{3.22 \times 10^{-19}}{1.60 \times 10^{-19}} \right) = 2.01 \text{ (eV)} \quad \checkmark$$

(3 marks)

- (c) Metal A is exchanged for metal B. The work function of metal B is half that of metal A. Add a line to the graph opposite to show how the maximum kinetic energy of electrons emitted from metal B varies with frequency. (2 marks)

(c) line parallel to the given line ✓
with half the value of the x - intercept ✓

- (d) At a fixed frequency of the incident radiation, the emitted electrons are collected and a current is recorded.

State and explain a change that may be made to increase this current.

You may be awarded marks for the quality of written communication in your answer.

(d) statement : increase the light intensity/brightness ✓

explanation : more incident photons (per second)
one photon interacts with one electron (any two)
more emitted electrons (per second)
greater rate of flow charge

.....
(3 marks)